

Game Theory (Applications)

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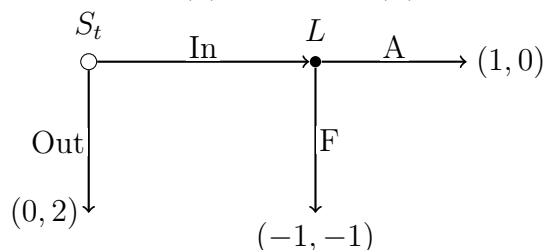
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1 Applications of Dynamic Games with Incomplete Information

1.1 Reputation Games

We investigate a general model of reputation through the chain store game.

Definition 1.1 (Chain Store Game). We have a sequence of short-run players ($S_t, t = 1, \dots, T$) which plays against a single long-run player (L). Each S_t decides whether to enter (In) or stay out (Out). Player L chooses fight (F) or acquiesce (A) whenever S_t enters. All earlier actions are observable.



For any finite T , the unique SPE by backward induction is (In, A). However, we want to see if it is possible for L to build up reputation so S_t does not enter.

Reputation Set Up:

- Suppose that L is either “sane” (with payoffs given in the game), or “crazy” (commitment type), always playing F against an entrant
- L knows whether he is sane or crazy but S_t does not
- Let ε_t denote the belief of S_t at the beginning of period t that L is crazy
 1. $\varepsilon_1 > 0$ is the prior probability of L being crazy (given, commonly known)
 2. in PBE, beliefs must be consistent with play, so ε_t is the expected probability of L being crazy at the beginning of stage t
- We can find the unique PBE by working backwards

We consider the PBE for any $T = 2$ by working backwards.

At $t = 2$, it is the last period so the only play consistent with PBE is

1. if S_2 enters, then sane L plays A, crazy L plays F
2. S_2 enters if $\varepsilon_2(-1) + (1 - \varepsilon_2)(1) > 0$ i.e. $\varepsilon_2 < \frac{1}{2}$
3. If $\varepsilon_2 > \frac{1}{2}$, S_2 stays out
4. If $\varepsilon_2 = \frac{1}{2}$, then S_2 may mix. Let m denote the probability that S_2 plays In

We can now determine the PBE at $t = 1$

Proposition 1.1. If $\varepsilon_1 \geq \frac{1}{2}$ then S_1 plays Out. However, if S_1 comes In, then L replies with F. In either case, we have $\varepsilon_2 = \varepsilon_1$ and so S_2 stays out

Proof. If the sane L's strategy is F at $t = 1$, then playing F at $t = 1$ induces $\varepsilon_2 = \varepsilon_1 \geq \frac{1}{2}$ (since playing F does not transmit information about whether L is crazy or sane and so posterior does not update on prior) and so S_2 stays out. Formally,

$$\varepsilon_2 = \mathbb{P}[\text{crazy}|\text{F}] = \frac{\mathbb{P}[\text{F}|\text{crazy}]\mathbb{P}[\text{crazy}]}{\mathbb{P}[\text{F}|\text{crazy}]\mathbb{P}[\text{crazy}] + \mathbb{P}[\text{F}|\text{sane}]\mathbb{P}[\text{sane}]}$$

Since $\mathbb{P}[\text{F}|\text{crazy}] = 1$ by hypothesis and since sane L's strategy is F, $\mathbb{P}[\text{F}|\text{sane}] = 1$ as well. So, $\varepsilon_2 = \mathbb{P}[\text{crazy}] = \varepsilon_1$. If the sane L's strategy at $t = 1$ is F with probability less than 1, then $\varepsilon_2 > \varepsilon_1 \geq \frac{1}{2}$ and so S_2 stays out after F is played at $t = 1$ as well. We know this since this implies $\mathbb{P}[\text{F}|\text{sane}] < 1$, making the denominator smaller than 1 and hence $\varepsilon_2 > \mathbb{P}[\text{crazy}] = \varepsilon_1$

Playing F is best for sane L at $t = 1$ because $-1 + 2 > 0$ i.e. the total payoffs in $t = 1, 2$ is higher than the unique SPE. Therefore, S_1 stays out at $t = 1$, since sane L's best reply is F at $t = 1$. No update on ε_2 , and therefore S_2 stays out too. ■

Proposition 1.2. If $\varepsilon < \frac{1}{2}$ and S_1 plays In, then sane L must mix F and A in response i.e. at $t = 1$

Proof. AFC that sane L's equilibrium strategy is (pure) F at $t = 1$, then $\varepsilon_2 = \varepsilon_1 < \frac{1}{2}$ (since ε_2 is not updated). Hence S_2 is not deterred. Then, sane L strictly prefers to play A at $t = 1$, which is a contradiction.

AFC that sane L's equilibrium strategy is (pure) A at $t = 1$, then deviating to F induces $\varepsilon_2 = 1$, which ‘fools’ and deters S_2 , which is preferred by sane L as $-1 + 2 > 0$ which is a contradiction again.

Therefore, sane L's best response to S_1 going In at $t = 1$ must be mixed by Nash's theorem. ■

The intuition here is that sane L must not imitate the crazy type ‘too perfectly’. If sane L plays F with probability 1 (like the crazy type), then playing F does not build a reputation for being crazy

Proposition 1.3. Suppose $\varepsilon_1 < \frac{1}{2}$ and S_1 goes In. If L plays A, then at $t = 2$, S_2 goes In as well. If L plays F at $t = 1$, then S_2 mixes In with probability $m = \frac{1}{2}$

Proof. If L plays A at $t = 1$, then S_2 learns L is sane and plays In.

If playing F at $t = 1$ deters S_2 from entering, then L strictly prefers F over A at $t = 1$. If S_2 definitely enters having seen (In, F) at $t = 1$, then the opposite is true. By **proposition 1.2**, L is indifferent between F and A at $t = 1$, therefore S_2 must mix after seeing (In, F) at $t = 1$. From $-1 + (1 - m)2 = 0$, we obtain $m = \frac{1}{2}$ ■

When sane L happens to play F at $t = 1$ (acting like crazy type), this move must not deter S_2 with probability 1, because then sane L would be tempted to overuse the deterrent.

Proposition 1.4. If $\varepsilon_1 < \frac{1}{2}$ and S_1 plays In, sane L plays F with probability $q = \frac{\varepsilon_1}{1 - \varepsilon_1}$. Furthermore, S_1 prefers to stay Out if $\varepsilon_1 > \frac{1}{4}$

Proof. First part: Sane L mixes at $t = 1$ to make S_2 indifferent between In and Out.

Let q be sane L's probability of playing F at $t=1$. By Bayes' rule

$$\varepsilon_2 = \mathbb{P}(\text{L crazy} | \text{F at } t = 1) = \frac{\varepsilon_1}{\varepsilon_1 + (1 - \varepsilon_1)q}$$

So S_2 is indifferent between In and Out iff $\varepsilon_2 = \frac{1}{2}$. Hence sane L must set

$$q = \frac{\varepsilon_1}{1 - \varepsilon_1}$$

Second part: If S_1 enters then crazy L plays F at $t = 1$ for sure and (as we have just derived), sane L plays F with probability $q = \frac{\varepsilon_1}{1 - \varepsilon_1}$.

The total probability that L (crazy or sane) plays F at $t = 1$ is

$$\varepsilon_1 + (1 - \varepsilon_1)q = 2\varepsilon_1$$

Hence, S_1 prefers to stay out if this probability exceeds $\frac{1}{2}$ since this is the inequality for the expected payoff

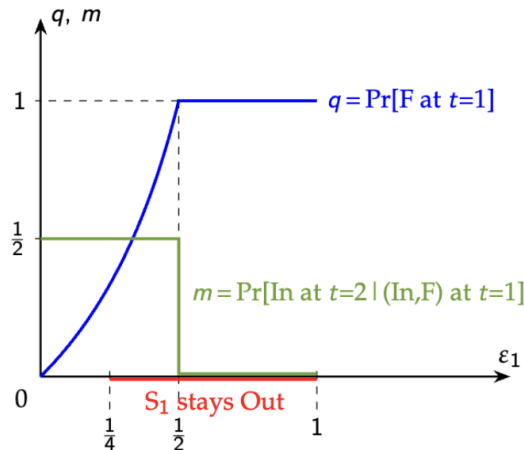
$$-2\varepsilon_1 + (1 - 2\varepsilon_1) > 0 \implies p < \frac{1}{2}$$

So, this happens if

$$\varepsilon_1 > \frac{1}{4}$$

This completes the derivation of the only PBE of the game ■

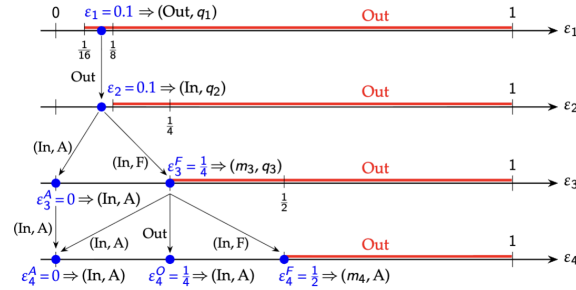
We can illustrate it as the following



Characteristics of the unique PBE for any $T \geq 2$

- S_1 stays Out if, and only if, $\varepsilon_1 > 1/2^T$. This threshold goes to 0 as $T \rightarrow \infty$.
- If S_1 comes In, then sane L plays F with probability 1 if $\varepsilon_1 > 1/2^{T-1}$
- For all $t > 1$: if L has fought all earlier entrants, then S_t stays Out with positive probability. If L has ever played A before t , then S_t goes In for sure
- If L ever plays A, just once, then L continues to play A until the end of the game. If L has always played F in the past then L keeps playing F with positive probability

Note: Once S_t plays In, the continuation payoff of L falls to zero. L starts to mix, therefore L is indifferent between A and F and therefore L's payoff is zero. Here we illustrate the case for $T = 4, \varepsilon_1 = 0.1$

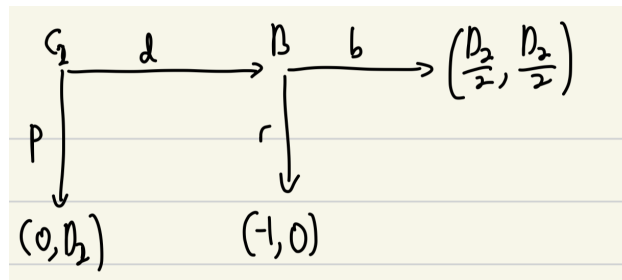


As $T \rightarrow \infty$, the average payoff of the sane, long-lived player L converges to 2, as if L could credibly commit to or threaten with F. Possibility of “crazy-type” allows L to get its Stackelberg payoff. Same is true in any game between a long run and myopic player.

Theorem 1.1 (Fudenberg and Levine, 1989). If the long-lived player (L) has each possible commitment type with positive prior probability, then in the unique PBE of the T -times repeated game against short-lived opponents (a sequence of S_t 's), L's average payoff converges to $\max_{a_L} \{u_L(a_L, BR_S(a_L))\}$ where $BR_S(a_L)$ is the short lived player's best response function

Example 1.1 (Credit Default). An international bank is facing two countries, separately and sequentially, at dates $t = 1, 2$. At date t , country t decides whether to repay its debt, D_t or default. If country t pays up, then its payoff is 0 and the bank gets D_t . If country t defaults, then the bank can either bail it out, in which case each side gets a payoff of $D_t/2$ or refuse the bailout, in which case the bank gets 0, and the country gets a final payoff of -1 . Assume that $D_t > 2$ for $t = 1, 2$.

- (a) Draw the game tree at $t = 2$. Find the SPE of this single-period game.



The subgame perfect equilibrium is (d, b) i.e. C_2 plays d and Bank plays b since if C_2 chooses d , Bank chooses b since $\frac{D_2}{2} > 0$ and so C_2 gets $\frac{D_2}{2} > 0$ i.e. higher than what C_2 gets if chose p

- (b) Assume the bank's total payoff is the sum of its payoffs in each period. What is the SPE of the two-period game?

The subgame perfect equilibrium is $(d, b; d, b)$ by analogous argument.

For the rest of the problem, assume that before the game starts, Nature picks the bank's type: either normal or tough, with 50-50% chance. A normal bank is rational with payoffs described above; a tough bank never bails out a country that defaults. The bank's type is its private information.

- (c) Suppose $D_1 < D_2$. Derive a PBE in which a normal bank refuses to bail out country 1 with probability $q \in (0, 1)$ in case it defaults, and if country 2 sees that country 1 has been refused a bailout, then it pays up with probability $p = D_1/D_2$. To do so, first compute country 2's updated belief that the bank is tough upon seeing that country 1 has been refused a bailout, and write down the condition that country 2 is indifferent between paying and defaulting; this pins down q . Under what condition on D_1 and D_2 does country 1 pay up in this equilibrium?

Suppose $D_1 < D_2$.

At $t = 2$, suppose C_2 sees Bank refuse C_1 default. Then,

$$\begin{aligned}\mu_2(\text{tough}) &= \mathbb{P}(\text{tough}|\text{refuse}) = \frac{\mathbb{P}(\text{refuse}|\text{tough})\mathbb{P}(\text{tough})}{\mathbb{P}(\text{refuse})} \\ &= \frac{0.5}{0.5 + \mathbb{P}(\text{refuse}|\text{normal})\mathbb{P}(\text{normal})} \\ &= \frac{0.5}{0.5 + 0.5q} \\ &= \frac{1}{1+q}\end{aligned}$$

Then, since C_2 pays w.p. $p = \frac{D_1}{D_2}$, he is indifferent between paying and defaulting i.e.

$$\begin{aligned}u(\text{pay}) &= 0 \\ u(\text{default}) &= -\left(\frac{1}{1+q}\right) + \left(1 - \frac{1}{1+q}\right) \frac{D_2}{2}\end{aligned}$$

This implies

$$\left(\frac{q}{1+q}\right) \frac{D_2}{2} = \frac{1}{1+q}$$

so $q = \frac{2}{D_2}$ which makes sense since $q \in (0, 1)$ as $D_2 > 2$.

If C_1 defaults, $u_B(\text{bail}) = \frac{D_1+D_2}{2}$ and $u_B(\text{refuse}) = 0 + pD_2 + (1-p)\frac{D_2}{2}$ since a rational bank would bail them out in the last period. Bank mixes w.p. q iff

$$\frac{D_1 + D_2}{2} = (1+p)\frac{D_2}{2} \implies p = \frac{D_1}{D_2}$$

We want to check when C_1 pays up i.e. $u_1(\text{pay}) \geq u_1(\text{default})$. We know that

$$\begin{aligned}u_1(\text{pay}) &= 0 \\ u_1(\text{default}) &= \mathbb{P}(\text{bail})\frac{D_2}{2} - (1 - \mathbb{P}(\text{bail})) = \frac{1}{2}(1-q)\frac{D_1}{2} - \frac{1+q}{2}\end{aligned}$$

This is because

$$\mathbb{P}(\text{bail}) = \mathbb{P}(\text{bail}|\text{tough})\mathbb{P}(\text{tough}) + \mathbb{P}(\text{bail}|\text{normal})\mathbb{P}(\text{normal}) = \frac{1}{2}(1-q)$$

with $\mathbb{P}(\text{bail}|\text{normal}) = 1 - q$. Hence, the condition becomes

$$0 \geq \frac{1-q}{4}D_1 - \frac{1+q}{2}$$

With $q = \frac{2}{D_2}$, we yield

$$0 \geq (1 - \frac{2}{D_2})D_1 - 2 - \frac{4}{D_2} \iff \frac{2(D_1 + 2)}{D_1 - 2} \geq D_2$$

If $D_1 = 4, D_2 = 6$, then

$$\frac{2(D_1 + 2)}{D_1 - 2} = 6 = D_2 \implies \text{indifferent}$$

If $D_2 > 6$, then C_1 will default; if $D_2 < 6$, then C_1 will pay up

- (d) Does reputation building (as seen in part c) benefit the normal bank? What goes wrong with the equilibrium if $D_1 > D_2$?

Helps if forces C_1 to pay up instead of defaulting since yields D_1 instead of $D_1/2$ and so it can defect later on and bail out C_1 and yield $D_1 + \frac{D_2}{2} > \frac{D_1 + D_2}{2}$.

If $D_1 > D_2$, then $p = \frac{D_1}{D_2} > 1$ and is no longer a valid probability so the PBE no longer exists.

Example 1.2 (Prisoner's Dilemma with Reputation). Consider the following stage game where Ann plays row and Bob plays column, with payoffs (u_1, u_2) in each cell. Assume $a > 1, b < 0$ and $a + b \leq 1$.

A/B	C	D
C	$1, 1$	b, a
D	a, b	$0, 0$

(1)

Bob is rational with payoffs listed in the table. Ann is rational with probability $(1 - p)$; however, if the game is repeated, she plays Tit-for-Tat (starts with C and then plays whatever Bob played in the previous period) with probability p . In what follows, assume $p + (1 - p)b \geq 0$.

- (a) Suppose that the game is played exactly twice. Derive the unique perfect Bayesian equilibrium. Hint: Start by describing what Bob and rational Ann do in the second period. Then determine what the different types of Ann do in period 1, and derive the best responses of Bob in period 1.

At $t = 2$, if Ann is rational, plays D since strictly dominant. Bob plays D no matter what since strictly dominant too. At $t = 2$, (D, D) with payoffs $(0, 0)$.

At $t = 1$, TFT Ann plays C by definition while rational Ann plays D no matter what since Bob plays D always at $t = 2$.

At $t = 1$, Bob believes Ann is rational w.p. $1 - p$ Bob prefers C iff

$$u_B(C) = (1 - p)b + (1 + a)p \geq u_B(D) = pa$$

Note: $(1 + a)p$ because 1 in $t = 1$ and a in $t = 2$ for a TFT Ann. Manipulation gives

$$p + (1 - p)b \geq 0$$

which holds by assumption. So the PBE is

- At $t = 1$, Bob plays C no matter what, Ann plays D w.p. $1 - p$ and C w.p. p
- At $t = 2$, Bob plays D and Ann plays D w.p. $1 - p$ and C w.p. p
- Beliefs: $\mu_2(\text{TFT}|C) = 1$ and $\mu_2(\text{TFT}|D) = 0$

- (b) Suppose the stage game is played three times. Prove that if $(1 - p)a \leq 1$ then there exists a perfect-Bayesian equilibrium in which both types of Ann as well as Bob play C for sure in period 1. Interpret in terms of “reputation”.

At $t = 3$, if Ann is rational, the strategy played by both is (D, D) with payoffs $(0, 0)$.

At $t = 1$, TFT Ann plays C by definition while rational Ann plays C . Bob plays C in the candidate equilibrium.

At $t = 2$, on the equilibrium path if Ann plays C in $t = 1$, Bob's posterior belief $\mu_2(\text{TFT}) = p$. So since

TFT copies last action, Ann plays C w.p. p and rational Ann plays D w.p. $1 - p$ since no signalling value at penultimate stage. Hence

$$\begin{aligned} u_B(C) &= (1 + a)p + (1 - p)b \\ u_B(D) &= ap + 0 \end{aligned}$$

So Bob prefers C to D iff $(1 + a)p + (1 - p)b - ap \geq 0$ which is $p + (1 - b)p \geq 0$ which holds by assumption. Hence at $t = 2$, Bob plays C for sure, with rational Ann playing D and TFT Ann playing C .

On the off-path, if Ann plays D , then $\mu_2(\text{TFT}) = 0$ so Bob plays D .

Consider $t = 1$, where we need to show that it is a best reply for both rational Ann and Bob to play C . Assuming Bob plays C at $t = 1$ then for rational Ann,

$$\begin{aligned} u_A(C) &= 1 + a + 0 \\ u_A(D) &= a < u_A(C) \end{aligned}$$

so she would prefer to play C (note that the continuation payoff is when rational Ann plays D at $t = 2$). Need to check if Bob playing C is mutual BR.

$$\begin{aligned} u_B(C) &= p(1 + 1 + a) + (1 - p)(1 + b) = p(a + 2) + (1 - p)(b + 1) \\ u_B(D) &= pa + (1 - p)a \\ &= a \end{aligned}$$

BR iff $u_B(C) \geq u_B(D)$ which gives

$$p(a + 2) + (1 - p)(b + 1) \geq a \iff p + (1 - b)b + 1 \geq a(1 - p)$$

since $p + (1 - p)b \geq 0$, $1 \geq a(1 - p)$ is sufficient for this inequality to hold which implies that Bob's BR is indeed C at $t = 1$. Hence, the PBE exists if $a(1 - p) \geq 1$ where

- On-path
 - Bob plays C at $t = 1, 2$ and D at $t = 3$
 - Rational Ann plays C at $t = 1$, D at $t = 2, 3$
 - TFT Ann plays C always
 - Bob's posterior is $\mu_1(\text{TFT}) = \mu_2(\text{TFT}) = p$
- Off-path:
 - Bob plays C at $t = 1$ but D at $t = 2, 3$, Ann plays D at $t = 1, 2, 3$. Bob's posterior is $\mu_2(\text{TFT}) = 0$
 - Bob plays D at $t = 1$, Ann plays D at $t = 1$, at $t = 2, 3$ we have (D, D) with $\mu_2(\text{TFT}) = 0$.
 - (C, C) at $t = 1$ and Bob plays D at $t = 2$ with rest same as on-path. Belief $\mu_2(\text{TFT}) = p$

The possibility that Ann would play TFT, once above a threshold would compel Bob to cooperate because defecting would cause TFT Ann to play D . Given this, rational Ann would want to mimic TFT Ann to preserve reputation.

1.2 Communication, Cheap Talk

Definition 1.2 (Crawford-Sobel Cheap Talk Game). The Crawford-Sobel Cheap-Talk game is

- Nature picks a random state $\theta \in [0, 1]$; P1 learns θ while P2 does not
- P1 (he) sends a message $m \in M$; the set M is rich and includes $[0, 1]$
- Having observed m , P2 (she) picks $y \in [0, 1]$

P1 and P2 have different goals – P2 wants to pick $y = \theta$ while P1 is biased for P2 to pick a larger y . Their payoffs are

$$\begin{aligned} u_1 &= -(\theta + b - y)^2 \\ u_2 &= -(\theta - y)^2 \end{aligned}$$

We can apply this to P1 being an expert and knowing the optimal policy and informs the PM (P2) but has a known bias $b > 0$.

A perfect Bayesian equilibrium here is

- P1's strategy $s_1 : [0, 1] \rightarrow \Delta(M)$ i.e. any mix over M
- in equilibrium, P2 relies $s_2(m) = \mathbb{E}[\theta | s_1(\theta) = m]$
- $s_1(\theta)$ is a mix over m 's that maximise $-(\theta + b - s_2(m))^2$

We only need to consider induced actions: y is induced in equilibrium if there is a state θ in which message m with $s_2(m) = y$ is sent with positive probability.

Example 1.3 (A two-action equilibrium under a uniform prior). Suppose $\theta \sim U[0, 1]$, and consider a candidate equilibrium in which P2 takes only two possible actions.

Specifically, suppose there is a cutoff $\theta^* \in (0, 1)$ such that:

$$\begin{aligned} \theta < \theta^* &\implies \text{P1 sends a message inducing action } y', \\ \theta > \theta^* &\implies \text{P1 sends a message inducing action } y''. \end{aligned}$$

Thus the message does not reveal the exact state. It only tells P2 whether the state is in the *lower interval* $[0, \theta^*]$ or the *upper interval* $[\theta^*, 1]$.

Since P2 wants to match y to the expected state, her best reply after each message is the conditional expectation of θ on the corresponding interval:

$$y' = \mathbb{E}[\theta \mid \theta \in [0, \theta^*]], \quad y'' = \mathbb{E}[\theta \mid \theta \in [\theta^*, 1]].$$

Because θ is uniformly distributed, the conditional expectation on an interval is just its midpoint. Hence

$$y' = \frac{0 + \theta^*}{2} = \frac{\theta^*}{2}, \quad y'' = \frac{\theta^* + 1}{2}.$$

Now consider P1's incentives. A sender of type θ prefers P2's action to be close to $\theta + b$, not to θ . So among the two available induced actions, type θ compares

$$-(\theta + b - y')^2 \quad \text{and} \quad -(\theta + b - y'')^2.$$

The cutoff type θ^* must be indifferent between the two messages, since types below θ^* choose the lower-action message and types above θ^* choose the higher-action message. Therefore $\theta^* + b$ must lie exactly halfway between y' and y'' :

$$\theta^* + b = \frac{y' + y''}{2}.$$

Substituting the expressions for y' and y'' gives

$$\theta^* + b = \frac{1}{2} \left(\frac{\theta^*}{2} + \frac{1 + \theta^*}{2} \right) = \frac{1 + 2\theta^*}{4}.$$

Solving,

$$4\theta^* + 4b = 1 + 2\theta^* \implies 2\theta^* = 1 - 4b \implies \theta^* = \frac{1}{2} - 2b.$$

This gives a valid two-action equilibrium only if $\theta^* \in [0, 1]$, which requires

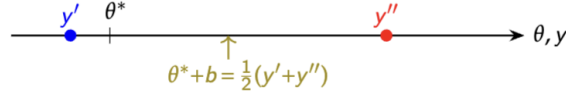
$$\frac{1}{2} - 2b \geq 0 \iff b \leq \frac{1}{4}.$$

The intuition is simple: the larger is the bias b , the more strongly P1 wants to push P2 upward. If b is too large, even relatively low types want to mimic high types, so a two-interval partition cannot be sustained. If b is small enough, then low types are willing to reveal that they are low, and high types are willing to separate into the upper interval.

Proposition 1.5. In any PBE for the Crawford and Sobel cheap talk game, the total number of induced y actions is finite.

Proof. Consider the uniform distribution example. Let $y' < y''$ be two induced actions and set θ^* so that $\theta^* + b = \frac{1}{2}(y' + y'')$. Clearly, $y' \leq \theta^* + b \leq y''$. The question is where is θ^* located. If $\theta > \theta^*$, then P1 prefers y'' over y' hence P1 won't induce y' in any state $\theta > \theta^*$.

P2 knows this, therefore, in equilibrium, $y' \leq \theta^*$. We conclude that $y' \leq \theta^* < \theta^* + b = \frac{1}{2}(y' + y'') < y''$ as seen



The gap between any two induced actions y' and y'' exceeds $b > 0$ and therefore there cannot be more than $1/b$ distinct induced actions in $[0, 1]$ ■

Basically, talk is cheap yet can be informative and we got infinitely many states but only finitely many induced actions.

Example 1.4 (PSet5 Q4 – costly verifiability). A Sender observes the state of nature ω uniformly distributed on $[0, 1]$. The Sender transmits a message $m \in [0, 1]$ to the Receiver, then the Receiver takes action $y \in \mathbb{R}$. The Sender's utility is $u_S(y, \omega) = -(\omega + b - y)^2$, the Receiver's utility is $u_R = -(\omega - y)^2$, where $b = \frac{1}{3}$ is the Sender's bias. The structure of the game (but not the value of ω) is commonly known between players.

- (a) When only cheap talk messages are available (that is, any $m \in [0, 1]$ can be sent in any state), prove that it is impossible for the Sender to transmit any information. That is, prove that in any PBE, the Receiver's action is the same irrespective of ω .

Solution.

I show that no informative equilibrium can exist.

In any PBE, after observing a message m , the Receiver chooses y to maximise

$$\mathbb{E}[-(\omega - y)^2 \mid m].$$

Since this is a quadratic loss problem, the Receiver's unique best response is

$$y(m) = \mathbb{E}[\omega \mid m].$$

Hence information is transmitted if and only if different messages induce different posterior means.

Suppose, for contradiction, that there is an informative PBE. Then there must exist at least two messages that induce two different actions, say

$$y_L < y_H.$$

Consider a Sender of type ω . His payoff from inducing action y is

$$u_S(y, \omega) = -(\omega + b - y)^2.$$

Thus his ideal action is $\omega + b$. In particular, for fixed ω , he prefers whichever induced action is closer to $\omega + b$.

If both y_L and y_H are induced in equilibrium, then there must be some cutoff type ω^* who is indifferent between them. This type satisfies

$$-(\omega^* + b - y_L)^2 = -(\omega^* + b - y_H)^2.$$

Since $y_L \neq y_H$, this is equivalent to

$$\omega^* + b = \frac{y_L + y_H}{2}.$$

So the indifferent type's ideal point must lie exactly midway between the two induced actions.

Now use the standard two-interval argument. Any informative equilibrium must induce at least two distinct actions, so in particular there would have to exist a two-action partition equilibrium as a special case. Suppose therefore that types in $[0, t]$ induce one action and types in $[t, 1]$ induce another. Because $\omega \sim U[0, 1]$, the Receiver's best responses are the conditional means on these intervals:

$$y_L = \mathbb{E}[\omega \mid \omega \in [0, t]] = \frac{t}{2}, \quad y_H = \mathbb{E}[\omega \mid \omega \in [t, 1]] = \frac{1+t}{2}.$$

The cutoff type t must be indifferent, so

$$t + b = \frac{y_L + y_H}{2}.$$

Substituting $b = \frac{1}{3}$ and the expressions for y_L, y_H gives

$$t + \frac{1}{3} = \frac{1}{2} \left(\frac{t}{2} + \frac{1+t}{2} \right) = \frac{1+2t}{4}.$$

Rearranging,

$$4t + \frac{4}{3} = 1 + 2t \implies 2t = -\frac{1}{3} \implies t = -\frac{1}{6},$$

which is impossible since $t \in [0, 1]$.

Therefore no two-action informative equilibrium exists. Since any informative equilibrium would have to induce at least two distinct actions, no informative equilibrium exists at all.

Hence in every PBE, the Receiver's action must be constant across messages. Equivalently, the Sender transmits no information: the Receiver's action is the same irrespective of ω .

- (b) Suppose that the Sender can verifiably disclose the value of ω to the Receiver at cost $c = 5/36$. That is, in state ω , the Sender can message m_ω which is only available in state ω ; if the Sender sends this special message then his utility becomes $u_S(y, \omega) - c$. Find a PBE in which ω is verifiably disclosed above a certain threshold.

Solution.

We look for a threshold equilibrium: there exists $\bar{\omega} \in [0, 1]$ such that the Sender discloses if and only if $\omega \geq \bar{\omega}$.

Consider the following candidate strategies and beliefs:

- if $\omega \geq \bar{\omega}$, the Sender sends the verifiable message m_ω ;
- if $\omega < \bar{\omega}$, the Sender sends a pooling cheap-talk message m_P ;
- upon receiving m_ω , the Receiver knows the true state and chooses

$$y(m_\omega) = \omega;$$

- upon receiving m_P , the Receiver believes that ω is uniformly distributed on $[0, \bar{\omega})$ and chooses the posterior mean

$$y(m_P) = \mathbb{E}[\omega \mid \omega < \bar{\omega}] = \frac{\bar{\omega}}{2}.$$

I now compute the Sender's payoff from each option.

If type ω pools, his payoff is

$$U_P(\omega) = -\left(\omega + \frac{1}{3} - \frac{\bar{\omega}}{2}\right)^2.$$

If type ω discloses, the Receiver chooses $y = \omega$, so the Sender gets

$$U_D(\omega) = -\left(\omega + \frac{1}{3} - \omega\right)^2 - \frac{5}{36} = -\left(\frac{1}{3}\right)^2 - \frac{5}{36} = -\frac{1}{9} - \frac{5}{36} = -\frac{1}{4}.$$

In a threshold equilibrium, the cutoff type $\bar{\omega}$ must be indifferent between pooling and disclosure. Hence

$$-\left(\bar{\omega} + \frac{1}{3} - \frac{\bar{\omega}}{2}\right)^2 = -\frac{1}{4}.$$

This simplifies to

$$\left(\frac{\bar{\omega}}{2} + \frac{1}{3}\right)^2 = \frac{1}{4}.$$

Taking the positive square root gives

$$\frac{\bar{\omega}}{2} + \frac{1}{3} = \frac{1}{2},$$

so

$$\frac{\bar{\omega}}{2} = \frac{1}{6} \quad \implies \quad \bar{\omega} = \frac{1}{3}.$$

Thus the candidate equilibrium is:

$$\text{disclose iff } \omega \geq \frac{1}{3}.$$

It remains to verify incentives. If $\omega < \frac{1}{3}$, then

$$U_P(\omega) = -\left(\omega + \frac{1}{3} - \frac{1}{6}\right)^2 = -\left(\omega + \frac{1}{6}\right)^2.$$

Compare this with $U_D(\omega) = -\frac{1}{4}$. Pooling is preferred whenever

$$-\left(\omega + \frac{1}{6}\right)^2 \geq -\frac{1}{4} \quad \iff \quad \left(\omega + \frac{1}{6}\right)^2 \leq \frac{1}{4} \quad \iff \quad \omega \leq \frac{1}{3}.$$

So all types below $\frac{1}{3}$ prefer to pool. Similarly, for $\omega > \frac{1}{3}$, disclosure yields the higher payoff, and at $\omega = \frac{1}{3}$ the Sender is indifferent.

The Receiver is also sequentially rational:

- after m_ω , choosing $y = \omega$ is optimal because the message is verifiable;
- after m_P , choosing $y = \frac{1}{6}$ is optimal because the posterior on the pooling region is uniform on $[0, \frac{1}{3})$.

Therefore a PBE is given by:

Sender: disclose ω iff $\omega \geq \frac{1}{3}$, otherwise send m_P ,

Receiver: $y = \omega$ after m_ω , $y = \frac{1}{6}$ after m_P .

- (c) Continue to assume that the Sender can verifiably disclose the state to the Receiver at cost c in part (b). Show that as bias b increases, there is more information transmission in the sense that in equilibrium the Sender verifiably discloses ω in more states.

Solution.

Now let the bias $b > 0$ vary, while the disclosure cost c remains fixed. I again look for a threshold equilibrium: disclose iff $\omega \geq \bar{\omega}$.

If the Sender pools, the Receiver's posterior after the pooling message is uniform on $[0, \bar{\omega})$, so

$$y_P = \frac{\bar{\omega}}{2}.$$

Hence a type ω who pools gets

$$U_P(\omega) = -(\omega + b - y_P)^2 = -\left(\omega + b - \frac{\bar{\omega}}{2}\right)^2.$$

If the Sender discloses, the Receiver chooses $y = \omega$, so the Sender gets

$$U_D(\omega) = -(\omega + b - \omega)^2 - c = -b^2 - c.$$

At the cutoff $\omega = \bar{\omega}$, indifference requires

$$-\left(\bar{\omega} + b - \frac{\bar{\omega}}{2}\right)^2 = -b^2 - c,$$

i.e.

$$\left(\frac{\bar{\omega}}{2} + b\right)^2 = b^2 + c.$$

Taking the positive square root yields

$$\frac{\bar{\omega}}{2} + b = \sqrt{b^2 + c},$$

so

$$\bar{\omega} = 2(\sqrt{b^2 + c} - b).$$

The disclosure region is therefore

$$[\bar{\omega}, 1].$$

To show that more bias leads to more disclosure, it is enough to show that $\bar{\omega}$ decreases with b .

Differentiate:

$$\frac{d\bar{\omega}}{db} = 2\left(\frac{b}{\sqrt{b^2 + c}} - 1\right).$$

Since $c > 0$, we have

$$\sqrt{b^2 + c} > b,$$

and so

$$\frac{b}{\sqrt{b^2 + c}} < 1.$$

Therefore

$$\frac{d\bar{\omega}}{db} < 0.$$

Thus $\bar{\omega}$ is decreasing in b . Equivalently, as b increases, the set of states in which the Sender discloses,

$$[\bar{\omega}, 1],$$

becomes larger.

Hence as bias increases, there is more information transmission in equilibrium, in the sense that the Sender verifiably discloses ω in more states.

Intuitively, when b is larger, the Sender wants a higher action relative to the Receiver's ideal action. Pooling then becomes less attractive, because the pooling action is too low from the Sender's perspective. By paying the disclosure cost, the Sender can induce the higher action $y = \omega$. As a result, a larger bias makes disclosure optimal for more types.

Variations of Cheap Talk

Multi-issue Cheap Talk with Single Expert

Definition 1.3 (Chakraborty and Harbaugh, Multi-issue cheap talk). The multi-issue cheap talk game is

- Nature picks $\theta_1, \dots, \theta_K \in [0, 1]$ distributed i.i.d
- P1 sends a message $m \in [0, 1]^K$
- having observed m , P2 picks $y \in [0, 1]^K$

The payoffs are

$$u_1 = -\sum_{k=1}^K (\theta_k + b - y_k)^2, u_2 = -\sum_{k=1}^K (\theta_k - y_k)^2$$

with $b > 0$.

If b is small then P1 and P2 can communicate via finitely many messages along each dimension (same as Crawford and Sobel) but there is no informative Crawford-Sobel partition equilibrium if b is large.

Proposition 1.6. No matter how large b is, there exists a PBE in which P1 induces

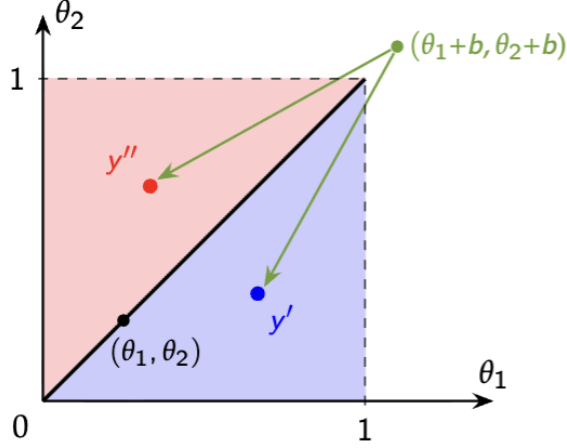
$$y' = \mathbb{E}[(\theta_1, \theta_2) | \theta_1 > \theta_2]$$

if $\theta_1 > \theta_2$ and

$$y'' = \mathbb{E}[(\theta_1, \theta_2) | \theta_2 > \theta_1]$$

if $\theta_1 < \theta_2$

We can illustrate this if



Generalises to any $K \geq 2$: if the bias is the same in every dimension, there is a PBE in which P1 conveys the true ordering of the coordinates of the state. As $K \rightarrow \infty$, P2 learns the state arbitrarily precisely.

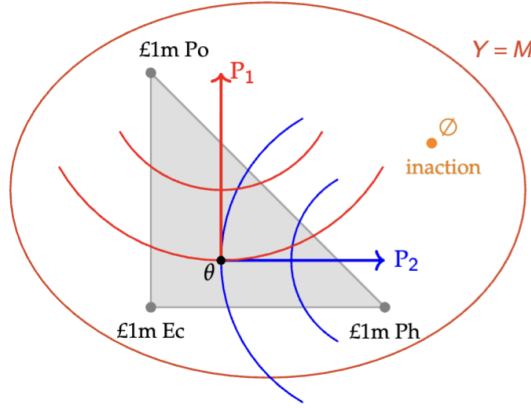
Multi-Expert Cheap Talk

Motivating Story: PPE gets anonymous donation of \$1m; programme director wants to split it among three departments (Politics, Philosophy, Economics) in the socially optimal way.

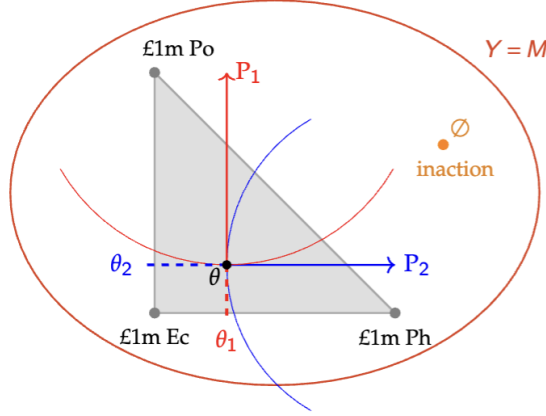
State of nature: $(\theta_E, \theta_{Po}, \theta_{Ph}) \geq 0$, with $\theta_E + \theta_{Po} + \theta_{Ph} = 1$; the director P_0 wants to set per action $(y_E, y_{Po}, y_{Ph}) = (\theta_E, \theta_{Po}, \theta_{Ph})$. Two experts know the state but P1 is biased towards Politics and P2 is towards philosophy.

The decision maker can gain more by comparing experts reports by asking each expert about whatever is orthogonal to their bias.

We can plot in the coordinate system $\theta_1 \equiv \theta_{Ph}, \theta_2 \equiv \theta_{Po}$ so that $\theta_E = 1 - \theta_1 - \theta_2$.



The message and action space may be richer than the state space i.e. have inaction. The plot shows the experts' direction of bias as well as indifference curves. We can ask P1 the horizontal coordinate θ_1 and P2 the vertical coordinate θ_2 since it is orthogonal to their biases



Each expert's bias is orthogonal to the coordinate to be reported so there is no incentive to lie. Set $y = (\theta_1, \theta_2)$ the true state. The main (practical) challenge is to find dimensions orthogonal to the bias.

2 Bargaining

2.1 Cooperative, Axiomatic Bargaining

Definition 2.1 (Cooperative Bargaining). A bargaining problem for $i = 1, 2$ is given by

- a bargaining set $U = \{u_1(x), u_2(x) : x \in X\}$, where X is the feasible physical allocations and u_1, u_2 are utility functions
- disagreement point, $d = (d_1, d_2)$ in utility terms

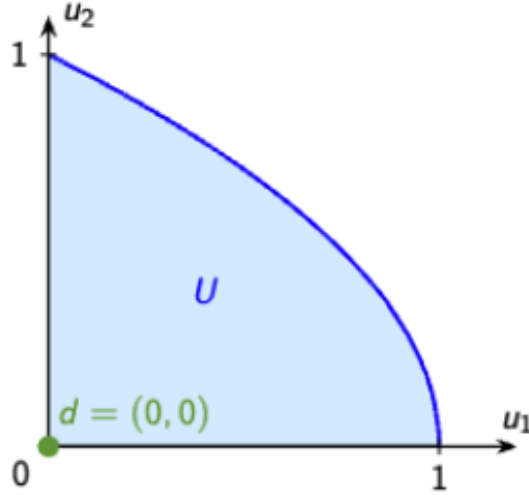
Remark. We usually assume that U is bounded, closed and convex and contains a point that is strictly better than disagreement for both players i.e. $(u_1, u_2) > (d_1, d_2)$. We usually also assume free disposal.

The intuition is simply that two parties negotiate over feasible allocations and if they fail to agree there is a default option.

Example 2.1 (Heterogenous Risk Attitudes). Players 1 and 2 split one unit of money and if they agree on $(x, 1 - x)$ then $u_1 = x$ and $u_2 = (1 - x)^{1-r}$ where $r \in [0, 1)$. If they disagree, then both players get 0. The pareto frontier is

$$u_2 = (1 - u_1)^{1-r}$$

since $u_1 = x$. The bargaining set is thus (for $r = 1/2$)

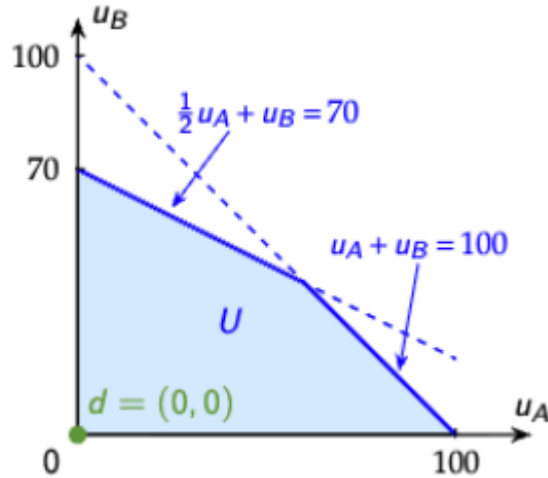


Example 2.2 (Heterogenous Bundles). Suppose Ann and Bob inherit 30 acres of land and 40 gold coins. Ann values 1 acre of land at 2 gold coins while Bob values 1 acre of land at 1 gold coins. So,

$$u_A(x, y) = x + 2y$$

$$u_B(x, y) = (40 - x) + (30 - y) = 70 - x - y$$

To construct the bargaining set,



Suppose Ann gets everything and so $u_A = 100$. Then, to stay on the pareto frontier, we start by transferring 1 gold coin from Ann to Bob and so the marginal rate of substitution is 1. Hence, $u_B + u_A = 100$ when $u_A \geq 60, u_B \leq 40$. Then, at $(u_A, u_B) = (60, 40)$, we start transferring land from Ann to Bob – for each util you transfer from Ann to Bob, Bob only gets half since he values land at 1 while Ann values land at 2. So, $\frac{1}{2}u_A + u_B = 70$ (since allocating everything to Bob gives him 70) for $u_A \leq 60, u_B \geq 40$.

NOTE: on the $\frac{1}{2}u_A + u_B = 70$ segment, the reason why Ann transferring land to Bob remains on the pareto frontier is that at any point (u_A, u_B) there is no transfer such that both can be made better off, since Ann cannot ask for land back and compensate Bob with gold coins since she, by construction, has no gold coins left (after transferring all to Bob when $u_B \geq 40$)

Remark. This is only for when $d = (0, 0)$. For a more general $d = (d_1, d_2)$, we have to begin from when $u_1 = d_1$ and u_2 has the rest of the feasible utility.

We want to find a bargaining solution $f(U, d) \in U$ that assigns a unique outcome to every problem (U, d) and to do this we need to look at properties that a good solution f should satisfy. These are Nash's 4 axioms

Axiom 2.1 (PAR). The solution must be on the Pareto-frontier of the bargaining set.

If $f(U, d) = (u_1^*, u_2^*)$, then there is no $(u_1, u_2) \in U$ such that $u_i \geq u_i^*$ for both $i = 1, 2$ and $u_i > u_i^*$ for some $i = 1, 2$

Axiom 2.2 (INV). The solution must be invariant to (i.e. shift and scale according to) increasing affine utility transformations.

Let $\hat{U} = \{(\alpha_1 + \beta_1 u_1, \alpha_2 + \beta_2 u_2) : (u_1, u_2) \in U\}$ and $\hat{d} = (\alpha_1 + \beta_1 d_1, \alpha_2 + \beta_2 d_2)$ where $\beta_1, \beta_2 > 0$.

Then

$$f(U, d) = (u_1^*, u_2^*) \iff f(\hat{U}, \hat{d}) = (\alpha_1 + \beta_1 u_1^*, \alpha_2 + \beta_2 u_2^*)$$

Axiom 2.3 (SYM). A symmetric bargaining problem must have a symmetric solution.

If $d_1 = d_2$ and $(u_1, u_2) \in U$ implies $(u_2, u_1) \in U$, then $f(U, d) \in (u^*, u^*)$

Axiom 2.4 (IIA). Independence from irrelevant alternatives: if outcomes other than $f(U, d)$ are dropped from U , then the solution (with the same d) remains the same.

If $\hat{U} \subseteq U$ and $f(U, d) \in \hat{U}$, then $f(\hat{U}, d) = f(U, d)$

Theorem 2.1. There is a unique bargaining solution that satisfies PAR, INV, SYM and IIA, namely $f(U, d) = (u_1^*, u_2^*)$ which solves

$$\begin{aligned} & \max_{u_1, u_2} (u_1 - d_1)(u_2 - d_2) \\ \text{s.t. } & (u_1, u_2) \in U \\ & u_1 \geq d_1, u_2 \geq d_2 \end{aligned}$$

The Nash Bargaining Solution maximises the 'Nash product' (an equal weight Cobb-Douglas social welfare function defined on the parties' gains relative to their disagreement payoffs) over U .

Proof. We prove this for the simple case of a two player bargaining problem for divide-a-dollar (homogenous bundle with identical utility functions). Consider a bargaining set $Z = \{(x, y) \geq (0, 0) : x + y \leq 1\}$ with $d = (0, 0)$.

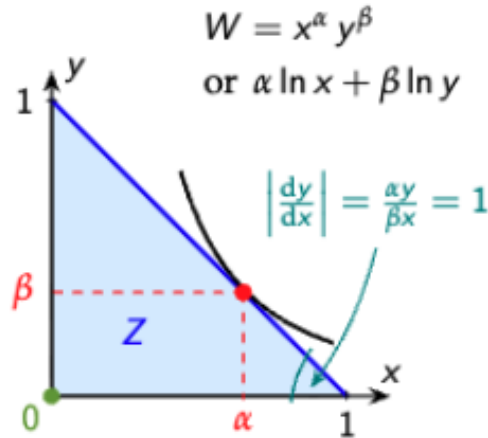
1. Use PAR: PAR requires that the bargaining solution be on the Pareto Frontier i.e. $f(Z, 0) = (\alpha, \beta)$ with $\alpha, \beta \geq 0, \alpha + \beta = 1$. Maximising Cobb-Douglas with weights α and β on x and y respectively gives

$$W = x^\alpha y^\beta$$

or

$$W = \alpha \ln x + \beta \ln y$$

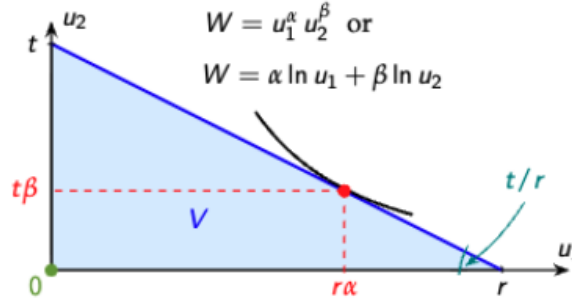
which picks out $x^* = \alpha$ and $y^* = \beta$ as desired



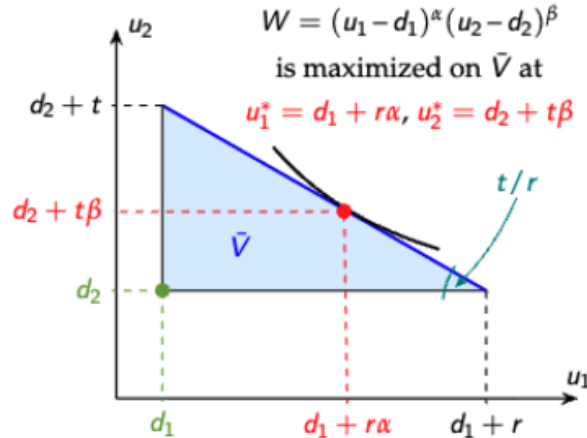
2. Using INV: rescale payoffs in Z so that $u_1(1) = r$ and $u_2(1) = t$ and call this bargaining set V . If $f(Z, 0) = (\alpha, \beta)$ then by INV, $f(V, 0) = (r\alpha, t\beta)$. Indeed, $W = u_1^r u_2^t$ picks $(r\alpha, t\beta)$ in $(V, 0)$ because

$$\left| \frac{du_1}{du_2} \right| = \frac{\partial W / \partial u_1}{\partial W / \partial u_2} = \frac{\alpha u_2}{\beta u_1} = \frac{t}{r} \iff r\alpha u_2 = t\beta u_1$$

which holds on the Pareto frontier at $u_1^* = r\alpha, u_2^* = t\beta$.



Shift V so that disagreement point becomes (d_1, d_2) ; with bargaining set $\tilde{V}$. By INV, if $f(Z, 0) = (\alpha, \beta)$ then $f(\tilde{V}, 0) = (d_1 + r\alpha, d_2 + t\beta)$. It is picked out by maximising $W = (u_1 - d_1)^\alpha (u_2 - d_2)^\beta$ over $\tilde{V}$.



3. Maximise $W = (u_1 - d_1)^\alpha (u_2 - d_2)^\beta$ on a general convex U : Consider a convex U with frontier g ,

$$U = \{(u_1, u_2) : u_2 \leq g(u_1)\}$$

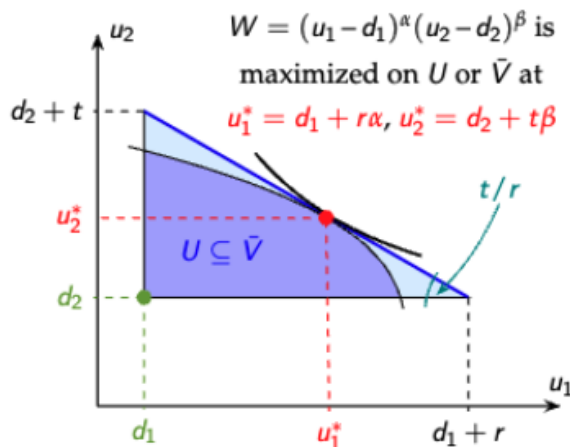
restricted to $u_1 \geq d_1, u_2 \geq d_2$. Find (u_1^*, u_2^*) that maximises $W = (u_1 - d_1)^\alpha (u_2 - d_2)^\beta$ over U . On frontier, the tangency condition is

$$\frac{\partial W/\partial u_1}{\partial W/\partial u_2} = \frac{\alpha(u_2^* - d_2)}{\beta(u_1^* - d_1)} = |g'(u_1^*)|$$

4. Expand U to include points below the tangent: Expand U to $\tilde{V}$ by adding all points below the tangent at (u_1^*, u_2^*) . Denote the highest u_1 by $d_1 + r$ and the highest u_2 by $d_2 + t$. Then, $W = (u_1 - d_1)^\alpha (u_2 - d_2)^\beta$ is maxed on U or $\tilde{V}$ at (u_1^*, u_2^*) where

$$\frac{\alpha(u_2^* - d_2)}{\beta(u_1^* - d_1)} = |g'(u_1^*)| = \frac{t}{r}$$

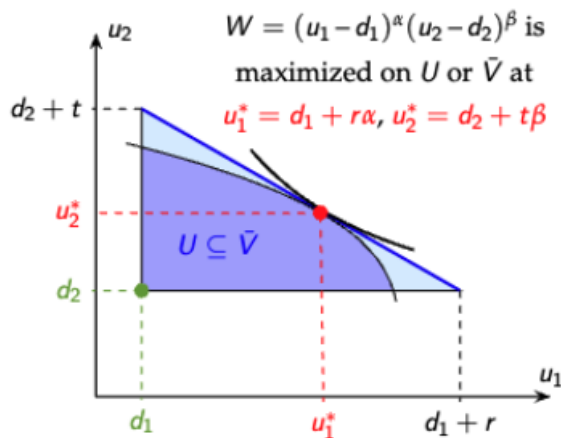
Hence, $u_1^* = d_1 + r\alpha, u_2^* = d_2 + t\beta$.



5. Use IIA to move between problems: Recall problems $Z, V, \tilde{V}, U$. Suppose that $f(Z, 0) = (\alpha, \beta)$ where $\alpha, \beta \geq 0, \alpha + \beta = 1$ by PAR. Then $f(V, 0) = (r\alpha, t\beta)$ and in fact $f(\tilde{V}, d) = (d_1 + r\alpha, d_2 + t\beta)$ by INV. Hence $f(U, d) = (d_1 + r\alpha, d_2 + t\beta)$ by IIA which is the solution to

$$\max W = (u_1 - d_1)^\alpha (u_2 - d_2)^\beta$$

on U .



6. Generalised Nash Solution + SYM: We have shown that if the bargaining solution f satisfies PAR, INV and IIA, then there exists $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$ such that $f(U, d) = (u_1^*, u_2^*)$ maximises over U the (α, β) weighted Cobb-Douglas social welfare function defined on players utilities above their respective outside options, $W = (u_1 - d_1)^\alpha (u_2 - d_2)^\beta$. This is the generalised Nash bargaining solution and the weighted CD social welfare function W is called the generalised Nash product. If SYM is imposed, then $\alpha = \beta = \frac{1}{2}$ i.e. a symmetric Nash bargaining solution. So, PAR, INV, IIA and SYM imply that $f(U, d)$ maximises $W = (u_1 - d_1)(u_2 - d_2)$ on U which completes the proof

Example 2.3 (PSet 5). We try to solve the following problem in three ways.

Problem: Ann and Bob have been willed equal shares of an estate consisting of \$200,000 cash and 100 acres

of farmland. Bob values land at \$3000 per acre whereas Ann has no such attachment and values it at \$1000 per acre. The utility functions are linear in money and land at these rates and if Ann and Bob fail to reach agreement then they divide the land and money equally.

We can draw the graph via the following: Suppose u_A gets everything in which case $u_A = 300,000$. Along the Pareto frontier, it is most efficient to transfer land to Bob (since it makes Ann lose less utility to give Bob the same amount of utility). For each dollar of land Ann gives, Bob gains three dollars of utility. This can continue until Ann gives all the land, which is when $u_A = 200,000, u_B = 300,000$. So $u_A = 300,000 - \frac{1}{3}u_B$ (since for every utility Bob gains from land, Ann loses only one third). Afterwards, Ann continues transferring money along the Pareto frontier, which implies $u_A + u_B = 500,000$. So, the problem is

$$\max(u_A - 150,000)(u_B - 250,000) \text{ s.t. } u_A + u_B \leq 500,000, u_A + \frac{1}{3}u_B \leq 300,000$$

Henceforth, we normalise to hundreds.

1. Solve via Kuhn-Tucker conditions.

Notice that the constraint qualification always holds because these are linear constraints with nonzero coefficients. Then,

$$\mathcal{L}(u_A, u_B, \lambda, \mu) = (u_A - 150)(u_B - 250) - \lambda(u_A + u_B - 500) - \mu(u_A + \frac{1}{3}u_B - 300)$$

Our FOCs are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial u_A} &= u_B - 250 - \lambda - \mu = 0 \\ \frac{\partial \mathcal{L}}{\partial u_B} &= u_A - 150 - \lambda - \frac{1}{3}\mu = 0 \end{aligned}$$

with the typical complementary slackness conditions. Then,

Case 1: $\mu, \lambda = 0$. This gives us that $u_A - 150 = u_B - 250 \implies u_A + u_B = 400$.

Case 2: Suppose $\mu = 0, \lambda > 0$. Then, $u_A + u_B = 500$ and $u_A - 150 = u_B - 250$. Substituting gives us $500 - u_A - 250 = u_A - 150$ which yields $2u_A = 400$ and hence $(u_A, u_B) = (200, 300)$.

Case 3: Suppose $\mu > 0, \lambda = 0$. Then, $3u_A + u_B = 900$ and $3u_A - 450 = u_B - 250$. Substituting gives $u_B = 350$ and $u_A = \frac{550}{3}$, which contradicts $u_A + u_B \leq 500$ and so $\lambda \geq 0$, a contradiction.

Clearly, $(u_A, u_B) = (200, 300)$ is the maximising value.

2. Solve via Nash axioms.

Let the bargaining problem be (S, d) , where S is the feasible utility set and d is the disagreement point. Measure utility in thousands of pounds.

If Ann receives cash m and x acres, then Bob receives cash $200 - m$ and $100 - x$ acres. Hence

$$u_A = m + x, \quad u_B = (200 - m) + 3(100 - x) = 500 - m - 3x.$$

If agreement fails, they divide both cash and land equally, so the disagreement point is

$$d = (100 + 50, 100 + 3 \cdot 50) = (150, 250).$$

By **Pareto optimality (PAR)**, the solution must be Pareto efficient. Moreover, since disagreement is always feasible, the solution must also be *individually rational*: each player must receive at least his or her disagreement utility. Here

$$d = (150, 250).$$

First find the Pareto-efficient allocations. If Ann holds any positive amount of land, then transferring one acre from Ann to Bob and simultaneously transferring £1000 of cash from Bob to Ann leaves Ann's utility unchanged (she loses land worth 1 and gains cash worth 1), while Bob's utility rises by

$3 - 1 = 2$. Hence any allocation in which Ann holds land is Pareto dominated. Therefore, at every Pareto-efficient allocation, Bob holds all 100 acres and only the cash is split.

Thus the Pareto frontier is

$$u_A + u_B = 500, \quad 0 \leq u_A \leq 200.$$

Since the bargaining solution must also weakly dominate the disagreement point $(150, 250)$, the only relevant part of this frontier is the individually rational segment

$$T = \{(u_A, u_B) : u_A + u_B = 500, 150 \leq u_A \leq 200\}.$$

By **independence of irrelevant alternatives (IIA)**, once we know the solution lies in T , we may discard all other feasible points and solve the reduced problem (T, d) .

Now translate utilities by the disagreement point. Let

$$g_A = u_A - 150, \quad g_B = u_B - 250.$$

By **invariance (INV)**, this affine change of utility does not affect the bargaining solution apart from transforming coordinates accordingly. In these new coordinates the disagreement point is $(0, 0)$, and on T we have

$$g_A + g_B = (u_A - 150) + (u_B - 250) = 500 - 400 = 100.$$

Also, since $150 \leq u_A \leq 200$, we have $0 \leq g_A \leq 50$, so the reduced feasible set becomes

$$G = \{(g_A, g_B) : g_A + g_B = 100, 0 \leq g_A \leq 50\}.$$

Consider now the larger symmetric bargaining problem

$$H = \{(g_A, g_B) : g_A + g_B = 100, g_A \geq 0, g_B \geq 0\},$$

with disagreement point $(0, 0)$. This problem is symmetric between the two players, so by **symmetry (SYM)** its solution must satisfy

$$g_A = g_B.$$

Since every point in H also satisfies $g_A + g_B = 100$, it follows that

$$g_A = g_B = 50.$$

Since $(50, 50) \in G \subseteq H$, **IIA** implies that removing the irrelevant alternatives in $H \setminus G$ does not change the solution. Hence the solution to the original reduced problem is also

$$(g_A, g_B) = (50, 50).$$

Transforming back to the original utilities gives

$$(u_A, u_B) = (150, 250) + (50, 50) = (200, 300).$$

Therefore the Nash bargaining solution is

$$(u_A, u_B) = (200, 300).$$

One allocation that implements this is: Ann receives all £200,000 of cash, and Bob receives all 100 acres of land.

3. Solve intuitively: Given that the disagreement point $d = (150, 250)$, it is the case that Ann can transfer more land to Bob. So, any pareto efficient allocation must give Bob all the land since Bob could transfer \$1000 to Ann and Ann could transfer 1 acre so that $\Delta u_A = 0, \Delta u_B = \2000 . Hence, this reduces the problem to $u_A = m, u_B = 500 - m$. So, it becomes

$$\max(m - 150)(250 - m) \text{ s.t. } 250 \geq m \geq 150$$

which yields $m = 200$ and the solution.

The alternative intuitive approach is to simply intuit which constraint binds – in this case, we conjecture that $u_A + u_B = 500$ binds while the other constraint remains slack (perhaps through some pareto frontier + disagreement point argument)

Discussion on Nash Bargaining Solution

1. NBS splits the surplus equally:

$$u_A^* = d_A + \frac{1}{2}(S - d_A - d_B), u_B^* = d_B + \frac{1}{2}(S - d_A - d_B)$$

when they share monetary surplus and both are risk-neutral

2. More risk aversion gets less share (more scared to disagree) since for $U = \{(u_1, u_2) : u_1 \in [0, 1], u_2 \leq (1 - u_1)^{1-r}\}$ with $r \in [0, 1]$ fixed and $d = (0, 0)$ then

$$u_A^* = \frac{1}{2-r}, u_B^* = \left(\frac{1-r}{2-r}\right)$$

so as r goes up, B gets less

3. NBS is not egalitarian: there are other feasible, more egalitarian allocations that the NBS does not yield
4. NBS is not monotonic: suppose we change inheritance to 60 gold coins but 25 acres and the rest of the problem is unchanged (heterogenous bundle game). Then, U expands to $u_A, u_B > 0, u_A + u_B \leq 110, \frac{1}{2}u_A + u_B \leq 85$. The solution yields

$$(u_A^*, u_B^*) = (55, 55)$$

which makes Ann worse off and does not lie on the kink even though U expands to U'

Advantages of Nash Bargaining Solution

Note that in a Nash demand game where there is a closed, bounded and convex bargaining set U and disagreement point (d_1, d_2) and players 1 and 2 simultaneously demand u_1 and u_2 respectively, all (u_1, u_2) on the pareto frontier with $(u_1, u_2) \geq (d_1, d_2)$ are nash equilibria. But Nash showed that the only NE robust to ‘small multiplicative payoff perturbations’ corresponds to the Nash bargaining solution

Example 2.4 (Perturbed Nash demand game). Suppose $(d_1, d_2) = (0, 0)$ and let $u_2 = g(u_1)$ be the pareto frontier of U . Hence $(u_1, u_2) \geq (0, 0)$ belongs to U iff $u_2 \leq g(u_1) \iff g(u_1) - u_2 \geq 0$. Each player pick u_1, u_2 and nature draws randomly $\varepsilon \geq 0$. Each player i gets payoff u_i if $g(u_1) - u_2 \geq \varepsilon$ and $d_i \equiv 0$ otherwise.

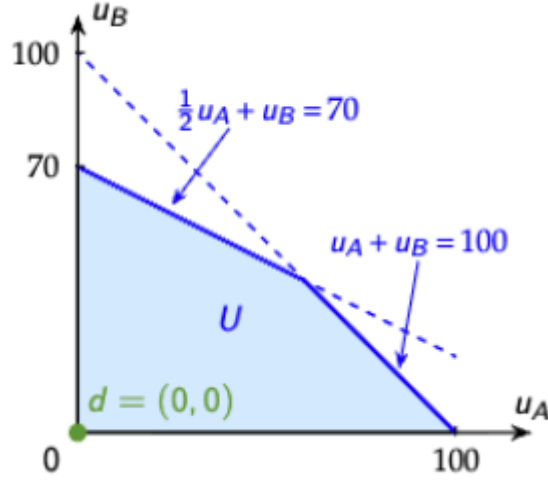
Solution: Let F denote the cdf of ε ; then player i ’s expected payoff is $F(g(u_1) - u_2) \times u_i$. Hence, $F(g(u_1) - u_2)$ is a multiplicative common-value payoff perturbation. The perturbation is small if F puts almost all probability weight near $\varepsilon = 0$.

Player $i = 1, 2$ maximise $F(g(u_1) - u_2)u_i$ in u_i given u_{-i} . The FOCs are

$$\begin{aligned} F'(g(u_1) - u_2)g'(u_1)u_1 + F(g(u_1) - u_2) &= 0 \\ -F'(g(u_1) - u_2)u_2 + F(g(u_1) - u_2) &= 0 \end{aligned}$$

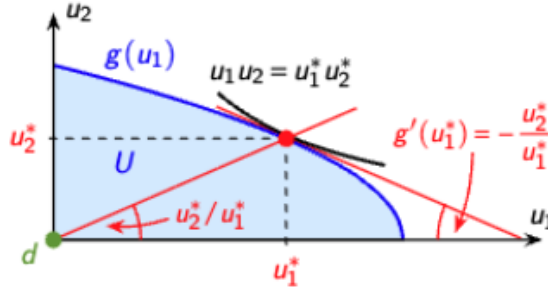
Combine and simplify yields

$$g'(\tilde{u}_1)\tilde{u}_1 + \tilde{u}_2 = 0$$



Note that $(\tilde{u}_1, \tilde{u}_2) \notin U$ makes sense. Now, we want to show that there is a convergence to the Nash Bargaining Solution as the perturbation vanishes.

As the distribution of ε approaches $\varepsilon = 0$ almost surely, the equilibrium point $(\tilde{u}_1, \tilde{u}_2)$ must converge to the boundary of U because players realise it is riskier and riskier to demand jointly infeasible utility levels. Hence, the equilibrium converges to (u_1^*, u_2^*) on the boundary of U where $g'(u_1^*)u_1^* + u_2^* = 0$.



which is the Nash bargaining solution

2.2 Non-Cooperative Bargaining, Dynamic Protocols

One-shot games

Definition 2.2 (Ultimatum Game). An ultimatum game can be characterised as

- P1 and P2 to share a set of goods Z ; allocations generate payoffs $(u_1, u_2) \in U$
- P1 makes a take-it-or-leave it offer to P2, $(u_1, u_2) \in U$
- if P2 accepts, then their payoffs are (u_1, u_2) ; if he rejects then (d_1, d_2)

Theorem 2.2. The unique subgame perfect equilibrium in the ultimatum game is

$$(u_1^*, u_2^*) = \operatorname{argmax}\{u_1 : (u_1, u_2) \in U, u_2 \geq d_2\}$$

Intuitively, P2 accepts all (u_1, u_2) with $u_2 \geq d_2$ and rejects any other offer.

Definition 2.3 (Divide and Choose). The Divide and Choose game can be characterised as

- P1 and P2 to divide a set of goods, Z

- P1 creates two bundles: X and $Z \setminus X$ and P2 picks one of them
- if P2 picks (say) bundle $Z \setminus X$ then payoffs are $u_1(X), u_2(Z \setminus X)$

Theorem 2.3. The unique subgame perfect equilibrium is P1 choosing X^* maximising $u_1(X)$ such that $u_2(X) = u_2(Z \setminus X)$

Proof. P2 picks $Z \setminus X$ from partition $\{X, Z \setminus X\}$ if either $u_2(Z \setminus X) > u(X)$ or $u_2(X) = u_2(Z \setminus X)$ and $u_1(X) > u_2(Z \setminus X)$. Otherwise, P2 takes X ■

On the equilibrium path, P1 makes P2 indifferent between the two bundle and P2 picks whichever bundle P1 likes less.

Example 2.5. See PSet 6 Q1

Alternating Offers (Dynamic)

Definition 2.4 (Alternating offers bargaining over a finite horizon). We can characterise the Alternating offers bargaining over a finite horizon as such

- at t odd, P1 proposes $(x, 1 - x)$ and P2 accepts or rejects
- at t even, P2 proposes $(y, 1 - y)$ and P1 accepts or rejects
- game ends with first acceptance and payoffs $(x, 1 - x)$ or $(y, 1 - y)$
- per period discount factors δ_1 and δ_2 respectively
- Assume that $t = 1, \dots, T$ where T is even (and P1 proposes first as a convention)

Theorem 2.4. In Alternating offers bargaining over a finite horizon, the unique subgame perfect equilibrium (on-path) is for P1 to set

$$x_1 = (1 - \delta_2) \left[1 + \dots + (\delta_1 \delta_2)^{\frac{1}{2}T-1} \right]$$

which is accepted by P2 on path

The proof is important because it shows us how to derive the result

Proof. We use backward induction. At $t = T$, P2 makes the offer in ultimatum game $(y, 1 - y) = (0, 1)$ with P1 accepting any $y \geq 0$ with probability 1.

At $T = T - 1$, P1 must set $1 - x_{T-1} \geq \delta_2$ to get it accepted and keep positive x (since P2 gets $\delta_2 \cdot 1$ in the next period). So, P1 sets $x_{T-1} = 1 - \delta_2$ and P2 accepts any offer $1 - x_{T-1} \geq \delta_2$ for sure.

At $T = T - 2$, P2 offers $y_{T-2} = \delta_1(1 - \delta_2)$ and P1 accepts this or any higher y . At $T = T - 3$, P1 needs to make $1 - x_{T-3} \geq \delta_2[1 - \delta_1(1 - \delta_2)]$ to ensure P2 accepts; P1 sets $x_{T-3} = 1 - \delta_2[1 - \delta_1(1 - \delta_2)] = (1 - \delta_2)(1 + \delta_1\delta_2)$ and so on.

Summarising P1's offers in odd periods and P2's responses

- At $t = T - 1$: P1 sets $x_{T-1} = 1 - \delta_2$; P2 accepts iff $x \leq x_{T-1}$
- At $t = T - 3$: P1 sets $x_{T-3} = (1 - \delta_2)(1 + \delta_1\delta_2)$; P2 accepts is $x \leq x_{T-3}$
- ...
- At $t = 1$: P1 sets $x_1 = (1 - \delta_2) \left[1 + \dots + (\delta_1 \delta_2)^{\frac{1}{2}T-1} \right]$ and P2 accepts iff $x \leq x_1$

■

Theorem 2.5. As $T \rightarrow \infty$, the unique SPE of finite-horizon alternating offers bargaining game converges to a stationary SPE of the infinite-horizon game, such that in odd periods P1 makes an offer with

$$x^* = \frac{1 - \delta_2}{1 - \delta_1 \delta_2}$$

whereas in even periods P2 makes an offer with

$$y^* = 1 - \frac{1 - \delta_1}{1 - \delta_1 \delta_2}$$

These offers are immediately accepted

This is because from the previous analysis, P1's own share in her offer at odd periods converges to

$$x^* = \frac{1 - \delta_2}{1 - \delta_1 \delta_2}$$

and in equilibrium, P2 must be indifferent between accepting $(1 - x^*)$ when offered and rejecting $1 - x^*$ to obtain $\delta_2(1 - y^*)$ via his counter offer (which implies x^* is maximised). So,

$$1 - x^* = 1 - \frac{1 - \delta_2}{1 - \delta_1 \delta_2} = \delta_2(1 - y^*) \iff 1 - y^* = \frac{1 - \delta_1}{1 - \delta_1 \delta_2}$$

Theorem 2.6 (Rubinstein (1982)). In the alternating bargaining infinite horizon game, there is no SPE other than the stationary one

Remark. We have the following takeaways

- Stationary SPE satisfies $1 - x^* = \delta_2(1 - y^*)$ and $y^* = \delta_1 x^*$
- no delay: in infinite horizon there is immediate acceptance
- patience pays: i 's share when offering $(1 - \delta_{-i})/(1 - \delta_i \delta_{-i})$ is increasing in δ_i but decreasing in δ_{-i}
- first mover advantage: if $\delta_1 = \delta_2 = \delta$, then $x^* = \frac{1}{1-\delta} > 1/2$

Theorem 2.7. Suppose that players evaluate $(x, 1 - x)$ as $u_1(x)$ and $u_2(1 - x)$ and let $\delta_1 = \delta_2 = \delta$. Then, as $\delta \rightarrow 1$, x^* and y^* converge to the same z^* which maximises $u_1(z)u_2(1 - z)$ on U i.e. to the Nash Bargaining Solution

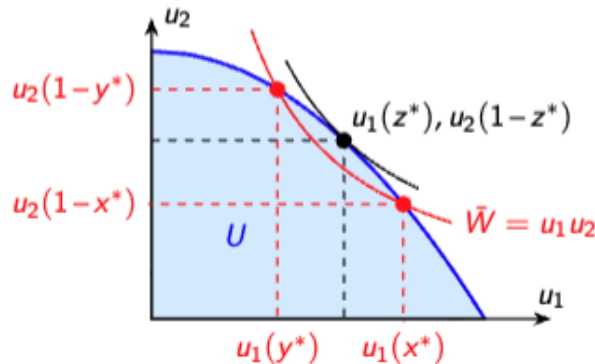
The intuition for this is that when we consider the infinite horizon stationary SPE, we have the following indifference conditions

$$u_2(1 - x^*) = \delta u_2(1 - y^*) \text{ and } u_1(y^*) = \delta u_1(x^*)$$

Multiplying them together and cancelling out δ yields

$$u_1(x^*)u_2(1 - x^*) = u_1(y^*)u_2(1 - y^*)$$

i.e. these two utility pairs are on the same indifference curve of the equal weight Cobb-Douglas $\bar{W}$.



It is provable that as $\delta \rightarrow 1$, x^* and y^* converge to the same z^* which maximises $u_1(z)u_2(1 - z)$ on U i.e. to the Nash Bargaining Solution

Bargaining Frictions (Dynamic)

In this section, we need to be familiar with the taylor series and continuous compounding.

Definition 2.5 (Frictionless Bargaining). A frictionless bargaining model may be characterised as

- Physical time is $t = \Delta, 2\Delta, \dots$ where $\Delta > 0$ is the length of a period; P1 and P2 alternate in offering $(x, 1 - x)$ and $(y, 1 - y)$ until first acceptance
- Player i 's between periods discount factor is $\delta_i = e^{-r_i\Delta} \approx 1 - r_i\Delta$ where $r_i > 0$ is i 's discount rate ('negative interest rate') per unit physical time
 - Notice that the discount factor δ_i is decreasing in the discount rate r_i
 - first order taylor approximation with $x = 0, \varepsilon = -r_i\Delta$ is applied to yield $e^{x+\varepsilon} \approx e^x + (e^x)'\varepsilon$
- As $\Delta \rightarrow 0$ the discount factor approach 1 even though players are impatient. This is the case of $T \rightarrow \infty$ and $\delta \rightarrow 1$

Theorem 2.8. Assuming risk neutrality, the stationary SPE in the frictionless bargaining model is

$$x^* = \frac{r_2}{r_1 + r_2 - r_1 r_2 \Delta}$$

which converges to

$$x^* = \frac{r_2}{r_1 + r_2}$$

as $\Delta \rightarrow 0$

This is because the stationary SPE is like in the alternating bargaining model with

$$x^* = \frac{1 - \delta_2}{1 - \delta_1 \delta_2}$$

and substituting $\delta_i = 1 - r_i\Delta$ yields

$$x^* = \frac{r_2\Delta}{1 - (1 - r_1\Delta)(1 - r_2\Delta)} = \frac{r_2}{r_1 + r_2 - r_1 r_2 \Delta}$$

Patience pays as in the alternating bargaining model and $r_1 = r_2 = \frac{1}{2} \implies x^* = \frac{1}{2}$

Definition 2.6 (Arbitrarily frequent interactions, impatience, flow cost or benefit). A bargaining model with frictions can be characterised as

- P1 and P2 (risk neutral) negotiate over 1 unit of money
- The protocol is infinite horizon alternating offers bargaining at $t = \Delta, 2\Delta, \dots$
- Period length $\Delta > 0$ is infinitesimal, i 's discount rate is r_i per unit time, hence the players' between periods discount factors are $\delta_i = 1 - r_i\Delta$ for $i = 1, 2$
- Player i also incurs additive cost c_i per unit time, i.e., $c_i\Delta$ per period
- Allow $c_i < 0$ interpreted as flow benefit from continued disagreement

Theorem 2.9. The unique stationary SPE in a bargaining model with frictions is

$$x^* = \frac{r_2 + c_2 - c_1}{r_1 + r_2}$$

as $\Delta \rightarrow 0$

Intuition: Suppose P1 offers $(x, 1 - x)$ in odd and P2 offers $(y, 1 - y)$ in even periods. The indifference conditions characterizing the stationary SPE are:

$$1 - x^* = -c_2\Delta + (1 - r_2\Delta)(1 - y^*) \quad (2)$$

$$y^* = -c_1\Delta + (1 - r_1\Delta)x^* \quad (3)$$

Solve the two equations (sub the latter into the former):

$$1 - x^* = -c_2\Delta + (1 - r_2\Delta)[1 + c_1\Delta - (1 - r_1\Delta)x^*]$$

$$\iff 0 = -c_2\Delta + c_1\Delta + r_1\Delta x^* - r_2\Delta - r_2c_1\Delta^2 + r_2\Delta x^* - r_1r_2\Delta^2 x^*.$$

Cross-divide by Δ , then take $\Delta \rightarrow 0$, finally express x^* :

$$x^* = \frac{r_2 + c_2 - c_1}{r_1 + r_2}$$

Compare with $x^* = r_2/(r_1 + r_2)$ of the frictionless bargaining model. If $r_1 = r_2 = r$, then:

$$x^* = -\frac{c_1}{r} + \frac{1}{2} \left(1 + \frac{c_1}{r} + \frac{c_2}{r} \right)$$

This is NBS with transferable utilities

- By disagreeing forever, P1 gets $-c_1/r$ in present value... hence $d_1 = -c_1/r$
- The joint surplus is $S = 1$, in NBS, P1 must get $x^* = d_1 + \frac{1}{2}(S - d_1 - d_2)$

3 Auctions and Efficient Mechanism Design

3.1 Basic Auction Theory

3.2 Basic Auction Theory

Definition 3.1 (Single-object auction environment). A single-object auction environment consists of:

- a finite set of bidders $N = \{1, \dots, n\}$, where $n \geq 2$;
- one indivisible object for sale;
- for each bidder $i \in N$, a type $t_i \in T_i$, representing bidder i 's private information;
- a type profile $t = (t_1, \dots, t_n) \in T = \prod_{i=1}^n T_i$, drawn from a commonly known prior distribution F on T ;
- for each bidder $i \in N$, a valuation function

$$v_i : T \rightarrow \mathbb{R},$$

where $v_i(t)$ is bidder i 's value for receiving the object when the realised type profile is t .

An auction mechanism specifies an allocation rule and a payment rule as functions of the bidders' reports or bids.

Definition 3.2 (Allocation and payment rules). Let B_i denote bidder i 's bid set and let $B = \prod_{i=1}^n B_i$. An auction mechanism is given by:

- an allocation rule $x = (x_1, \dots, x_n) : B \rightarrow \{0, 1\}^n$ such that

$$\sum_{i=1}^n x_i(b) \leq 1 \quad \text{for all } b \in B;$$

- a payment rule $p = (p_1, \dots, p_n) : B \rightarrow \mathbb{R}_+^n$, where $p_i(b)$ is bidder i 's payment.

Bidder i 's payoff is

$$u_i(b, t) = x_i(b)v_i(t) - p_i(b).$$

Definition 3.3 (Private-values environment). The environment is a private-values environment if, for each bidder i ,

$$v_i(t) = v_i(t_i),$$

so bidder i 's valuation depends only on her own type.

Definition 3.4 (Independent private-values environment). The environment is an independent private-values environment if it is a private-values environment and the types $t_1, \dots, t_n$ are independently distributed. A standard symmetric benchmark assumes:

- $T_i = [\underline{v}, \bar{v}]$ for all i ;
- $t_1, \dots, t_n$ are i.i.d. with cdf F ;
- $v_i(t_i) = t_i$ for each bidder i .

Definition 3.5 (Common-values environment). The environment is a common-values environment if there exists a function $v : T \rightarrow \mathbb{R}$ such that

$$v_i(t) = v(t) \quad \text{for all } i \in N.$$

Thus the ex post value of the object is the same for all bidders, though different bidders may have different signals about it.

Definition 3.6 (First-price sealed-bid auction). In a first-price sealed-bid auction:

- each bidder $i \in N$ simultaneously submits a sealed bid $b_i \in \mathbb{R}_+$;
- the highest bidder wins the object;
- the winner pays her own bid.

Formally, with a fixed tie-breaking rule,

$$p_i(b) = x_i(b)b_i,$$

so bidder i 's payoff is

$$u_i(b, t) = x_i(b)(v_i(t) - b_i).$$

Definition 3.7 (Second-price sealed-bid auction). In a second-price sealed-bid auction:

- each bidder $i \in N$ simultaneously submits a sealed bid $b_i \in \mathbb{R}_+$;
- the highest bidder wins the object;
- the winner pays the second-highest bid.

If bidder i wins, her payment is

$$p_i(b) = \max_{j \neq i} b_j.$$

Hence her payoff is

$$u_i(b, t) = x_i(b) \left(v_i(t) - \max_{j \neq i} b_j \right).$$

Definition 3.8 (Dutch auction). A Dutch auction is a dynamic auction in which the auctioneer starts from a high price and continuously lowers it. Each bidder chooses when to stop the clock; the first bidder to do so wins the object and pays the current price.

Definition 3.9 (English auction). An English auction is an ascending-price auction in which the price is gradually raised and bidders successively drop out. The last remaining bidder wins and pays the final standing price.

Proposition 3.1 (Strategic equivalence of Dutch and first-price sealed-bid auctions). For a single-object auction, the Dutch auction and the first-price sealed-bid auction are strategically equivalent.

Proof. In a first-price sealed-bid auction, bidder i chooses a bid b_i . In a Dutch auction, bidder i chooses the price at which she is willing to stop the clock. In either format, the bidder with the highest chosen cutoff wins and pays exactly that cutoff. Thus the strategic problem is the same. ■

Proposition 3.2 (Weak dominance of truthful bidding in the second-price auction). In an independent private-values environment, bidding one's true valuation is a weakly dominant strategy in the second-price sealed-bid auction.

Proof. Fix bidder i with valuation v_i , and fix the bids of all other bidders. Let

$$m = \max_{j \neq i} b_j$$

be the highest rival bid.

If bidder i bids $b_i = v_i$, then:

- if $v_i < m$, bidder i loses and gets payoff 0;
- if $v_i > m$, bidder i wins and gets payoff $v_i - m > 0$.

Now compare any deviation $b'_i \neq v_i$.

If $b'_i < v_i$, then the only possible change is that bidder i may lose in some case where

$$b'_i < m < v_i.$$

But in that case winning would have yielded positive payoff

$$v_i - m > 0,$$

so bidding below v_i cannot improve her payoff.

If $b'_i > v_i$, then the only possible change is that bidder i may win in some case where

$$v_i < m < b'_i.$$

But then winning yields payoff

$$v_i - m < 0,$$

so bidding above v_i cannot improve her payoff.

Therefore bidding $b_i = v_i$ yields a payoff at least as high as any other bid, for every profile of rival bids. Hence truthful bidding is weakly dominant. ■

In the private-values benchmark, the English auction has the same essential logic as the second-price auction: remaining active until the price reaches one's valuation is optimal, and the winner pays a price determined by rivals' willingness to stay in rather than by her own bid directly.

Proposition 3.3 (Bid shading in first-price auctions). In a first-price auction, bidders generally bid below their valuations.

The reason is that a higher bid increases the probability of winning, but also increases the payment conditional on winning. Unlike the second-price or English auction, the bidder cannot separate the "win/lose" decision from the price she pays, so she optimally shades below value.

Theorem 3.1 (Revenue Equivalence). Consider a symmetric independent private-values environment with risk-neutral bidders. Suppose the auction mechanism always allocates the object to the bidder with the highest valuation, and suppose a bidder with the lowest possible valuation obtains expected payoff 0. Then any two such auction mechanisms give each type the same expected payoff and therefore yield the same expected seller revenue.

Under these benchmark assumptions, the standard efficient auction formats — first-price sealed-bid, Dutch, second-price sealed-bid, and English — have the same expected revenue.

Example 3.1 (Why revenue equivalence is subtle). In a first-price auction, bidders shade below value. In a second-price auction, truthful bidding is weakly dominant. Thus the strategic form is very different across the two auctions. Revenue equivalence says that despite these different bidding rules, expected revenue coincides under the benchmark assumptions.

Definition 3.10 (Winner's curse). The winner's curse is the phenomenon that in a common-values auction, conditional on winning, a bidder should infer that her estimate of the object's value was relatively optimistic.

This matters only in common-values environments. If the object has one true underlying value and bidders merely observe different signals about it, then winning itself is informative: it means one's estimate was highest among the bidders. A rational bidder therefore conditions on the event of winning and bids less aggressively than she would if she ignored this inference.

Example 3.2 (Winner's curse intuition). Suppose several firms bid for drilling rights to an oil field. The tract has one true value, but each firm has only a noisy estimate of that value. The firm with the highest estimate wins. Precisely because it had the highest estimate, the winner should suspect that its estimate was too optimistic. A bidder who ignores this effect risks overpaying.

Example 3.3 (PSet6 Q3 – uniform price multi-unit auction game). Two bidders competing are two identical objects. Buyer i 's privately-known valuation is X_i for the first unit and Y_i for the second unit, where $0 \leq Y_i \leq X_i \leq 1$. The valuations are determined as follows. Each buyer i draws two independent random variables from the uniform distribution on $[0, 1]$. Then i 's valuation for the first unit is the maximum, whereas her valuation for the second unit is the minimum of these two draws. Therefore, the cdf of X_i is $F(x) = \mathbb{P}[X_i \leq x] = x^2$: this is the probability that both uniform draws are less than x . The corresponding density is $f(x) = 2x$ for all $x \in [0, 1]$. The auction is a uniform-price auction: Buyers submit two bids each. The highest two bids win; the winner(s) must pay the third highest (i.e. the highest losing) bid. Buyer i 's payoff is the sum of the valuation(s) for the object(s) that she has won less the amount she is required to pay.

- (a) Suppose that buyer 1 submits bids $\{X_1, 0\}$, that is, her valuation for the first unit and zero. Explain that the other buyer's expected payoff from bidding $\{a, b\}$ with $0 \leq b \leq a \leq 1$ can be written as $(1 - b^2)(X_2 - b) + \int_0^b (X_2 + y_2 - 2x)2xdx$.

Our bids are $\{X_1, 0, a, b\}$. We have two cases.

Case 1: $X_1 > b$. So, top two bids are X_1, a . So buyer 2 wins 1 unit with third highest bid b and she pays b with payoff $X_2 - b$.

Case 2: $b > X_1$. So, top two bids are a, b and buyer 2 wins 2 units with third highest bid X_1 and so payoff is $X_2 + Y_2 - 2X_1$.

The total expected payoff is

$$\begin{aligned} \mathbb{P}(X_1 > b)(X_2 - b) + \mathbb{P}(X_1 < b)\mathbb{E}[X_2 + Y_2 - 2X_1 | X_1 < b, X_2, Y_2] &= (1 - b^2)(X_2 - b) + \int_0^b (X_2 + Y_2 - 2x)f_X(x)dx \\ &= (1 - b^2)(X_2 - b) + \int_0^b (X_2 + Y_2 - 2x)2xdx \end{aligned}$$

- (b) Hence prove that it is a Bayes-Nash equilibrium in the uniform price auction for each buyer i to submit bids $\{X_i, 0\}$.

We want to show that $\{X_1, 0\}$ and $\{X_2, 0\}$ are mutual best responses. Compute Player 2's expected payoff

$$\begin{aligned} u_2 &= (1 - b^2)(X_2 - b) + \int_0^b (X_2 + Y_2 - 2x)2xdx \\ &= X_2 - bY_2b^2 - \frac{1}{3}b^3 \end{aligned}$$

Differentiating gives

$$\frac{du_2}{db} = -1 + 2Y_2b - b^2 \leq 0$$

since $2Y_2b - b^2 \leq 2b - b^2 = 1 - (1 - b)^2 \leq 1$ as $Y_2 \in [0, 1]$. So $u_2(b)$ is strictly decreasing wrt to b and so $b^* = 0$. Thus, buyer 2's expected payoff is $u_2(0) = X_2$. Any choice of $a > 0$ does not affect payoffs so X_2 is a best response.

For Player 1, we can similarly construct (a, b) to $\{X_2, 0\}$ where

$$u_1 = (1 - b^2)(X_1 - b) + \int_0^b (X_1 + Y_1 - 2x)2xdx$$

Likewise, $b^* = 0, a^* = X_1$. Hence, $\{X_1, 0\}$ and $\{X_2, 0\}$ is a BNE.

- (c) Explain why the outcome of this auction is inefficient. Can you think of a modification of the payment rule that would induce truthful bidding?

Both buyers get exactly one unit since $\{X_1, X_2, 0, 0\}$ are the bids. But if $X_1 > X_2$ and $Y_1 > X_2$, then buyer 1 should get both bids.

We can use a VCG mechanism where the buyer pays externality they impose on others.

Remark. Note that

$$\mathbb{P}(A)\mathbb{E}[g(X)|A] = \int_A g(x)dP = \int_A g(x)f_X(x)dx$$

since $\mathbb{E}[g(X)|A]$ is the average value of $g(X)$ given that A happens and $f_X(x)$ provides the accurate weighting of how much each value of x contributes to this average

Example 3.4 (Pset6 Q4 – two sided private info). It costs seller (S) an amount c to produce a single, indivisible good. The valuation of a buyer (B) for this good is v . Both v and c are expressed in the same monetary units and are independently drawn from the uniform distribution on $[0, 1]$. S only observes c , B only observes v .

The two players are committed to the following trading game: S will announce a reservation level r and B will simultaneously announce a price p . The good will be traded at the announced price p iff $p \geq r$.

- (a) Assuming the cost c is paid only when the good is successfully traded, carefully write down the expected payoffs to the players in this game.

If buyer B announces price p , then she obtains payoff $v - p$ if trade occurs and 0 otherwise. Since trade occurs exactly when $p \geq r$, her expected payoff conditional on her valuation v is

$$\mathbb{E}[u_B | v] = (v - p) \mathbb{P}(p \geq r | v).$$

Now r is chosen by the seller as a function of her private cost c , so the event $\{p \geq r\}$ is equivalently the event $\{r(c) \leq p\}$. Hence we can rewrite this as

$$\mathbb{E}[u_B | v] = (v - p) \mathbb{P}(r(c) \leq p).$$

Since $c \sim U[0, 1]$, this probability can be written as an integral over the seller's type space:

$$\mathbb{E}[u_B | v] = (v - p) \int_0^1 \mathbf{1}\{r(c) \leq p\} dc.$$

This integral form is useful because the indicator function $\mathbf{1}\{r(c) \leq p\}$ picks out exactly those seller types for which trade takes place.

If in addition $r(\cdot)$ is strictly increasing, then the condition $r(c) \leq p$ is equivalent to $c \leq r^{-1}(p)$. Therefore the set of seller types with whom trade occurs is the interval $[0, r^{-1}(p)]$, and so

$$\mathbb{E}[u_B | v] = (v - p) \int_0^{r^{-1}(p)} dc.$$

Evaluating the integral gives

$$\mathbb{E}[u_B | v] = (v - p) r^{-1}(p),$$

provided $r^{-1}(p) \in [0, 1]$.

More carefully, allowing for boundary cases, we obtain

$$\mathbb{E}[u_B | v] = (v - p) \begin{cases} 0 & \text{if } p < r(0), \\ r^{-1}(p) & \text{if } r(0) \leq p \leq r(1), \\ 1 & \text{if } p > r(1). \end{cases}$$

The interpretation is straightforward: buyer B 's expected payoff is her gain from trade, $v - p$, multiplied by the probability that the seller's reservation level is low enough for trade to occur.

- (b) Hence show that there is a Bayes-Nash equilibrium in which the buyer announces a price $p = v/2$. What does the seller announce?

We want to show that there is a BNE with $p = v/2$. First, consider the seller's Best Response: trade occurs iff $v/2 \geq r \iff v \geq 2r$. If $r \leq \frac{1}{2}$, since $v \sim U[0, 1]$, $\mathbb{P}(v \geq 2r | r) = \int_{2r}^1 1 dv = 1 - 2r$. If $r > \frac{1}{2}$, $2r > 1$ but $v \sim U[0, 1]$ so $\mathbb{P}(v \geq 2r | r) = 0$. Hence,

$$\mathbb{E}[p | \text{trade}] = \mathbb{E}\left[\frac{v}{2} | v \geq 2r\right] = \frac{1}{2} \cdot \frac{2r + 1}{2} = \frac{2r + 1}{4}$$

So seller chooses r over $r \in [0, 1/2]$ to maximise

$$u_S = (1 - 2r) \left(\frac{2r + 1}{4} - c \right) = \frac{2r + 1}{4} - c - \frac{2r^2 + r}{2} + 2rc$$

Taking the FOC gives us

$$\frac{1}{2} - \frac{4r + 1}{2} + 2c = 0$$

Hence, $c = r$ for $c \leq 1/2$. If $c > \frac{1}{2}$, any $r \geq 1/2$ yields payoff 0 which is optimal so $r = c$ is optimal. Hence, $r = c$ is best response to $p = \frac{v}{2}$.

Want to show that $p = \frac{v}{2}$ is best response to $r = c$.

Trade occurs iff $\mathbb{P}(c \leq p) = p$. Then,

$$u_B = (v - p)p = vp - p^2$$

Taking FOC

$$\frac{du_B}{dv} = v - 2p = 0 \implies p = \frac{v}{2}$$

Hence, $(v/2, c)$ is BNE

- (c) Is this equilibrium efficient i.e. does a trade take place whenever $v > c$? Comment.

Trade happens just when

$$p > r \iff \frac{v}{2} \geq c \iff v \geq 2c$$

but possible for $v \geq 2c$ even when not $v > c$ so trade might not occur even if efficient to do so.

- (d) Now suppose there is full information (so that c and v are observed by both players before play). Show that there are (many) efficient pure-strategy Nash equilibria. Are there any inefficient Nash equilibria? Since c and v are common knowledge, this is now a complete-information game.

Step 1: Best responses.

- Fix the seller's reservation level r . The buyer trades iff $p \geq r$. If trade occurs, her payoff is $v - p$; otherwise it is 0. Hence:
 - if $r \leq v$, the buyer wants trade, but at the lowest possible price, so a best response is $p = r$;
 - if $r > v$, trade would give negative payoff, so any $p < r$ is a best response, since this guarantees no trade and payoff 0.
- Fix the buyer's announced price p . The seller gets $p - c$ if trade occurs and 0 otherwise. Hence:
 - if $p \geq c$, trade is weakly beneficial, so any $r \leq p$ is a best response, since this ensures trade and payoff $p - c \geq 0$;
 - if $p < c$, trade would give negative payoff, so any $r > p$ is a best response, since this blocks trade and gives payoff 0.

Step 2: Efficient pure-strategy Nash equilibria.

If $v \geq c$, trade is efficient, since the buyer values the object at least as much as it costs the seller to produce it. Consider any pair

$$p = r = t \quad \text{for some } t \in [c, v].$$

Then trade occurs, the buyer gets

$$u_B = v - t \geq 0,$$

and the seller gets

$$u_S = t - c \geq 0.$$

We verify that this is a Nash equilibrium.

- Given $r = t \leq v$, the buyer's best response is $p = t$, since she wants trade at the lowest price that still secures trade.
- Given $p = t \geq c$, the seller's best response is any $r \leq t$, and in particular $r = t$ is a best response.

Hence every pair $(p, r) = (t, t)$ with $t \in [c, v]$ is a pure-strategy Nash equilibrium. Since there are infinitely many such t , there are many efficient pure-strategy Nash equilibria.

If $v < c$, then no trade is efficient. In that case, any profile with

$$p < c \quad \text{and} \quad r > p$$

is a Nash equilibrium: the seller blocks trade, which is optimal since trade would yield negative payoff, and the buyer does not want to raise her price to induce trade at a loss. These equilibria are also efficient, since efficient allocation requires no trade when $v < c$.

Step 3: Are there inefficient Nash equilibria?

Yes, when $v > c$. In that case trade is efficient, but there are also Nash equilibria with no trade.

For example, let

$$p < c \quad \text{and} \quad r > v.$$

Then no trade occurs. The buyer's payoff is 0, and since $r > v$, any attempt to match or exceed r would force her to pay more than her valuation and so give negative payoff. Thus choosing $p < r$ is a best response. The seller's payoff is also 0, and since $p < c$, allowing trade would give negative payoff, so choosing $r > p$ is a best response.

Therefore any profile satisfying

$$p < c \quad \text{and} \quad r > v$$

is a pure-strategy Nash equilibrium with no trade.

When $v > c$, such equilibria are inefficient, because trade should occur in any efficient outcome.

In particular,

$$(p, r) = (0, 1)$$

is an inefficient Nash equilibrium whenever $v < 1$ and $c > 0$, with in particular $v > c$. The buyer will not deviate upward because trading at price 1 would give payoff $v - 1 < 0$, and the seller will not deviate downward because trade at price 0 would give payoff $-c < 0$.

Conclusion. There are many efficient pure-strategy Nash equilibria: when $v \geq c$, every $(p, r) = (t, t)$ with $t \in [c, v]$ is one. There are also inefficient pure-strategy Nash equilibria whenever $v > c$, namely no-trade equilibria such as any (p, r) with $p < c$ and $r > v$.

3.3 Problems in Auction Design

Auction design concerns are:

collusion/coordination, entry deterrence, and, subject to these, information revelation.

Definition 3.11 (Auction design problem). An auction design problem consists of choosing a feasible mechanism so as to allocate one or more objects while taking account of:

- the informational environment;
- bidders' market power and ability to coordinate;
- incentives for entry by weaker or disadvantaged bidders;
- the extent to which price discovery and information aggregation are desirable.

Proposition 3.4 (First-order design priorities). In many practically relevant auctions, the main priorities are:

1. preventing collusion or tacit coordination;
2. preventing entry deterrence;
3. conditional on (1) and (2), choosing a format that reveals useful information and allocates efficiently.

When standard auction theory does and does not matter

Theorem 3.2 (Revenue equivalence benchmark, informal statement). Under the standard benchmark assumptions of symmetric independent private values, risk neutrality, efficient allocation, and zero expected payoff for the lowest type, the seller's expected revenue is the same across the standard efficient auction formats.

Thus, for a single object and absent entry or collusion problems, auction rules often matter much less than one might initially think.

The revenue-equivalence benchmark becomes a poor guide when:

- more than one object is sold;
- bidders have market power or can coordinate;
- bidder participation itself depends on the mechanism;
- the mechanism must aggregate information across different goods or varieties.

Ascending versus sealed-bid auctions

The general trade off is that ascending auctions often perform well when information revelation is valuable and competition is robust, but they can perform poorly when public bidding facilitates coordination, retaliation, or entry deterrence. Sealed-bid auctions suppress these dynamic responses, but at the cost of less price discovery.

The essential design trade-off is therefore: price discovery and information aggregation versus exposure to coordination and strategic intimidation.

Collusion, coordination, and demand reduction

Definition 3.12 (Collusion and tacit coordination in auctions). Collusion or tacit coordination in an auction refers to bidder behaviour that softens competition relative to the competitive benchmark, for example by dividing the market, reducing demand, or avoiding aggressive bidding against one another.

Proposition 3.5 (Conditions that support coordination). A group of bidders can more easily coordinate in an auction when they can:

1. identify a market-sharing arrangement;
2. detect deviation from it;
3. punish deviation credibly;
4. deter entry by outsiders.

Public and dynamic auction formats can help bidders satisfy these conditions, especially when bidders can monitor one another's behaviour round by round.

Example 3.5 (Austrian 3G auction: demand reduction). In the Austrian 3G auction, there were 12 lots and 6 bidders, and bidders were permitted to win multiple lots. A public statement by Telekom Austria effectively indicated willingness to settle for 2 lots provided rivals did likewise, together with a threat to compete for more if others did not. The bidding ended almost immediately. The economic interpretation is demand reduction: bidders softened competition by accommodating one another's desired quantities instead of bidding aggressively.

This example is important because it shows that the danger is not only explicit illegal collusion; auction rules can also support tacit market-sharing outcomes that look very much like ordinary oligopolistic coordination.

Entry deterrence

Definition 3.13 (Entry deterrence in auctions). Entry deterrence occurs when potential bidders refrain from participating, or bid only weakly, because the auction format or the competitive environment makes winning sufficiently unlikely or unprofitable.

Why entry deterrence can be severe Suppose one or more incumbent bidders are known to be especially strong. If weaker bidders must incur costs to participate, and especially if the auction is ascending so incumbents can publicly top their bids, then weak bidders may rationally choose not to enter at all.

The crucial point is that a bidder need not be literally unbeatable to deter rivals. It is enough that rivals expect low winning probabilities and low profits conditional on winning.

Example 3.6 (Netherlands versus UK 3G). The contrast between the UK and Dutch 3G auctions illustrates entry deterrence sharply. In the UK, there were 5 licences and only 4 strong incumbents, so at least one licence had to go to an entrant; this attracted entry and sustained strong competition. In the Netherlands, there were 5 licences and 5 strong incumbents, so an entrant could reasonably expect that any bid in an ascending format would simply be topped by an incumbent. Entry was therefore much less attractive, and the auction performed much worse.

Why sealed bids can help entry When entry deterrence is the main concern, a sealed-bid format may outperform an ascending format. The reason is that strong bidders cannot react publicly to entrants' bids, while strong bidders in a first-price environment also do not bid their full values. Hence weaker bidders retain some chance of winning.

Example 3.7 (Why Denmark did better). Denmark faced a setting in which sealed bidding was better suited than a public ascending auction. Because incumbents could not openly and dynamically top entrants, entry was more viable, and the auction performed better than an ascending format likely would have in that environment.

When ascending auctions are attractive

An ascending auction is particularly attractive when:

- information aggregation is important;
- bidders can learn from the bidding process;
- there is little scope for demand reduction;
- entry is secure and collusion is not a major concern.

Example 3.8 (Why the UK chose an ascending auction). In the UK 3G auction, bidders were allowed to win at most one licence, so there was little scope for dividing the spoils by reducing demand. Moreover, with 5 licences and only 4 strong incumbents, entry was not a major worry. Under those conditions, a simultaneous ascending format was likely to perform well because it could reveal information and support efficient allocation without the usual collusion and entry problems becoming dominant.

Auctions versus administrative allocation

Proposition 3.6 (Why auctions are usually preferable to beauty contests). Relative to administrative allocation or “beauty contests,” auctions are typically:

- more transparent;
- quicker to implement;
- less contestable ex post;
- better at allocating to high-value users;
- better at transferring value to the seller without deadweight loss.

The lecture's practical conclusion is not that every auction is well designed, but that even a bad auction is often better than the main alternative of discretionary administrative allocation.

Multiple goods and the need for richer mechanisms

Once multiple varieties of goods are sold, standard single-object auction formats become much less satisfactory. The auctioneer may care about the mix of goods sold, bidders may care about substitution across varieties, and running separate auctions can fragment competition.

A multi-variety auction design problem arises when:

- there are several substitutable or heterogeneous goods;
- bidders' demands depend on the relative prices of those goods;
- the seller may also care about the composition of total supply across goods.

Why separate auctions may fail Running separate auctions for different varieties may perform badly because:

- the seller must decide quantities of each variety before observing demand;
- bidders must decide ex ante which auction to enter;
- competition is diluted across auctions, increasing market power.

Why simultaneous ascending auctions may also fail in this setting

A simultaneous ascending auction may be unsuitable in a multi-variety setting because:

- the process may take too long;
- public bidding may facilitate collusion or predation;
- it is hard to make the mix of varieties supplied depend flexibly on the bids.

The Product-Mix Auction

Definition 3.14 (Product-Mix Auction). A Product-Mix Auction is a mechanism for allocating multiple varieties of a good in which:

- the auctioneer expresses preferences over the composition of supply, typically through a supply function;
- bidders simultaneously report their preferences over quantities and varieties;
- the mechanism computes prices and an allocation consistent with a competitive equilibrium.

The key design idea is to treat the auction not as a sequence of separate sales, but as a competitive market-clearing exercise in which both the seller's and buyers' preferences over the product mix are explicitly incorporated.

Example 3.9 (Bank of England motivation). In the Bank of England application, the Bank wished to lend against different qualities of collateral. A borrower might be willing to borrow the same amount against either good or poor collateral, but only at different interest rates, and might also have a preference over which collateral to use depending on the spread between those rates. This is naturally a product-mix problem rather than a sequence of independent single-good auctions.

Definition 3.15 (Basic bid in the Product-Mix Auction). A bid in the Product-Mix Auction can specify willingness to trade off different varieties at different prices. For example, a bid such as

(60bp, 50bp; 200m)

may be interpreted as:

“I would like to borrow £200m; I would pay up to 50 basis points using good collateral, or up to 60 basis points using poor collateral; and at lower interest rates I prefer poor collateral provided the rate difference is less than 10 basis points.”

This kind of bid is richer than a standard scalar bid, but still simple enough to be interpreted geometrically and implemented computationally.

Proposition 3.7 (Competitive-equilibrium interpretation). The Product-Mix Auction is designed to implement a competitive equilibrium:

- bidders submit preference information;
- the auctioneer submits or fixes a supply schedule;
- the mechanism determines prices that clear the market;

- the final allocation is the one each agent would choose at those prices.

Theorem 3.3 (Efficiency principle for the Product-Mix Auction). Under appropriate conditions, the Product-Mix Auction achieves an efficient allocation in the sense that each bidder, and the auctioneer, receives exactly the bundle or composition that would be chosen at the final equilibrium prices.

This is the key theoretical result highlighted in the lecture: the mechanism implements the competitive allocation rather than an arbitrary administrative one.

Example 3.10 (Interpretation of the Bank of England graphics). In the Bank of England diagrams, the auctioneer supplies a curve relating the relative price difference

$$P_A - P_B$$

to the fraction of total quantity allocated to one collateral type. Bidders' submitted preferences generate a corresponding relative demand curve. The intersection determines the equilibrium relative price and the equilibrium product mix.

Example 3.11 (Normal versus stressed demand). A particularly useful comparative-statics example in the slides is the shift from "normal" to "stressed" demand. Under stressed conditions, bidders demand a greater proportion of the weaker collateral type. In the Product-Mix Auction, this is reflected in a shift of the relative demand curve, and the mechanism automatically adjusts both prices and the composition of allocated goods accordingly.

Why the Product-Mix Auction is attractive

Proposition 3.8 (Advantages of the Product-Mix Auction). The Product-Mix Auction is attractive when:

- bidders need to express substitution patterns across goods;
- the auctioneer cares about the composition of supply;
- market power is reduced by pooling goods into one competitive mechanism rather than splitting them across separate auctions;
- simplicity and transparency remain important.

A major design achievement of the mechanism is that it allows rich preferences while remaining conceptually interpretable through ordinary supply-and-demand geometry.

General lesson: There is no universally optimal auction format. The appropriate mechanism depends on the interaction between market structure, entry conditions, collusion risks, informational needs, and the number and variety of goods being sold.

3.4 Mechanism Design with transfers

Set Up:

- Set X where $x \in X$ are various social decisions
- Agent i 's utility measured in money is $u_i(x, \theta_i) + p_i$ for $i \in N$
 - θ_i is i 's type, private information (can be complicated and multi-dimensional)
 - p_i is her transfer received (transferrable utility i.e. quasilinear preferences)

Definition 3.16 (Socially Efficient Decision Rule). Given every realised state $\theta = (\theta_i)_{i \in N}$, the socially efficient decision rule is

$$x^*(\theta) = \operatorname{argmax}_{x \in X} \sum_{i \in N} u_i(x, \theta_i)$$

The fundamental question is whether we can implement this without observing θ directly (efficient mechanism design under private values and transferable utilities).

Remark. Notice that even though the payoff is inclusive of transfers i.e. $u_i(x, \theta_i) + p_i$, we may ignore transfers when considering the socially efficient allocation since **there are no income effects with quasilinear utility**. In other words,

$$\text{efficiency} \iff \max_{x \in X} \sum_{i \in N} u_i(x, \theta_i)$$

More precisely, notice that even though payoffs are inclusive of transfers, i.e. $v_i = u_i(x, \theta_i) + p_i$, we can ignore transfers when determining the socially efficient allocation. Under budget balance, $\sum_{i \in N} p_i = 0$, so

$$\sum_{i \in N} v_i = \sum_{i \in N} u_i(x, \theta_i) + \underbrace{\sum_{i \in N} p_i}_{=0} = \sum_{i \in N} u_i(x, \theta_i).$$

Quasilinearity ensures transfers are welfare-neutral in aggregate—the marginal utility of money is constant at 1 for all agents, so a dollar transferred from i to j is an exact wash. Consequently, the choice of x and the choice of $\mathbf{p} = (p_i)_{i \in N}$ are separable, and efficiency reduces to

$$x^*(\theta) = \arg \max_{x \in X} \sum_{i \in N} u_i(x, \theta_i).$$

Definition 3.17 (Vickery-Clarke-Groves Mechanism). The Vickery-Clarke-Groves (VCG) Mechanism utilises the following procedure

1. Each $i \in N$ reports a type, $\hat{\theta}_i$; let $\hat{\theta} = (\hat{\theta}_i)_{i \in N}$ denote the reported types
2. Carry out decision $x^*(\hat{\theta})$, and set i 's transfer equal to

$$p_i^*(\hat{\theta}) = \sum_{j \neq i} u_j(x^*(\hat{\theta}), \hat{\theta}_j) - \pi_i(\hat{\theta}_{-i})$$

Each i reports her type, we carry out the efficient action at the reported types, then each i receives the sum of the other agents' gross (pre-transfer) payoffs calculated at the efficient allocation with the reported types, less a transfer contingent on the others' reports.

Remark. The function $\pi_i(\hat{\theta}_{-i})$ is an arbitrary function of the other agents' reported types $\hat{\theta}_{-i} = (\hat{\theta}_j)_{j \neq i}$, independent of i 's own report $\hat{\theta}_i$. It serves as a baseline correction: since it does not depend on $\hat{\theta}_i$, agent i cannot manipulate it, so it drops out of i 's optimisation and incentive compatibility is preserved. The social planner is therefore free to choose π_i to satisfy additional desiderata. Two canonical choices are:

- $\pi_i(\hat{\theta}_{-i}) = 0$: the simplest Groves mechanism.
- $\pi_i(\hat{\theta}_{-i}) = \max_{x \in X} \sum_{j \neq i} u_j(x, \hat{\theta}_j)$: the Clarke pivot rule, where the baseline is the welfare achievable without agent i .

Intuition: Each agent receiving the sum of everyone else's utilities (the first term of the transfer term) aligns individual preferences with those of society, and the second term adjusts budget.

We can restrict our attention to direct mechanisms where each agent reports their type to a central computer which decides on x and transfers according to a pre-set rule due to the revelation principle (see below). Briefly, the revelation principle can be characterised as: for any equilibrium of any (practical) mechanism there exists a direct revelation mechanism with an equilibrium in truthful reports that has the same outcome.

This is the most important result

Theorem 3.4 (Truth-telling in VCG). In a VCG mechanism, it is weakly dominant for each i to report $\hat{\theta}_i = \theta_i$; the outcome is efficient if we ignore the net sum of transfers (which is lost)

Proof. Rewrite efficiency of x^* as follows. For all $\theta_i, \hat{\theta}_{-i}$ and x ,

$$u_i(x^*(\theta_i, \hat{\theta}_{-i}), \theta_i) + \sum_{j \neq i} u_j(x^*(\theta_i, \hat{\theta}_{-i}), \hat{\theta}_j) \geq u_i(x, \theta_i) + \sum_{j \neq i} u_j(x, \hat{\theta}_j).$$

In the VCG mechanism, if i 's true type is θ_i and the reports are $\hat{\theta}_i$ and $\hat{\theta}_{-i}$, then i 's payoff is

$$u_i(x^*(\hat{\theta}_i, \hat{\theta}_{-i}), \theta_i) + \sum_{j \neq i} u_j(x^*(\hat{\theta}_i, \hat{\theta}_{-i}), \hat{\theta}_j) - \pi_i(\hat{\theta}_{-i}),$$

which is maximised at $\hat{\theta}_i = \theta_i$ by the efficiency of x^* . Therefore, irrespective of the others' reports, i has no incentive to lie. ■

Sometimes we want to ensure that there is budget balance.

Definition 3.18 (Pivot Mechanism). In a VCG mechanism, the pivot mechanism is

$$\pi_i(\hat{\theta}_{-i}) = \max_{x \in X} \sum_{j \neq i} u_j(x, \hat{\theta}_j)$$

Under the Vickery payment scheme, in allocation $x_i(\hat{\theta})$, player i pays

$$-p_i^*(\hat{\theta}) = \max_{x \in X} \sum_{j \neq i} u_j(x, \hat{\theta}_j) - \sum_{j \neq i} u_j(x^*(\hat{\theta}), \hat{\theta}_j)$$

which is the external effect of i 's preference on the welfare of others. Hence i 's net payoff is her marginal contribution:

$$m_i(\hat{\theta}) = \sum_{j \in N} u_j(x^*(\hat{\theta}), \hat{\theta}_j) - \max_{x \in X} \sum_{j \neq i} u_j(x, \hat{\theta}_j)$$

When there is a single unit for sale, it is like a second price auction where the winner pays the highest bid.

Definition 3.19 (Vickery's K -unit auction). Suppose N agents are interested in buying K identical goods. Each agent has decreasing, privately known marginal valuations

- For $i \in N$, $v_1^i > \dots > v_K^i$ are i 's valuations for the k -th unit
- Ex-ante distribution of v_k^i are ignored since it won't matter

The goal is to allocate the goods to agents who value them the most. Monetary transfers from and to agents are possible.

- Risk aversion is allowed, but no budget constraint
- Each agent participates for a non-negative payoff

We have the following rules and properties of the K -unit Vickery auction

1. Each agent $i \in N$ submits K bids, say $\hat{v}_1^i > \dots > \hat{v}_K^i$
2. Highest K bids (pooled from all other bidders) win
3. Each i pays the sum of the bids submitted by others that would have won had i not participated in the auction

The price a winner pays is not the “highest rejected” or “lowest winning” bid; indeed, each i may pay difference prices for the same good. Agent i 's bids do not affect how much she pays, only whether she wins; the principle is to ‘pay your externality’ i.e. the bids you crowded out.

Corollary 3.5. It is weakly optimal for each $i \in N$ to submit $\hat{v}_k^i = v_k^i, \forall k \in K$

Example 3.12 (Simple example). Suppose A,B,C and there are 3 apples. The valuations are

- A: 8, 2, 0
- B: 7, 6, 5
- C: 4, 0, -6

If all bid honestly, A gets 1 apple and B gets 2.

- A pays 5 (B's rejected bid)
- B pays 4+2 in total (since C would have gotten 1 apple valued at 4 and A would have gotten a second apple valued at 2 if B did not participate)

Pivotal agents pay their externality

Ausubel invented an open-bid, ascending price auction that is the counterpart of Vickery's auction

Definition 3.20 (Ausubel's Auction). Price clock is set to 0. Each i indicates her demand by showing at most K fingers. As p rises agents may reduce their demand.

If at price p the total number of fingers held up by agents other than i falls one short of the number of still available units, then we say that agent i has clinched a unit at price p . The unit clinched by i as well as i 's demand for it are removed.

The price clock resumes unless all units have been allocated.

Clarke's version of the pivot mechanism is for binary decisions where each agent i has private benefit v_i (net of cost) for 'yes' and 0 for 'no' – this is usually used for public goods.

Definition 3.21 (Clarke's pivot mechanism). Clarke's Pivot Mechanism has the following rules

1. Each $i \in N$ announces a net benefit $\hat{v}_i \in \mathbb{R}$
2. 'Yes' carried out iff $\sum_{i \in N} \hat{v}_i \geq 0$
3. Agent j is called pivotal if either $\sum_{i \neq j} \hat{v}_i \geq 0$ and $\sum_{i \in N} \hat{v}_i < 0$, or $\sum_{i \neq j} \hat{v}_i < 0$ and $\sum_{i \in N} \hat{v}_i \geq 0$.
Each pivotal j must pay $t_j = |\sum_{i \neq j} \hat{v}_i|$

Proposition 3.9. It is a weakly dominant strategy for each agent $i \in N$ to report v_i truthfully

Proof. Fix agent $j \in N$, and let

$$S_{-j} \equiv \sum_{i \neq j} \hat{v}_i.$$

We consider two cases.

Case 1: Suppose $S_{-j} \geq 0$.

Agent j is pivotal exactly when she reports $\hat{v}_j$ such that

$$\hat{v}_j + S_{-j} < 0,$$

that is, when

$$\hat{v}_j < -S_{-j}.$$

Hence agent j 's payoff is

$$u_j(\hat{v}_j) = \begin{cases} v_j, & \text{if } \hat{v}_j \geq -S_{-j}, \\ -S_{-j}, & \text{if } \hat{v}_j < -S_{-j}. \end{cases}$$

Notice that the payoff level in each region does not depend on the precise value of $\hat{v}_j$; the report $\hat{v}_j$ only determines which region applies.

Therefore agent j obtains the higher payoff by choosing the first region whenever

$$v_j \geq -S_{-j}.$$

In particular, truthful reporting $\hat{v}_j = v_j$ is optimal.

Case 2: Suppose $S_{-j} < 0$.

Agent j is pivotal exactly when she reports $\hat{v}_j$ such that

$$\hat{v}_j + S_{-j} \geq 0,$$

that is, when

$$\hat{v}_j \geq -S_{-j} = |S_{-j}|.$$

Hence agent j 's payoff is

$$u_j(\hat{v}_j) = \begin{cases} v_j - |S_{-j}|, & \text{if } \hat{v}_j \geq |S_{-j}|, \\ 0, & \text{if } \hat{v}_j < |S_{-j}|. \end{cases}$$

Again, the exact value of $\hat{v}_j$ matters only through which region it places the agent in.

Thus agent j prefers to choose the first region whenever

$$v_j \geq |S_{-j}|.$$

Hence truthful reporting $\hat{v}_j = v_j$ is optimal in this case as well.

Since reporting $\hat{v}_j = v_j$ is optimal in both cases, truthful reporting is optimal regardless of the other agents' reports, whether those reports are truthful or not. $\blacksquare$

Remark. However, the Clarke mechanism is not (i) collusion proof or (ii) there may be budget imbalance

We want to ensure that there is budget balance so the social planner need not spend anything. More generally, we want the agents' transfers to sum up to the same constant in every state. The VCG mechanism is unable to do this.

Definition 3.22 (AGV Mechanism). Let $\xi_i(\theta_i) = \mathbb{E}[\sum_{j \neq i} u_j(x^*(\theta), \theta_j) | \theta_i]$ is agent i 's expected externality on all others.

Let $x(\hat{\theta}) = x^*(\hat{\theta})$, $p_i(\hat{\theta}) = \bar{p}_i(\theta)$, where

$$\bar{p}_i(\theta_i) := \xi_i(\hat{\theta}_i) - \frac{1}{N-1} \sum_{j \neq i} \xi_j(\hat{\theta}_j) - C_i$$

Theorem 3.6. If the agents' types θ_i are statistically independent, then in the AGV mechanism it is a Bayes-Nash equilibrium for each i to report θ_i truthfully; the outcome is efficient and the transfers sum up to $\sum_i C_i$

Proof. If all agents other than i report truthfully, then agent i 's expected utility from reporting $\hat{\theta}_i$ is

$$\mathbb{E} \left[u_i(x^*(\hat{\theta}_i, \theta_{-i}), \theta_i) + \sum_{j \neq i} u_j(x^*(\hat{\theta}_i, \theta_{-i}), \theta_j) - \tau_i(\theta_{-i}) \mid \theta_i \right],$$

where

$$\tau_i(\theta_{-i}) = \frac{1}{N-1} \sum_{j \neq i} \xi_j(\theta_j) + C_i$$

does not depend on $\hat{\theta}_i$.

By independence, the conditional expectation above does not depend on θ_i except through the payoff term itself. Since x^* is efficient, for every realization of θ_{-i} the chosen allocation

$$x^*(\hat{\theta}_i, \theta_{-i})$$

maximizes total surplus when $\hat{\theta}_i = \theta_i$. Hence the expression inside the expectation is pointwise maximized by truthful reporting, $\hat{\theta}_i = \theta_i$.

Therefore truthful reporting is optimal for agent i .
Finally, the transfers satisfy

$$\sum_i t_i = \sum_i C_i.$$

■

Some mechanisms are ‘almost VCG’

Example 3.13. Consider three buyers A, B, C and two licences.

Buyers A and B each want only one licence, and each values a single licence at 2. Buyer C wants the bundle of both licences and values the two together at 3.

The efficient allocation is to give one licence to A and one to B , since this yields total surplus

$$2 + 2 = 4,$$

whereas allocating both licences to C yields only

$$3.$$

Under Vickrey payments, each winner pays the externality she imposes on others. If A were absent, then C would receive both licences and total surplus would be 3. With A present, the other agents’ surplus is only 2 (from B ’s licence). So A ’s participation reduces others’ surplus by

$$3 - 2 = 1.$$

Hence A pays 1. By symmetry, B also pays 1.

However, this creates a potential ex post challenge: the total payment from A and B is only 2, while C would be willing to offer 2.5 for the bundle. To prevent such blocking deviations, Ofcom imposed *core payments*, requiring A and B jointly to pay at least 3.

Thus core payments protect the outcome against ex post objections, but once Vickrey payments are modified in this way, truthful reporting is no longer generally a dominant strategy.

Optional Two users want to use a facility that can serve only one at a time.

User 1 has a job at every date $t = 1, 2, \dots$, and each job is worth v_1 .

User 2 needs to be served only once, and her single job is worth v_2 .

Future payoffs are discounted by the factor δ .

Definition 3.23 (Dynamic Pivot Mechanism). The *dynamic pivot mechanism* (or dynamic VCG mechanism) asks the agents to report v_1 and v_2 , and then allocates the facility efficiently:

serve User 2 first if $v_2 > v_1$,

and otherwise never serve User 2.

Transfers are given by pivot payments:

- if $v_2 > v_1$, then User 2 is pivotal and pays v_1 once;
- if $v_1 > v_2$, then User 1 is pivotal and pays $(1 - \delta)v_2$ every period.

Truthful reporting of v_1 and v_2 is induced, since each agent pays exactly when she is pivotal.

This is non-trivial: it is not the same as simply running a static auction, such as a second-price auction, in every period.

3.5 Matching

Preliminaries

We discuss matching in the context of school choice.

Definition 3.24 (School Choice Model). For a school choice model, we have the following primitives

- A finite set of students $A = \{a_1, \dots, a_n\}$
- A finite set of schools $S = \{s_1, \dots, s_n\}$
- Each student a has complete and transitive (and strict) preferences $\succ_a$ over schools and the option of being unmatched
- Each school s has a quota q_s of students and its strict priority list π_s over students

Remark. This is an example of an object allocation model because only students are strategic. There are two strategic sides models such as the National Resident Matching programme

Definition 3.25 (Matching Function). A matching is a function $\mu : A \cup S \rightarrow 2^A \cup S$ such that for any $a \in A$ and any $s \in S$ we have

- $\mu(a) \in S \cup \{a\}$: a student is either matched to one school or unmatched
- $\mu(s) \in 2^A$ and $|\mu(s)| \leq q_s$: schools are matched to a number of students up to their quota
- $a \in \mu(s) \iff \mu(a) = s$: if a school s is matched to a student a , then student a must be matched to school s

Definition 3.26 (Direct Mechanism). A direct mechanism $\psi \in \Psi$ is a mapping from the set of preference profiles to the set of matchings

Theorem 3.7 (Revelation Principle). For any mechanism φ and any equilibrium strategy profile $\sigma^* = (\sigma_a^*)_{a \in A}$ under φ , there exists a direct mechanism $\psi \in \Psi$ such that:

1. **(Incentive Compatibility)** Truth-telling is an equilibrium of ψ , i.e., for all $a \in A$ and all $\succ'_a \in \mathcal{P}$,

$$\psi_a(\succ) \succeq_a \psi_a(\succ'_a, \succ_{-a})$$

2. **(Equivalence)** The equilibrium outcome of φ under σ^* coincides with the truth-telling outcome of ψ , i.e.,

$$\psi(\succ) = \varphi(\sigma^*(\succ))$$

Remark. Hence, by the Revelation Principle, we can restrict our attention to direct mechanisms

Definition 3.27 (Strategyproof). A mechanism ψ is strategyproof if, for all $a \in A$ and any preference profile $\succ$, there is no preference profile $\succ'_a$ such that $\psi(\succ'_a, \succ_{-a})$ is strictly preferred to $\psi(\succ)$ by a

Definition 3.28 (Stable Matching). A matching μ is **stable** if it satisfies:

- **(Individual Rationality)** For all $a \in A$, $\mu(a) \succeq_a a$ and for all $s \in S$, every $a \in \mu(s)$ is acceptable to s
- **(No Blocking Pair)** There is no pair $(a, s) \in A \times S$ such that:

$$s \succ_a \mu(a) \quad \text{and} \quad (|\mu(s)| < q_s \text{ or } a \succ_s a' \text{ for some } a' \in \mu(s))$$

i.e., no student a and school s both prefer each other to their current match

Basically, to check for stability, you want to find a blocking pair that mutually prefers each other to their current partner.

We want to develop axioms to understand what makes a good matching outcome

Definition 3.29 (Pareto-efficient). A matching μ is pareto efficient if there is no other matching μ' that makes all students weakly better off and at least one student strictly better off

Remark. Notice that in object allocation, we only care about the students' welfare

Definition 3.30 (Individually Rational). A matching μ is individually rational if for all students $a \in A$, $\mu(a)$ is weakly preferred to being unmatched

We do not want to match students against their will

Definition 3.31 (Non-wasteful). A matching μ is non-wasteful if for all $a \in A$ and $s \in S$, $s \succ_a \mu(a)$ implies $|\mu(s)| = q_s$

If a student prefers a school to her match, then that school must have filled its capacity

Definition 3.32 (No justified envy). A matching μ has no justified envy if for all $a, a' \in A$ with $\mu(a') = s \in S$, $s \succ_a \mu(a)$ implies $a' \pi_s a$

If a student prefers a school s to his match, then all students matched to a school s must have a higher priority at s than student a

Proposition 3.10. A matching μ is stable if it satisfies

- individual rationality
- non-wastefulness
- elimination of justified envy

Immediate Acceptance Algorithm

Algorithm 1 Immediate Acceptance / Boston Algorithm

```

1: Input: Students  $A$ , Schools  $S$ , preference lists, school priorities and quotas  $q_s$ 
2: Initialize: All students unmatched; each school's remaining capacity =  $q_s$ 
3: Step  $t = 1$ :
4: for each student  $a \in A$  do
5:    $a$  proposes to her first-choice school
6: end for
7: for each school  $s \in S$  do
8:   Immediately and permanently accept highest-priority proposing students up to  $q_s$ 
9:   Reject all other proposing students
10: end for
11: Step  $t \geq 2$ :
12: while some student was rejected in Step  $t - 1$  do
13:   for each rejected student  $a \in A$  do
14:      $a$  proposes to her next highest choice
15:   end for
16:   for each school  $s \in S$  do
17:     Remaining capacity  $\leftarrow q_s - |\text{permanently accepted students}|$ 
18:     Immediately and permanently accept highest-priority proposing students up to remaining capacity
19:     Reject all others
20:   end for
21:    $t \leftarrow t + 1$ 
22: end while
23: Termination: No more proposals are made
24: Output: Permanent matches

```

Basically this just says

- At $t = 1$,
 - Each student proposes to her first choice school
 - Each student immediately accepts the highest priority proposing students up to its quota and rejects all other students
- At $t \geq 2$,
 - Each student rejected in Step $t - 1$ applies to her next highest choice
 - Each school immediately accepts the highest priority students proposing in Step t up to its remaining capacity (= its quota minus the sum of admitted students from previous steps)
- Termination: No more proposals are made by the students

Theorem 3.8. The IA algorithm terminates at a pareto-efficient matching

Proposition 3.11. The IA algorithm is easily manipulable

Clearly, if your favourite school is definitely oversubscribed, but your second favourite might have a place left, you can list your second favourite school as your top choice

Top Trading Cycle Algorithm

Algorithm 2 Top Trading Cycles (TTC)

```

1: Input: Students  $A$ , Schools  $S$ , preference lists, school priorities and quotas  $q_s$ 
2: Initialize: All students unmatched; remaining capacity =  $q_s$  for all  $s \in S$ 
3: Step  $t \geq 1$ :
4: for each student  $a \in A$  do
5:   Draw an arrow from  $a$  to her most preferred acceptable and non-full school
6: end for
7: for each non-full school  $s \in S$  do
8:   Draw an arrow from  $s$  to its highest priority student
9: end for
10: Identify all cycles (there must exist at least one)
11: for each cycle do
12:   Assign each student in the cycle to the school she is pointing at
13:   Remove all students in the cycle from the problem
14:   Reduce the remaining capacity of each school in the cycle by 1
15: end for
16:  $t \leftarrow t + 1$ 
17: Termination: All students have been removed or all acceptable schools are full
18: Output: Permanent matches

```

Remark. Basically, at each step $t \geq 1$, for each student we draw an arrow from the student to her most preferred school (among those that are acceptable and not full).

For each school (that is not full), we draw an arrow from that school to its highest priority student.

There must be at least one cycle: if we start from an agent in the cycle and follow the arrows, we eventually visit that agent again.

Each student in a cycle is assigned to the school she is pointing at.

Students in a cycle are removed from the problem and each school in a cycle reduces its remaining quota by 1.

The algorithm stops when all students have been removed or all schools that are still acceptable are full.

Theorem 3.9. The TTC algorithm is a pareto-efficient and strategyproof mechanism

Deferred Acceptance Algorithm

Algorithm 3 Deferred Acceptance (DA) — Student-Proposing

```

1: Input: Students  $A$ , Schools  $S$ , preference lists, school priorities and quotas  $q_s$ 
2: Initialize: All students unmatched; remaining capacity  $= q_s$  for all  $s \in S$ 

3: Step  $t = 1$ :
4: for each student  $a \in A$  do
5:    $a$  proposes to her most preferred acceptable school
6: end for
7: for each school  $s \in S$  do
8:   Hold the highest-priority proposing students up to quota  $q_s$ 
9:   Reject all other proposing students
10: end for

11: Step  $t \geq 2$ :
12: for each student  $a$  rejected in Step  $t - 1$  do
13:    $a$  proposes to her next highest acceptable choice (if any remains)
14: end for
15: for each school  $s \in S$  do
16:   Consider all new applicants together with previously held students
17:   Hold the highest-priority students from the combined set up to quota  $q_s$ 
18:   Reject all others
19: end for

20: Termination: No more proposals are made by any student
21: Output: Permanent matches

```

Theorem 3.10. The Deferred Acceptance Algorithm is a stable and strategyproof mechanism

Definition 3.33. A stable matching μ is student-optimal if it is weakly preferred by all students to any other stable matching μ'

Theorem 3.11. The DA algorithm terminates at the student-optimal stable matching

Proposition 3.12. The student-optimal stable matching may be Pareto-inefficient

Deferred Acceptance can be extremely inefficient in theory

Theorem 3.12 (Kesten, 2010). Given any set of schools S and capacity vector $q = (q_s)_{s \in S}$, there always exists a set of students A and a school choice problem $(S, (\succ_a)_{a \in A})$ for which the student-optimal DA algorithm assigns each student at either his or her worst choice or his or her next worst choice

The basic upshot is that there are a multiplicity of trade-offs

Proposition 3.13. We have the following tradeoffs for TTC and student-proposing DA algorithms:

- Student proposing DA is not pareto-efficient
- TTC does not eliminate justified envy
- For general school priorities, it may not be possible to eliminate justified envy in any pareto-efficient assignment (however, if school priorities are sufficiently correlated, this conflict disappears (Ergin, 2002))

We also have the following impossibility theorem

Theorem 3.13 (Kesten, 2010). There is no pareto-efficient and strategyproof mechanism ψ that selects the pareto-efficient and stable matching whenever it exists There is no pareto-efficient and strategyproof mechanism that pareto dominates the student stable mechanism

The second result indicates that it might be worth improving on the efficiency of the DA since it is not pareto dominated.

4 Evolutionary Game Theory

4.1 Evolutionary Stable Equilibrium

Fundamental question: can players learn to play NE? And correspondingly, are there NE that are robust enough such that mutant strategies cannot invade them.

Definition 4.1 (Evolutionarily Stable Equilibrium). We say $\alpha^* \in \Delta(A)$ is an evolutionarily stable strategy if

1. (α^*, α^*) is a symmetric Nash equilibrium, i.e., for all $\beta \in \Delta(A)$

$$u(\alpha^*, \alpha^*) \geq u(\beta, \alpha^*)$$

2. if $\beta \neq \alpha^*$ is a best-reply to α^* , then α^* does strictly better against β than β against β

$$u(\beta, \alpha^*) = u(\alpha^*, \alpha^*) \implies u(\alpha^*, \beta) > u(\beta, \beta)$$

Example 4.1 (Hawk-Dove). Consider the following game (where the payoffs are symmetric) where $V < C$

	H	D
H	$\frac{1}{2}(V - C)$	V
D	0	$\frac{1}{2}V$

Three Nash equilibria: (H, D) and (D, H) (asymmetric and cannot be ESS) and a symmetric NE $(qH + (1 - q)D, qH, (1 - q)D)$ where $q = \frac{v}{c}$. Need to check condition (2).

By construction of a mixed Nash, any $\beta = pH + (1 - p)D$ is a best reply to $\alpha^* = qH + (1 - q)D$ with $q = v/c$. Thus

$$\begin{aligned} u(H, \beta) &= \frac{p}{2}(V - C) + (1 - p)V = V - \frac{p}{2}V - \frac{p}{2}C \\ u(D, \beta) &= \frac{1 - p}{2}V \end{aligned}$$

Hence, the difference between payoff of α^* against β and β against β is

$$\begin{aligned} u(\alpha^*, \beta) - u(\beta, \beta) &= (q - p)(V - \frac{p}{2}V - \frac{p}{2}C) + [(1 - q) - (1 - p)]\frac{1 - p}{2}V \\ &= (q - p)(V - \frac{p}{2}V - \frac{p}{2}C - \frac{1 - p}{2}V) \\ &= \frac{q - p}{2}(V - pC) \\ &= \frac{v/c - p}{2}(V - pC) \\ &= \frac{1}{2}C(v/c - p)^2 > 0 \end{aligned}$$

Therefore $\alpha^* = (v/c)H + (1 - v/c)D$ is ESS.

Example 4.2 (Simplified Stag Hunt). Consider the following game

	Stag	Hare
Stag	5, 5	0, 3
Hare	3, 0	2, 2

The NE are (Stag, Stag), (Hare, Hare), $(\frac{1}{2}\text{Stag} + \frac{1}{2}\text{Hare}, \frac{1}{2}\text{Stag} + \frac{1}{2}\text{Hare})$, payoff 2.5 each.

Each pure strategy is a strict best reply to itself, therefore no mutant can invade when the population plays a single pure strategy. So (Stag, Stag), (Hare, Hare) are ESS.

The symmetric mixed strategy is not ESS

- Stag is (equally weak) best reply against 50-50 mix (by construction)
- Stag against Stag pays 5 utils, 50-50 mix against Stag pays 4 utils

Hence Stag successfully invades a population that plays 50-50 mix

Proposition 4.1. If a symmetric Nash equilibrium is strict, then it is an ESS

Remark. Mixed Nash equilibrium is never strict (indifferent when mixing); for symmetric mixed Nash we must verify (ii) to see if it is ESS

Theorem 4.1. Every 2x2 symmetric game has an evolutionarily stable equilibrium

Proposition 4.2. There is no ESS in some 3x3 games

Example 4.3 (Rock, Paper, Scissors). Consider the following game

	R	P	S
R	x, x	$-1, 1$	$1, -1$
P	$1, -1$	x, x	$-1, 1$
S	$-1, 1$	$1, -1$	x, x

Unique symmetric Nash is $\alpha = \frac{1}{3}R + \frac{1}{3}P + \frac{1}{3}S$ for a payoff of $x/3$. ESS iff $x < 0$. If $x \geq 0$, then any pure strategy can invade α since

- Any pure strategy is equally best reply against $\alpha = \frac{1}{3}R + \frac{1}{3}P + \frac{1}{3}S$
- payoff of a pure strategy against itself is x
- payoff of α against a pure strategy is $\frac{1}{3}xx$ as $x \geq 0$ hence not an ESS

4.2 Replicator Dynamic

Set up for Symmetric Two player games with large population and anonymous random matching

- payoff of strategy i played against opponent's strategy k is u_{ik} for $i, k = 1, \dots, m$
- proportion of agents playing i at time t is $p_i(t)$
- expected payoff from playing strategy i is $\pi_i(t) = \sum_{k=1}^m p_k(t)u_{ik}$
- average payoff of all agents in the population is $\Pi(t) = \sum_{i=1}^m p_i(t)\pi_i(t)$
- relative fitness of strategy i at $\pi_i(t) - \Pi(t)$

Definition 4.2 (Replicator Dynamic). Consider a symmetric Two player games with large population and anonymous random matching. Then, the replicator dynamic is

$$\frac{dp_i(t)}{dt} = \lambda p_i(t) [\pi_i(t) - \Pi(t)]$$

where $\lambda > 0$

Remark. The interpretation of this is that

$$\dot{p}_i/p_i = \lambda [\pi_i(t) - \Pi(t)]$$

is the growth of strategy i adjusted for some proportion parameter

Proposition 4.3. Every evolutionarily stable strategy is a locally stable rest point of the replicator dynamic

However, the replicator dynamic may converge to a strategy that is not ESS.

Replicator Dynamic for Binary action games

- We can normalise $\lambda > 0$ to 1 WLOG
- Let p denote the proportion of strategy 1 so $\pi_1 = pu_{11} + (1-p)u_{12}$, and $\pi_2 = pu_{21} + (1-p)u_{22}$ and $\Pi = p\pi_1 + (1-p)\pi_2$. Intuitively, this means the payoff of playing strategy 1 = probability of meeting someone else playing strategy 1 times playing strategy 1 against them plus probability of meeting someone else playing strategy 2 times playing strategy 1 against them. Same for strategy 2
- therefore $\pi_1 - \Pi = \pi_1 - [p\pi_1 + (1-p)\pi_2] = (1-p)(\pi_1 - \pi_2)$
- and so $\dot{p} \equiv \frac{dp(t)}{dt} = p(1-p)(\pi_1 - \pi_2)$ i.e. p increases iff $\pi_1 > \pi_2$

Cookbook approach for replicator dynamic analysis for symmetric 2x2 games

1. Denote the proportion of agents playing strategy 1 by p then compute the payoffs are playing strategies 1 and 2 respectively i.e.

$$\begin{aligned}\pi_1 &= pu_{11} + (1-p)u_{12} \\ \pi_2 &= pu_{21} + (1-p)u_{22}\end{aligned}$$

2. The replicator dynamic is an ordinary differential equation

$$\dot{p} \equiv \frac{dp(t)}{dt} = p(1-p)(\pi_1 - \pi_2)$$

where $p = 0, 1$ are always rest points (or fixed points, steady states) of the process; any symmetric (mixed) Nash equilibrium strategy is a rest point too

3. Draw the phase diagram and identify whether any of the rest points is locally or globally stable and what the basins of attraction are

Here is a guide for the last point

- Let

$$\dot{p} = f(p) = p(1-p)(\pi_1(p) - \pi_2(p)).$$

The rest points are the values $p^* \in [0, 1]$ such that $f(p^*) = 0$. Hence $p = 0$ and $p = 1$ are always rest points, and any interior $p^* \in (0, 1)$ satisfying

$$\pi_1(p^*) = \pi_2(p^*)$$

is also a rest point.

- For $p \in (0, 1)$, we have $p(1-p) > 0$, so the sign of $\dot{p}$ is determined entirely by the sign of

$$\pi_1(p) - \pi_2(p).$$

Thus:

$$\begin{aligned}\pi_1(p) > \pi_2(p) &\implies \dot{p} > 0 \quad (\text{motion to the right}), \\ \pi_1(p) < \pi_2(p) &\implies \dot{p} < 0 \quad (\text{motion to the left}).\end{aligned}$$

- The phase diagram is obtained by checking the sign of $\pi_1(p) - \pi_2(p)$ on each interval between rest points and drawing arrows accordingly.
- A rest point p^* is *locally stable* if arrows point toward it from nearby points on both sides, and *unstable* if arrows point away from it on both sides. Equivalently, for an interior rest point,

$$f'(p^*) < 0 \implies p^* \text{ is locally stable,} \quad f'(p^*) > 0 \implies p^* \text{ is unstable.}$$

- The *basin of attraction* of a stable rest point p^* is the set of initial conditions $p(0)$ such that $p(t) \rightarrow p^*$.
- A rest point is *globally stable* if every admissible initial condition converges to it. In practice, once the phase diagram is drawn, global stability and the basins of attraction can be read off from the direction of the arrows.

Example 4.4 (Hawk-Dove). Consider the following game (where the payoffs are symmetric) where $V < C$

	H	D
H	$\frac{1}{2}(V - C)$	V
D	0	$\frac{1}{2}V$

The payoffs are

$$\pi_H = \frac{p}{2}(V - C) + (1 - p)V$$

$$\pi_D = \frac{1 - p}{2}V$$

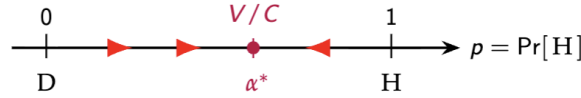
If p is the proportion of H then p increases iff $\pi_H > \pi_D$ i.e.

$$\frac{p}{2}(V - C) + (1 - p)V > \frac{1 - p}{2}V \iff p < \frac{v}{c}$$

We claim this is globally stable since for any starting point/initial point $p \in (0, 1)$, it converges to $\alpha^* = (v/c)H + (1 - v/c)D$. We see this since when

$$\dot{p} > 0 \iff p < \frac{v}{c}$$

$$\dot{p} < 0 \iff p > \frac{v}{c}$$



Notice that we exclude the rest points $p = 0, 1$. Also note that these are not locally stable points because any perturbations $p > 0$ or $p < 1$ in some neighbourhood around 0 or 1 move to $p = v/c$

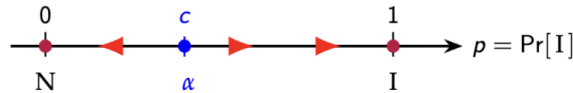
Example 4.5 (Simplified Stag Hunt). Each player either invests c or nothing; if both invest they get 1 each

	I	N
I	1	0
N	c	c

The payoffs are $\pi_I = p$ and $\pi_N = c$, with the proportion of players playing I being p . Hence, $\pi_I > \pi_N \iff p > c$. Therefore, $p = c$ is unstable since

- At $p \in (c, 1]$, $\dot{p} > 0$ so the proportion of players playing I increases away from c to 1
- At $p \in [0, c)$, $\dot{p} < 0$ so the proportion of players playing I decreases away from c to 0

Likewise, $p = 0, 1$ are locally stable with $[0, c)$ and $(c, 1]$ being respective basins of attraction.



Example 4.6 (Prisoner's Dilemma with Three Strategies). Consider the repeated Prisoner's Dilemma with three supergame strategies:

$$C^\infty \text{ (always cooperate), } \quad D^\infty \text{ (always defect), } \quad T \text{ (tit-for-tat).}$$

A pair continues to the next round with probability $1 - s > 1/2$, so $s < 1/2$. The induced payoff matrix is

	C^∞	D^∞	T
C^∞	$3/s$	0	$3/s$
D^∞	$5/s$	$1/s$	$4 + 1/s$
T	$3/s$	$1/s - 1$	$3/s$

where only the row player's payoff is shown.

Let p_T denote the proportion of tit-for-tat players and p_D the proportion of always-defect players. Then the proportion of always-cooperators is

$$p_C = 1 - p_T - p_D.$$

The state space is therefore the simplex

$$\{(p_T, p_D) : p_T \geq 0, p_D \geq 0, p_T + p_D \leq 1\}.$$

The expected payoffs to the three strategies are

$$\begin{aligned} \pi_C &= p_C \frac{3}{s} + p_D \cdot 0 + p_T \frac{3}{s} = \frac{3}{s}(p_C + p_T) = \frac{3}{s}(1 - p_D), \\ \pi_D &= p_C \frac{5}{s} + p_D \frac{1}{s} + p_T \left(4 + \frac{1}{s}\right) \\ &= \frac{5}{s}(1 - p_T - p_D) + \frac{p_D}{s} + 4p_T + \frac{p_T}{s} \\ &= \frac{1}{s}(5 - 4p_D - 4(1 - s)p_T), \\ \pi_T &= p_C \frac{3}{s} + p_D \left(\frac{1}{s} - 1\right) + p_T \frac{3}{s} \\ &= \frac{3}{s}(1 - p_D) + p_D \left(\frac{1}{s} - 1\right) = \frac{1}{s}(3 - (2 + s)p_D). \end{aligned}$$

Hence

$$\begin{aligned} s\pi_C &= 3 - 3p_D, \\ s\pi_D &= 5 - 4p_D - 4(1 - s)p_T, \\ s\pi_T &= 3 - (2 + s)p_D. \end{aligned}$$

Let

$$\Pi = p_C\pi_C + p_D\pi_D + p_T\pi_T$$

denote average population payoff. Under the replicator dynamic,

$$\dot{p}_i = p_i(\pi_i - \Pi), \quad i \in \{C, D, T\}.$$

In particular, for $i = D, T$,

$$\dot{p}_i > 0 \iff \pi_i > \Pi.$$

Step 1: The condition for p_T to rise. We want to determine when $\dot{p}_T > 0$. Since

$$\dot{p}_T > 0 \iff \pi_T > \Pi,$$

and

$$\Pi = p_T \pi_T + p_D \pi_D + (1 - p_T - p_D) \pi_C,$$

we obtain

$$\begin{aligned} \pi_T > \Pi &\iff \pi_T - [p_T \pi_T + p_D \pi_D + (1 - p_T - p_D) \pi_C] > 0 \\ &\iff (1 - p_T) \pi_T - p_D \pi_D - (1 - p_T - p_D) \pi_C > 0. \end{aligned}$$

Multiplying through by s and substituting the formulas for $s\pi_C, s\pi_D, s\pi_T$, this becomes

$$\begin{aligned} (1 - p_T)[3 - (2 + s)p_D] - p_D[5 - 4p_D - 4(1 - s)p_T] \\ - (1 - p_T - p_D)(3 - 3p_D) > 0. \end{aligned}$$

Expanding and simplifying,

$$\begin{aligned} 0 &< 3 - 3p_T - (2 + s)p_D + (2 + s)p_T p_D - 5p_D + 4p_D^2 + 4(1 - s)p_T p_D \\ &\quad - 3 + 3p_D + 3p_T - 3p_T p_D + 3p_D - 3p_D^2 \\ &= p_D^2 - (1 + s)p_D + 3(1 + s)p_T p_D \\ &= p_D[p_D - (1 + s)(1 - 3p_T)]. \end{aligned}$$

Thus

$$\dot{p}_T > 0 \iff p_D > (1 + s)(1 - 3p_T).$$

So the curve

$$p_D = (1 + s)(1 - 3p_T)$$

separates the region where the share of tit-for-tat rises from the region where it falls:

$$\dot{p}_T > 0 \text{ above the line,} \quad \dot{p}_T < 0 \text{ below the line.}$$

Step 2: The condition for p_D to rise. Similarly,

$$\dot{p}_D > 0 \iff \pi_D > \Pi.$$

Using

$$\Pi = p_T \pi_T + p_D \pi_D + (1 - p_T - p_D) \pi_C,$$

we get

$$\begin{aligned} \pi_D > \Pi &\iff \pi_D - [p_T \pi_T + p_D \pi_D + (1 - p_T - p_D) \pi_C] > 0 \\ &\iff (1 - p_D) \pi_D - p_T \pi_T - (1 - p_T - p_D) \pi_C > 0. \end{aligned}$$

Multiplying through by s and substituting the payoff formulas yields

$$\begin{aligned} (1 - p_D)[5 - 4p_D - 4(1 - s)p_T] - p_T[3 - (2 + s)p_D] \\ - (1 - p_T - p_D)(3 - 3p_D) > 0. \end{aligned}$$

Expanding and simplifying,

$$\begin{aligned} 0 &< 5 - 9p_D + 4p_D^2 - 4(1 - s)p_T + 4(1 - s)p_T p_D - 3p_T + (2 + s)p_T p_D \\ &\quad - 3 + 3p_D + 3p_T - 3p_D - 3p_T p_D + 3p_D^2 \\ &= 2 - 3p_D + p_D^2 - (4 - 4s)p_T + (-3s + 3sp_T + 3p_T)p_D \\ &= p_D^2 - 3(1 - p_T + sp_T)p_D + 2 - 4(1 - s)p_T. \end{aligned}$$

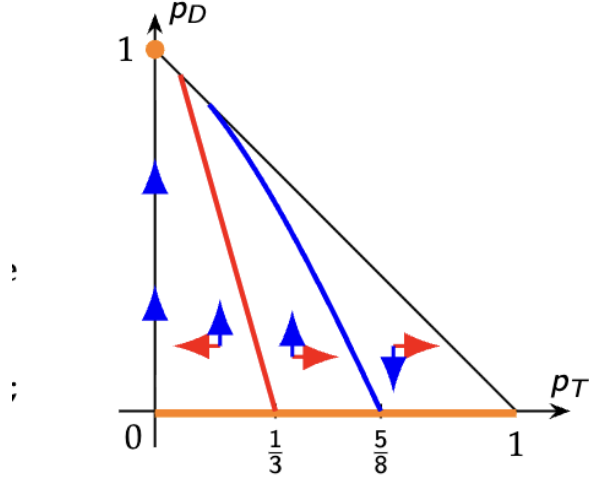
Therefore

$$\dot{p}_D > 0 \iff p_D^2 - 3(1 - p_T + sp_T)p_D + 2 - 4(1 - s)p_T > 0.$$

So the curve

$$p_D^2 - 3(1 - p_T + sp_T)p_D + 2 - 4(1 - s)p_T = 0$$

separates the region where the share of always-defect rises from the region where it falls.



After stating the two boundary equations above, insert the figure showing the simplex, the line

$$p_D = (1 + s)(1 - 3p_T),$$

the curve

$$p_D^2 - 3(1 - p_T + sp_T)p_D + 2 - 4(1 - s)p_T = 0,$$

and the arrows indicating the signs of $\dot{p}_T$ and $\dot{p}_D$. This is the natural place for the diagram, since the figure simply visualises the two conditions just derived.

Rest points. A rest point is a state (p_T, p_D) such that

$$\dot{p}_T = 0 \quad \text{and} \quad \dot{p}_D = 0.$$

From the formulas above, there are two types of rest points.

(i) The vertex

$$(p_T, p_D) = (0, 1),$$

corresponding to the population state D^∞ .

(ii) Every point on the edge

$$p_D = 0, \quad 0 \leq p_T \leq 1,$$

corresponding to mixtures of C^∞ and T .

To see why the whole edge $p_D = 0$ consists of rest points, note that when $p_D = 0$,

$$\pi_C = \pi_T = \frac{3}{s}.$$

Hence C^∞ and T earn the same payoff, so there is no force changing their relative population shares.

Local stability. A rest point is locally stable if every trajectory starting sufficiently near it remains near it and converges back to it.

- The point $(0, 1)$, corresponding to D^∞ , is locally asymptotically stable: in a neighbourhood of this vertex, the arrows in the phase diagram point toward it, so trajectories starting sufficiently close converge to it.
- The edge $p_D = 0$ is a continuum of rest points. It is therefore better described as a stable rest set rather than as a collection of isolated asymptotically stable points. Trajectories starting in the relevant region converge to some point on this edge, but not necessarily to the same point from which they were perturbed.

Basins of attraction. The basin of attraction of a stable rest point or rest set is the set of initial conditions from which the trajectory converges to it.

- The basin of attraction of D^∞ is the region of the simplex from which the arrows point toward $(0, 1)$.
- The basin of attraction of the edge $p_D = 0$ is the region of the simplex from which the arrows lead downward toward $p_D = 0$, so that always-defect eventually disappears.

In particular, the line

$$p_D = (1 + s)(1 - 3p_T)$$

shows where the direction of motion of p_T changes. To the right of this line, $\dot{p}_T > 0$, so the share of tit-for-tat increases. As $s \rightarrow 0$, this line approaches

$$p_D = 1 - 3p_T,$$

and the region from which the population moves toward the edge $p_D = 0$ occupies two-thirds of the simplex.

Conclusion. The replicator dynamic does not generally converge to the pure tit-for-tat state $(p_T, p_D) = (1, 0)$. Instead, the long-run outcome depends on the initial condition. Some initial conditions converge to the always-defect state D^∞ , while others converge to the edge $p_D = 0$, consisting of mixtures of C^∞ and T . Thus tit-for-tat is not globally attracting in this model, although it survives and expands for a substantial set of initial conditions.

Example 4.7 (PSet7 Q3). Consider the Hawk-Dove game

	H	D
H	$-1, -1$	$2, 0$
D	$0, 2$	$1, 1$

- (a) Find the Nash equilibria of the game and show that it has an ESS.

Pure NE: $\{(D, H), (H, D)\}$. The mixed Nash is $\{(\frac{1}{2}H + \frac{1}{2}D, \frac{1}{2}H + \frac{1}{2}D)\}$ (easy to show).

Since pure NE are symmetric and all 2x2 symmetric games have ESS then the mixed Nash is an ESS.

Since mixed NE, by construction

$$u(H, \alpha^*) = u(D, \alpha^*)$$

Hence any $\beta = pH + (1 - p)D$ is a best reply to $\alpha^* = \frac{1}{2}H + \frac{1}{2}D$ and so $u(\beta, \alpha^*) = u(\alpha^*, \alpha^*)$ since we can choose $p = \frac{1}{2}$. Then,

$$u(H, \beta) = -p + 2(1 - p) = 2 - 3p$$

$$u(D, \beta) = 1 - p$$

Calculating the difference yields

$$\begin{aligned} u(\alpha^*, \beta) - u(\beta, \beta) &= (\frac{1}{2} - p)(2 - 3p) + (\frac{1}{2} - 1 + p)(1 - p) \\ &= (\frac{1}{2} - p)(2 - 3p) + (p - \frac{1}{2})(1 - p) \\ &= \frac{1}{2} - 2p + 2p^2 \end{aligned}$$

Check $\Delta = (-2)^2 - 4(2)(\frac{1}{2}) = 4 - 4 = 0$ / But the root is when $p = \frac{1}{2}$ and so for all $p \in [0, 1] \setminus \{\frac{1}{2}\}$,

$$u(\alpha^*, \beta) - u(\beta, \beta) > 0$$

Hence $\{(\frac{1}{2}H + \frac{1}{2}D, \frac{1}{2}H + \frac{1}{2}D)\}$ is an ESS

- (b) Suppose that all players are drawn from a single large population and are randomly matched in pairs to play the game. Let p be the proportion of players choosing H . Discuss how p evolves under the replicator dynamic.

Let $p \in [0, 1]$ be the proportion playing strategy H . Then a random opponent plays H w.p. p . So, the expected payoffs for each strategy are

$$\begin{aligned}\pi_H(t) &= p(-1) + 2(1-p) = 2 - 3p \\ \pi_D(t) &= p(0) + (1-p) = 1 - p\end{aligned}$$

The average payoff is

$$\Pi(t) = p\pi_H(t) + (1-p)\pi_D(t)$$

The replicator dynamic yields (where we normalise $\lambda = 1$)

$$\frac{dp_H(t)}{dt} = p_H(t)[\pi_H(t) - \Pi(t)] = p(1-p)[\pi_H(t) - \pi_D(t)] = p(1-p)(1-2p)$$

Notice that $\pi_H > \pi_D$ iff $p \leq 1/2$ and so H grows when $p < \frac{1}{2}$. $\pi_H < \pi_D$ iff $p \geq 1/2$ so H falls when $p > 1/2$. This means that the ESS $p = \frac{1}{2}$ is globally stable for $p \in (0, 1)$.

- (c) Suppose now that row and column players form separate populations and that they are randomly matched against each other to play the game. Let p be the proportion of row players playing H and q be that of column players playing H .

Assume that p grows in proportion to the difference between the payoff of a row player from playing H against the column population and the average payoff in the row population playing against the column population. A similar story applies for q . Explain why this yields

$$\begin{aligned}\dot{p} &= p(1-p)(1-2q) \\ \dot{q} &= q(1-q)(1-2p)\end{aligned}$$

How would you expect (without detailed calculations) p and q to evolve if (i) $p > \frac{1}{2}$ and $q < \frac{1}{2}$ initially, (ii) $p = q$ initially, (iii) $q < p < \frac{1}{2}$ initially? Which strategies are played in the long run. For Row player, plays an opponent playing H w.p. q . Hence

$$\begin{aligned}\pi_H(t) &= q(-1) + 2(1-q) = 2 - 3q \\ \pi_D(t) &= q(0) + (1-q) = 1 - q\end{aligned}$$

Since binary actions,

$$\dot{p} = p(1-p)(\pi_H(t) - \pi_D(t)) = p(1-p)(1-2q)$$

Likewise for column,

$$\begin{aligned}\pi_H(t) &= p(-1) + 2(1-p) = 2 - 3p \\ \pi_D(t) &= p(0) + (1-p) = 1 - p\end{aligned}$$

Hence

$$\dot{q} = q(1-q)(1-2p)$$

Then

- If $p > \frac{1}{2}$ and $q < \frac{1}{2}$ initially, then $\dot{p} > 0$ and row population uses H more; $\dot{q} < 0$ and column population uses D more. Hence, $(p, q) \rightarrow (1, 0)$
- If $p = q$ initially then, $\dot{p} = \dot{q} = \dot{x} - x(1-x)(1-2x)$. This is exactly the same as in (b) where the values converge to $(1/2, 1/2)$
- if $q < p < \frac{1}{2}$, then both $\dot{p} > 0$ and $\dot{q} > 0$. Since $q < p$ it follows that $1-2q > 1-2p$ and $p(1-p) > q(1-q)$ and so $\dot{p} > \dot{q}$ and so the difference between p and q widens.

4.3 Stochastic Best Reply

Example 4.8 (Cournot Myopic Best-Reply). Consider a Cournot duopoly in which firms $i = 1, 2$ simultaneously choose quantities $q_i \geq 0$. The payoff of firm i is

$$u_i(q_i, q_{-i}) = (1 - q_i - q_{-i})q_i.$$

Thus inverse demand is $P = 1 - q_1 - q_2$, so each firm's profit is price times own output.

For a given q_{-i} , firm i solves

$$\max_{q_i \geq 0} (1 - q_i - q_{-i})q_i.$$

Expanding the objective gives

$$u_i(q_i, q_{-i}) = q_i - q_i^2 - q_{-i}q_i.$$

The first-order condition is

$$\frac{\partial u_i}{\partial q_i} = 1 - 2q_i - q_{-i} = 0,$$

so the best reply of firm i is

$$R_i(q_{-i}) = \frac{1 - q_{-i}}{2}.$$

Cournot's adjustment process is a simple model of *myopic best-reply learning*. Each firm begins from some arbitrary initial quantity q_i^0 , and then at each date $t = 0, 1, 2, \dots$, each firm chooses a best reply to the rival's previous-period output:

$$q_i^{t+1} = R_i(q_{-i}^t) = \frac{1 - q_{-i}^t}{2}.$$

Hence the dynamic system is

$$q_1^{t+1} = \frac{1 - q_2^t}{2}, \quad q_2^{t+1} = \frac{1 - q_1^t}{2}.$$

We now show that this process converges to the Cournot-Nash equilibrium.

First, subtract the two update equations:

$$q_1^{t+1} - q_2^{t+1} = \frac{1 - q_2^t}{2} - \frac{1 - q_1^t}{2} = \frac{q_1^t - q_2^t}{2}.$$

So the difference between the two firms' outputs evolves according to

$$q_1^{t+1} - q_2^{t+1} = \frac{1}{2}(q_1^t - q_2^t).$$

Iterating this relation yields

$$q_1^t - q_2^t = \left(\frac{1}{2}\right)^t (q_1^0 - q_2^0).$$

Since $(1/2)^t \rightarrow 0$ as $t \rightarrow \infty$, it follows that

$$\lim_{t \rightarrow \infty} (q_1^t - q_2^t) = 0.$$

Thus the two firms' outputs become arbitrarily close over time.

In the limit, therefore, both firms must choose the same quantity. Let this common limit be q . Passing to the limit in the updating rule gives

$$q = \frac{1 - q}{2}.$$

Solving,

$$2q = 1 - q \implies 3q = 1 \implies q = \frac{1}{3}.$$

Hence

$$(q_1^t, q_2^t) \rightarrow \left(\frac{1}{3}, \frac{1}{3}\right).$$

Therefore the myopic best-reply process converges to the unique Cournot-Nash equilibrium

$$(q_1^*, q_2^*) = \left(\frac{1}{3}, \frac{1}{3}\right).$$

The key intuition is that each firm best responds to the rival's last output, and this causes the gap between the firms' quantities to shrink by a factor of 1/2 each period. Once the quantities become arbitrarily close, the only fixed point of the best-reply mapping is (1/3, 1/3), so the process converges there.

There is also the idea of fictitious play in finite games

- Each player best replies to the opponent's mixing corresponding to the cumulative frequency of the opponent's pure strategies played in the past
- converges to Nash in ISD-solvable games, also in zero sum games
- but not in general, and even if play converges, payoffs might not

Definition 4.3 (Foster and Young Stochastic partial best reply processes). Consider stochastic partial best reply processes

- n agents play with anonymous random pairwise matching
- each player is stuck playing a given action until asked to update
- in every period, a few players are picked at random to update: either choose best reply to the population with probability $1 - \varepsilon$, or pick a random action otherwise
- Players are re-matched at random, payoffs realise and process repeats

The resulting process is a Markov chain on finitely many states

Definition 4.4 (Simple Foster and Young stochastic best reply/learning process). The simple variant is

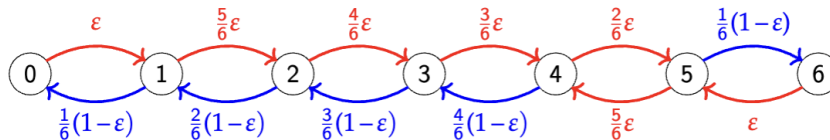
- In each period, exactly 1 agent is chosen at random (with probability $1/n$)
- if $i/n < c$, this agent adopts strategy 2 with probability $1 - \varepsilon$ and strategy 1 with probability ε
- if $i/n > c$, she adopts strategy 1 with probability $1 - \varepsilon$ and strategy 2 with ε

Example 4.9 (Investment Game). Consider the following investment game with $0 < c < 1$

	I	N
I	1, 1	0, c
N	c , 0	c , c

The three NE are (I, I) , (N, N) and $(cI + (1-c)N, cI + (1-c)N)$. Suppose that n agents play this repeatedly, in random and anonymous pairings. State of the system at t is defined as $t : i(t) \in \{0, \dots, n\}$ = number of agents playing action I .

Suppose $n = 6$ and $c = 0.7$. This is a markov chain with 7 states, $i = 0, \dots, 6$



We can interpret the arrows as follows

- When $i = 0$, everyone is playing N . So the probability of selecting an N player (to change to I) is 1 and the probability that this agent selects I is ε (since $0/n < c$). The probability that the agent adopts N (i.e. plays the same) is $1 - \varepsilon$
- When $i = 1$, the probability to select a person playing N is $5/6$ and the probability that person switches to I is ε . Furthermore, the probability that we pick the single I player is $1/6$ and the probability he adopts N is $1 - \varepsilon$. The probability that the state stays at 1 is: The probability that the agent that plays i is picked and continues with I which is $\frac{1}{6}\varepsilon$ and the probability that the agent that plays N is picked and continues with N is $\frac{5}{6}(1 - \varepsilon)$ and hence is

$$\frac{1}{6}\varepsilon + \frac{5}{6}(1 - \varepsilon)$$

Verify that these add to one

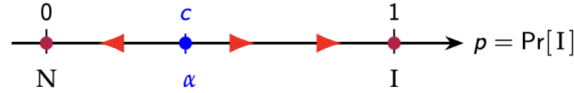
$$\frac{1}{6}\varepsilon + \frac{5}{6}(1 - \varepsilon) + \frac{5}{6}\varepsilon + \frac{1}{6}(1 - \varepsilon) = \frac{1}{6} + \frac{5}{6} + \frac{4}{6}(\varepsilon - \varepsilon) = 1$$

- Same for the rest

In the investment game, there is a stationary distribution $\alpha_0, \dots, \alpha_n$ with $\sum_i \alpha_i = 1$ as $\varepsilon \rightarrow 0$, the process is almost always at state $i = 0$ if $c > \frac{1}{2}$ and $i = n$ if $c < \frac{1}{2}$ i.e. at the state corresponding to the risk dominant Nash equilibrium of the game. Consider

- If $c > \frac{1}{2}$, then the deviation product at $(I, I) = (1 - c)(1 - c)$ which is less than the product at (N, N) which is c^2
- If $c < \frac{1}{2}$, then the deviation product at $(N, N) = c^2$ which is less than $(1 - c)(1 - c)$

If $c > \frac{1}{2}$, $i = 0$ is the absorbing state with the largest basin of attraction – see the image in the original game



and so is easy to jump into and hard to jump out of

Theorem 4.2. If each state is reached from any state with positive probability, then

- There is a unique distribution over the states that is invariant to the process (i.e. transition applied to this distribution leaves the distribution unchanged)
- starting from any state distribution, the process as well as the time-averaged frequencies of states converge to this stationary limit distribution

Proposition 4.4. The state that the Foster-Young stochastic partial best reply learning process converges to is called stochastically stable. It is always the absorbing state (at $\varepsilon = 0$) with the largest basin of attraction. Furthermore, in 2x2 coordination games, it corresponds to the risk dominant Nash equilibrium.

Remark. A risk dominant equilibrium is an equilibrium with higher deviation payoff product

Example 4.10 (PSet7 – Q4). Consider the following game

	L	R
L	4, 4	0, 1
R	1, 0	1, 1

- (a) Find all ESS. The row player's payoff is listed first.
Pure NE is $\{(L, L), (R, R)\}$ and mixed NE is $\{\frac{1}{4}L + \frac{3}{4}R, \frac{1}{4}L + \frac{3}{4}R\}$. All are symmetric NE so need to check each.

- For (L, L) – notice this is a strict nash equilibrium i.e. there is no best reply other than itself. Hence, the antecedent is false and (L, L) is ESS
- (R, R) – same reasoning and is an ESS
- Let $\beta = qL + (1 - q)R$ which is a BR against $\alpha^* = \frac{1}{4}L + \frac{3}{4}R$ for any $q \in [0, 1]$. Then calculating the payoffs

$$\begin{aligned} u(L, \beta) &= 4q \\ u(R, \beta) &= 1 \end{aligned}$$

and so

$$\begin{aligned} u(\alpha^*, \beta) - u(\beta, \beta) &= q + \frac{3}{4} - 4q^2 + 1 - q \\ &= -(2q - \frac{1}{2})^2 \leq 0 \end{aligned}$$

and strict when $\beta \neq \alpha^*$

Hence, ESS only the pure NE.

- (b) Write down the single-population replicator dynamic for the game above and discuss how play evolves under it.

Let p be the proportion playing L . Then,

$$\begin{aligned} \pi_L(t) &= 4p \\ \pi_R(t) &= 1 \end{aligned}$$

So,

$$\dot{p} = p(1 - p)(\pi_L(t) - \pi_R(t)) = p(1 - p)(4p - 1)$$

The rest points are $p = 0, \frac{1}{4}, 1$. If $p \in (0, \frac{1}{4})$ then $4p - 1 < 0$ and $\dot{p} < 0$ and so $p \rightarrow 0$. If $p \in (\frac{1}{4}, 1)$, then $4p - 1 > 0$ and $\dot{p} > 0$ and so $p \rightarrow 1$ (i.e. more play L). Hence, $p = 0, 1$ are locally stable for their respective basins of attraction but $p = 1/4$ is unstable.

- (c) Suppose that the above game is played by a finite population of fixed size. Each player in the population plays either L or R. Each period, one of the players has the opportunity to revise their strategy, with all players equally likely to be chosen for this opportunity. With probability $1 - \varepsilon$ the chosen player chooses a best reply to the distribution of strategies in the population, with probability ε she chooses each strategy with probability $1/2$. If ε is small but positive, explain carefully how you would expect the system to evolve in the long run. Interpret and comment.

Let p be the proportion of players playing strategy L . Given that (L, L) is the risk dominant Nash Equilibrium and has the larger basin of attraction from $(\frac{1}{4}, 1)$. Consider states $i(t) = 0, 1$

- At $i = 0$, the probability that a player playing R is picked is 1 and so the probability that he changes his strategy is $1 - \varepsilon + \frac{1}{2}\varepsilon$. The probability that he does not change his strategy is $\frac{1}{2}\varepsilon$ i.e. state remains the same
- At $i = 1$, the probability that a player playing R is picked is $N - 1/N$ and so the probability that the state changes is

$$\frac{N-1}{N}(1 - \frac{1}{2}\varepsilon)$$

and the probability that the state changes back to 0 is precisely

$$\frac{1}{N}(1 - \frac{1}{2}\varepsilon)$$

i.e. that single player playing L is picked and changes his strategy i.e. with $1 - \varepsilon$ or $\varepsilon/2$. Lastly, the probability that the state does not change is

$$\frac{1}{N} \frac{\varepsilon}{2} + \frac{N-1}{N} \frac{\varepsilon}{2}$$

and they all add to 1

The markov chain has a stationary distribution as $\varepsilon \rightarrow 0$ the process is almost always as state $i = 0$ since it has the largest basin of attraction. In other words, as the probability of making a mistake decreases (plays the best reply with probability 1), the population converges to everyone playing L .

5 Repeated Games

5.1 Folk Theorems

Repeated games consists of a stage game played at $t = 1, \dots, T$ where $T < \infty$ or $T = \infty$. Usually, each player simultaneously picks an action $\alpha_i \in \Delta(A_i)$ for stage payoff $u_i(a) \in \mathbb{R}$. We take past actions to be observed by everyone. A repeated game strategy for player i specifies an action (pure or mixed) in each period t as a function of the entire history at t . We usually discount future payoffs by $\delta \in (0, 1)$ to yield the present value.

How do we compare payoffs in repeated games with payoffs in stage-games? If $T < \infty$ (i.e. finite games), then $\delta = 1$ (without loss of generality), so we may simply use the average payoff for comparability between repeated and stage games. If $T = \infty$, we need to use the average discounted value.

Definition 5.1 (Average Discounted Value). Consider an infinitely repeated game where the present value for player i is

$$\text{Present Value}_i = \sum_{t=1}^{\infty} \delta^t u_i(\alpha_i^t)$$

Then, the average discounted value (ADV) for player i is

$$\text{ADV}_i = \text{PV} \times (1 - \delta)$$

We can think of the average discounted value as the weighted average of today's payoff and continuation payoff. We can use a recursive formulation of ADV

$$\text{ADV}_i = (1 - \delta) \times (\text{today's payoff}) + \delta \times (\text{ADV of continuation play})$$

Furthermore, the average discounted value of v per period for k periods then $\hat{v}$ per period for ever is

$$(1 - \delta^k)v + \delta^k \hat{v}$$

The question to ask is what kind of payoffs can we generate in a repeated game? We must consider both feasibility and individual rationality.

Definition 5.2 (Feasibility). $(v_1, \dots, v_n)$ is a feasible average discounted payoff of the repeated stage game if

$$(v_1, \dots, v_n) \in \text{co}\{(w_1, \dots, w_n) : w_i = u_i(a) \text{ for some } a \in A\}$$

where 'co' means convex hull i.e. smallest convex superset

Note that feasible payoffs means that players can generate them not that they are willing to. Players can generate such payoffs through (i) a public randomisation device or (ii) coordinated cycling over pure outcomes of the stage game: any feasible $(v_1, \dots, v_n)$ can be approximated arbitrarily closely in ADV using sufficiently many repetitions and δ close enough to 1.

Definition 5.3 (Minmax Payoffs). Let player i 's minmax payoff be

$$\underline{v}_i = \min_{\alpha_{-i}} \max_{\alpha_i} u_i(\alpha_i, \alpha_{-i})$$

The basic intuition of this is that other players coordinate to minimise your payoffs from your best replies. One way to figure out what this number is is via coinciding lower and upper bounds by guessing

Example 5.1. Consider the matching pennies game

	H	T
H	1, -1	-1, 1
T	-1, 1	1, -1

To find the minmax payoffs, we can guess the lower and upper bounds

- P1 can hold P2 to zero payoff by P1 mixing 50-50 which implies $\underline{v}_2 \leq 0$
- P2 can guarantee zero payoff by P2 mixing 50-50 which implies $\underline{v}_2 \geq 0$

So, $\underline{v}_2 = 0$

Proposition 5.1 (Individual Rationality). If $(v_1, \dots, v_n)$ are average discounted Nash equilibrium payoffs of a repeated game then $v_i \geq \underline{v}_i$ for all $i \in N$

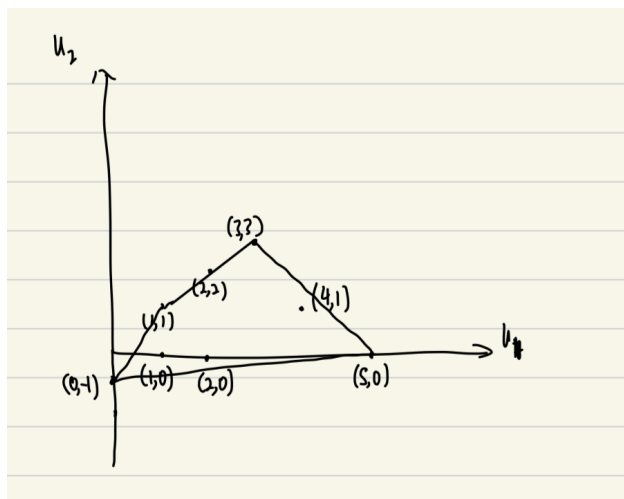
Proof. If i best responds to α_{-i}^1 and then given the history that the play generates, best responds to $\alpha_{-i}^t(h^t)$ for all $t > 1$ and h^t , then her payoff will be at least $\underline{v}_i$ in each period. Hence i won't get (accept) any less in equilibrium ■

We denote V^* for the set of all average or average-discounted payoff profiles $(v_1, \dots, v_n)$ that are feasible and strictly individually rational.

Example 5.2. Consider the following normal form game. P1 chooses row, P2 chooses column.

	l	m	r
T	4, 1	2, 2	1, 0
M	5, 0	3, 3	1, 1
B	1, 0	2, 0	0, -1

We find both the feasible payoffs and the individually rational set.



First we want to find P1 minmax payoffs. Notice that M strictly dominates B and weakly dominates T so M is always a best reply to any mix for $\alpha_2 \in \Delta(A_2)$. Let P2 mix with probabilities (q_l, q_m, q_r) so that

$$\mathbb{E}[M] = 5q_l + 3q_m + q_r$$

So, P2 solves

$$\min_{(q_l, q_m, q_r)} (5q_l + 3q_m + q_r) \text{ s.t. } q_l + q_m + q_r = 1$$

which is minimised by $(q_l, q_m, q_r) = 1$. So, $v_1 = 1$.

Now find P2's minmax payoffs. Notice as well that m strictly dominates r and weakly dominates l so m is always a best reply. Hence, P1 solves

$$\min_{(p_T, p_M, p_B)} (2p_T + 3p_M) \text{ s.t. } p_T + p_M + p_B = 1$$

which is solved by $(p_T, p_M, p_B) = (0, 0, 1)$. Hence $v_2 = 0$. The strict individually rational payoff set is thus

$$\{(u_1, u_2) \in \mathbb{R}^2 : u_1 > 1, u_2 > 0\}$$

And with $V = \text{co}\{(0, -1), (5, 0), (3, 3), (1, 1)\}$, we have

$$\begin{aligned} V^* &= V \cap \{(u_1, u_2) \in \mathbb{R}^2 : u_1 > 1, u_2 > 0\} \\ &= \text{co}\{(1, 0), (5, 0), (3, 3), (1, 1)\} \end{aligned}$$

Example 5.3. Consider the following stage game

	L	R
U	1, 1	0, 0
M	0, 3	2, 2
D	0, 0	2, -1

We find both the minmax payoffs and the feasible and individually rational payoffs. First find P1's minmax values. Let P2 mix $(q, 1 - q)$ over (L, R) . Then,

$$\begin{aligned} u_1(U) &= q \\ u_1(M) &= 2(1 - q) \\ u_1(D) &= 2(1 - q) \end{aligned}$$

Notice that the problem reduces to

$$\min_{q \in [0, 1]} \max\{q, 2(1 - q)\}$$

Since q is increasing in q and $2(1 - q)$ is decreasing in q , clearly, the max function is decreasing in q until $q = 2(1 - q)$ and then increases thereafter. So the solution to this is

$$q = 2(1 - q) \implies q = \frac{2}{3}$$

Alternatively, we can check the when the best responses switch i.e. when

$$u_1(U) = q \geq 2(1 - q) = u_1(M) = u_1(D)$$

where U is BR when $q \geq 2/3$ and M, D are best responses when $q < 2/3$. So, minmax is when $q = \frac{2}{3}$. In the same vein, when q falls from 1 until $2/3$, $u_1(U)$ as the BR falls and after which it increases as $u_1(M)$ or $u_1(D)$ become best responses. So, the minmax is when

$$q = 2(1 - q) \implies q = \frac{2}{3}$$

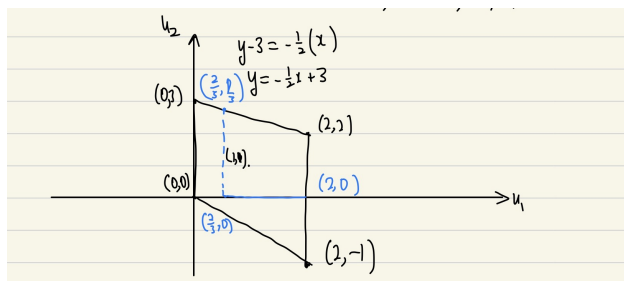
So, $v_1 = \frac{2}{3}$.

Let P1 mix w.p. (p_U, p_M, p_D) s.t. $p_U + p_M + p_D = 1$. This yields

$$\begin{aligned} u_2(L) &= p_U + 3p_M \\ u_2(R) &= 2p_M - p_D = p_U + 3p_M - 1 = u_2(L) - 1 \end{aligned}$$

The feasible payoff set is

where the strict IR and feasible set can be graphically drawn as


$$V^* = V \cap \{(u_1, u_2) \in \mathbb{R}^2 : u_1 > \frac{2}{3}, u_2 > 0\}$$

Theorem 5.1 (Folk Theorem for Nash Equilibria with infinite repetition). If $v = (v_1, \dots, v_n) \in V^*$, then there exists $\underline{\delta} < 1$ such that for all $\delta > \underline{\delta}$, the infinitely repeated game has a Nash equilibrium in which each player i 's ADV equals v_i .

If player i deviates, she gets at most $\bar{v}_i$ (her maximal payoff) once, and then receives $\underline{v}_i$ (her minmax payoff) forever. Hence she does not deviate if

Rearranging,

$$\delta \geq \frac{\bar{v}_i - v_i}{\bar{v}_i - \underline{v}_i}.$$

$$\delta \geq \bar{\delta} := \max_i \left\{ \frac{\bar{v}_i - v_i}{\bar{v}_i - v_i} \right\},$$

Example 5.4. Consider the following game

	L	R
U	2, 2	1, 3
D	-1, 1	-1, 1

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We want to sustain subgame perfect equilibrium. Basically, we need to play Nash equilibrium in every subgame, even in the continuation after a deviation. This subgame (continuation after t) is not the same as a stage game (game played AT t). Playing a stage game Nash at every t (or cycling over some) is trivially an SPE.

Here is a useful existence theorem for SPE.

Theorem 5.2. Suppose $v = (v_1, \dots, v_n)$ is a feasible ADV payoff vector such that for all i , v_i exceeds player i 's payoff in some Nash equilibrium of the stage game.

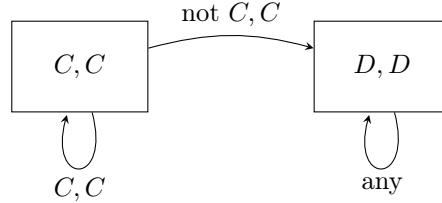
For δ sufficiently close to 1, there is a subgame-perfect equilibrium of the infinitely repeated game with average discounted payoffs v

Before we get into the subgame-perfect equilibrium strategies, we want to (i) simplify our representation of these strategies (using automaton), (ii) simplify our proofs that these are subgame perfect equilibrium.

Definition 5.4 (Strategies as automaton). An automaton for a player i in an infinitely repeated game has the following components

- A set Q_i (of states)
- An element $q_i^0 \in Q_i$ (the initial state)
- A function $f_i : Q_i \rightarrow \Delta(A_i)$ that assigns an action to every state (the output function)
- A function $\tau_i : Q_i \times A_i \rightarrow Q_i$ that assigns a state to every pair consisting of a state and an action (the transition function)

For instance, a grim trigger strategy for prisoner's dilemma can be represented as



Theorem 5.3 (One-shot deviation principle). In a repeated game with discounting, a strategy profile is a subgame-perfect equilibrium iff the following holds for all histories, at all t , for all i :

As long as players other than i play their proposed strategies at and after t , player i cannot gain by deviating at t and then reverting back to her proposed strategy (which prescribes what to do after her own deviation as well) from $t + 1$ on.

Proof. Suppose that player i 's repeated-game strategy s_i cannot be improved by any one-step deviation after any history, yet there exists another strategy s'_i that strictly improves upon s_i at some date t , after some history h^t , perhaps in an arbitrarily complicated way. We derive a contradiction.

Key idea 1. Due to discounting, payoffs far in the future make little difference: for any $\Delta > 0$, there exists T_Δ such that payoffs after period $t + T_\Delta$ contribute less than Δ to player i 's average discounted value.

Therefore, if s'_i increases player i 's payoff relative to s_i by at least Δ , it must do so within finitely many periods, namely within T_Δ periods. Hence, without loss of generality, we may assume that s'_i differs from s_i only for some histories in periods

$$t, \dots, t + T_\Delta,$$

and thereafter coincides with s_i from period $t + T_\Delta + 1$ onward. If $T_\Delta = 1$, then s'_i is a one-step deviation, contradicting the hypothesis. So suppose $T_\Delta > 1$.

Backward induction. Consider a new strategy s_i'' defined as follows: it agrees with s_i' for all histories in periods up to and including $t + T_\Delta - 1$, but from period $t + T_\Delta$ onward it agrees with s_i . That is, s_i'' reverts to s_i one period earlier than s_i' .

Since s_i cannot be improved upon by a one-period deviation, and since $s_i'' \equiv s_i$ from $t + T_\Delta$ onward, it follows that s_i'' is weakly better than s_i' after any history from $t + T_\Delta$ onward. Because s_i'' and s_i' coincide up to period $t + T_\Delta - 1$, this implies that s_i'' is weakly better than s_i' overall.

Therefore, if s_i' strictly improves upon s_i , then so does s_i'' . But s_i'' differs from s_i for only $T_\Delta - 1$ periods rather than T_Δ .

Repeating this argument, we can shorten the duration of the profitable deviation by one period at a time. After finitely many steps, we obtain a strategy that differs from s_i in only one period and still strictly improves upon it. This contradicts the assumption that no one-step deviation can improve upon s_i .

Hence no such strategy s_i' can exist. Therefore, if s_i cannot be improved upon by any one-step deviation after any history, then it cannot be improved upon at all. ■

Remark. What this means is that we only need to check unilateral, one shot deviations in every continuation i.e. at every state of the automaton.

Theorem 5.4 (Perfect Folk Theorem). Assume V^* is the n -dimensional set of feasible and strictly individually rational payoffs. For any $(v_1, \dots, v_n) \in V^*$ there is $\underline{\delta} \in (0, 1)$ such that for all $\delta > \underline{\delta}$, the infinitely repeated game has a subgame-perfect equilibrium with average discounted payoffs $(v_1, \dots, v_n)$

Proof by the one-shot deviation principle. Let $v = (v_1, \dots, v_n)$ be the payoff vector that we want to support in subgame-perfect equilibrium of the infinitely repeated game.

For each player i , let $m^i = (m_1^i, \dots, m_n^i)$ be a stage-game action profile that minmaxes player i . Normalize payoffs so that

$$u_i(m^i) = 0,$$

while for every $k \neq i$,

$$u_k(m^i) \geq 0.$$

Denote player i 's maximal stage-game payoff by $\bar{v}_i$.

Pick a feasible payoff vector w such that

$$0 < w < v$$

componentwise. Choose $T^* \in \mathbb{N}$ large enough that

$$T^* w_i > \bar{v}_i \quad \text{for all } i.$$

Then choose $\varepsilon > 0$ small enough so that, for every player i , the vector

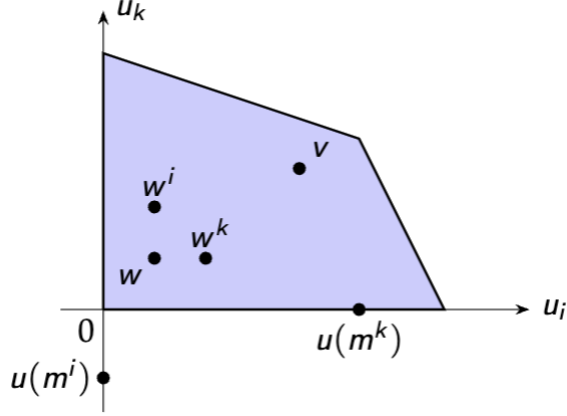
$$w^i := (w_i, w_{-i} + \varepsilon)$$

is feasible and still satisfies

$$w^i < v.$$

That is, $w_i^i = w_i$ and $w_k^i = w_k + \varepsilon$ for all $k \neq i$.

We can illustrate the notation in the following



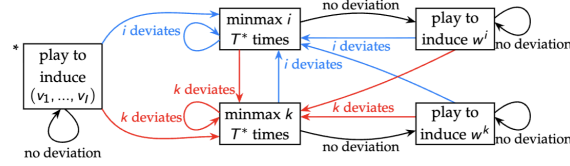
We consider the following strategy profile with three phases.

Phase I (collaboration). Play stage-game actions that generate v . Continue doing so unless some player i deviates, in which case move to Phase II^i .

Phase II^i (punishment of player i). Play m^i for exactly T^* periods. If no player deviates during this punishment phase, move to Phase III^i . If instead player k deviates during Phase II^i , restart punishment with Phase II^k .

Phase III^i (reconciliation after punishing player i). Play stage-game actions that generate w^i . Continue forever unless some player k deviates, in which case move to Phase II^k .

We can illustrate this as an automaton (not really but roughly one)



We now verify, by the one-shot deviation principle, that for δ sufficiently close to 1, no player has a profitable one-shot deviation after any history.

Phase I. On path, player i receives

$$v_i = (1 - \delta^{T^*+1})v_i + \delta^{T^*+1}v_i.$$

If player i deviates once in Phase I, then in the deviation period she can get at most $\bar{v}_i$, and thereafter she is punished for T^* periods and then receives continuation payoff w_i . Hence her deviation payoff is at most

$$(1 - \delta)\bar{v}_i + \delta^{T^*+1}w_i.$$

To compare these expressions, note that

$$(1 - \delta^{T^*+1})v_i = (1 - \delta)(1 + \delta + \dots + \delta^{T^*})v_i.$$

For δ sufficiently close to 1,

$$(1 - \delta)(1 + \delta + \dots + \delta^{T^*})v_i > (1 - \delta)T^*v_i > (1 - \delta)T^*w_i > (1 - \delta)\bar{v}_i,$$

where the middle inequality uses $v_i > w_i$, and the last inequality uses the definition of T^* . Also,

$$\delta^{T^*+1}v_i > \delta^{T^*+1}w_i$$

because $v_i > w_i$. Therefore,

$$v_i = (1 - \delta^{T^*+1})v_i + \delta^{T^*+1}v_i > (1 - \delta)\bar{v}_i + \delta^{T^*+1}w_i.$$

So a one-shot deviation in Phase I is not profitable.

Phase IIⁱ. Consider first player i , the player currently being punished. Since m^i minmaxes player i , she can get at most 0 during the punishment phase. A one-shot deviation can only restart punishment and hence postpone the transition to the more favorable reconciliation phase. Therefore player i cannot profit by a one-shot deviation during Phase IIⁱ.

Now consider a player $j \neq i$ during Phase IIⁱ, with $K \leq T^*$ punishment periods remaining. If player j follows the prescribed strategy, her continuation payoff is

$$(1 - \delta^K)u_j(m^i) + \delta^K(w_j + \varepsilon),$$

since after the punishment phase ends the game moves to Phase IIIⁱ, where player $j \neq i$ receives $w_j + \varepsilon$. If instead player j makes a one-shot deviation, then in the deviation period she can get at most $\bar{v}_j$, and thereafter the game moves to Phase II^j, so after T^* punishment periods she receives continuation payoff w_j . Thus her deviation payoff is at most

$$(1 - \delta)\bar{v}_j + \delta^{T^*+1}w_j.$$

Hence player j does not deviate if

$$(1 - \delta^K)u_j(m^i) + \delta^K(w_j + \varepsilon) \geq (1 - \delta)\bar{v}_j + \delta^{T^*+1}w_j.$$

Rearranging,

$$(1 - \delta^K)u_j(m^i) - (1 - \delta)\bar{v}_j + (\delta^K - \delta^{T^*+1})w_j + \delta^K\varepsilon \geq 0.$$

As $\delta \rightarrow 1$, the first three terms converge to 0, while the last term converges to $\varepsilon > 0$. Therefore, for δ sufficiently close to 1, no player $j \neq i$ has a profitable one-shot deviation in Phase IIⁱ.

Phase IIIⁱ. In this phase, player i receives w_i , while each player $k \neq i$ receives $w_k + \varepsilon$. Consider any player j .

If player j follows the prescribed strategy, her payoff is at least w_j . If instead she makes a one-shot deviation, then in the current period she can get at most $\bar{v}_j$, and from the next period onward she is punished for T^* periods and then receives continuation payoff w_j . Thus her deviation payoff is at most

$$(1 - \delta)\bar{v}_j + \delta^{T^*+1}w_j.$$

As in the analysis of Phase I,

$$(1 - \delta^{T^*+1})w_j = (1 - \delta)(1 + \delta + \dots + \delta^{T^*})w_j > (1 - \delta)T^*w_j > (1 - \delta)\bar{v}_j$$

for δ sufficiently close to 1. Therefore,

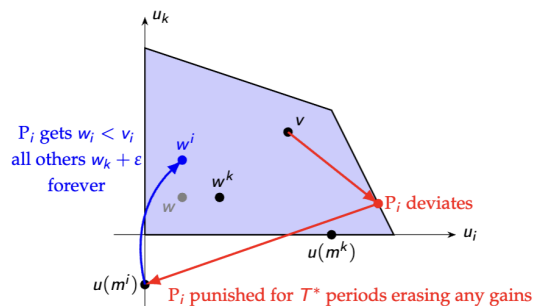
$$w_j = (1 - \delta^{T^*+1})w_j + \delta^{T^*+1}w_j > (1 - \delta)\bar{v}_j + \delta^{T^*+1}w_j.$$

So a one-shot deviation in Phase IIIⁱ is not profitable either.

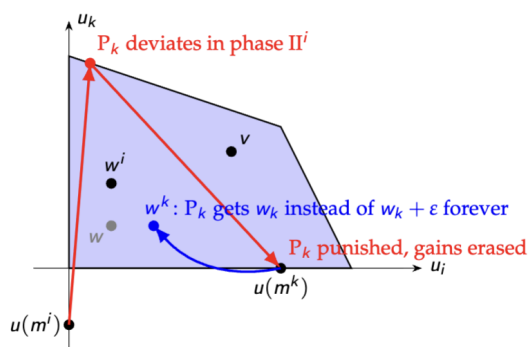
Since after every history and in every phase no player has a profitable one-shot deviation for δ sufficiently close to 1, the one-shot deviation principle implies that this strategy profile is subgame perfect. In particular, it supports the target payoff vector v . ■

We can further illustrate the perfect folk theorem graphically for more intuition

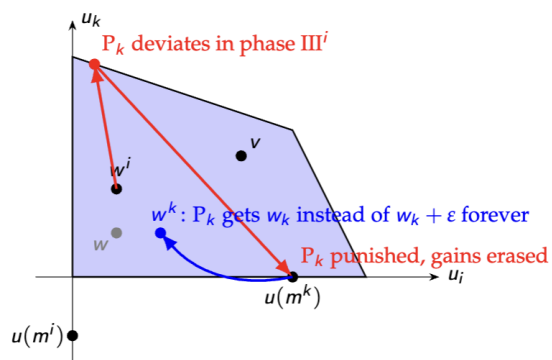
1. What if player i deviates in Phase I



2. What if player i deviates in Phase II ^{i}



3. What if player i deviates in Phase III ^{i}



The dimensionality condition is important (non-trivial) in the perfect folk theorem

Example 5.5. Consider the following three-player game. Player 1 chooses row, Player 2 column, and Player 3 matrix; payoffs are (u_1, u_2, u_3) .

		ℓ	r
matrix A	T	$(1, 1, 1)$	$(0, 0, 0)$
	B	$(0, 0, 0)$	$(0, 0, 0)$

		ℓ	r
matrix B	T	$(0, 0, 0)$	$(0, 0, 0)$
	B	$(0, 0, 0)$	$(1, 1, 1)$

- (a) What is the set of feasible and individually rational payoffs?

The set of feasible payoffs is

$$V = \text{co}\{(0, 0, 0), (1, 1, 1)\} = \{(v, v, v) \in \mathbb{R}^3 : 0 \leq v \leq 1\}$$

IR payoffs:

- For P1: if (r, A') or (l, B') then P1 gets 0 no matter what so $\underline{v}_1 = 0$
- For P2: if (B, A') or (T, B') then P2 gets 0 no matter what so $\underline{v}_2 = 0$
- For P3: if (B, l) or (A, r) then P3 gets 0 no matter what so $\underline{v}_3 = 0$

Hence

$$V^* = \{(v, v, v) \in \mathbb{R}^3 : 0 \leq v \leq 1\}$$

- (b) Show that average discounted payoffs (v, v, v) with $v < 1/4$ cannot be sustained in a subgame perfect equilibrium for any discount factor δ .

Hint: First show that for any SPE payoff v we have

$$v \geq (1 - \delta) \cdot \frac{1}{4} + \delta\alpha,$$

where α is the lowest symmetric payoff sustainable in any SPE. Then apply this inequality to $v = \alpha$. Let α be the lowest symmetric payoff sustained in some SPE. We want to show that $\alpha \geq 1/4$. Suppose players mix independently s.t.

$$p_1 = \mathbb{P}(T), p_2 = \mathbb{P}(l), p_3 = \mathbb{P}(A')$$

Then

- P1 problem is

$$u_1(T) = p_2 p_3, u_1(B) = (1 - p_2)(1 - p_3)$$

and so $\max\{p_2 p_3, (1 - p_2)(1 - p_3)\}$

- P2 problem is

$$u_2(T) = p_1 p_3, u_2(B) = (1 - p_1)(1 - p_3)$$

and so $\max\{p_1 p_3, (1 - p_1)(1 - p_3)\}$

- P3 problem is

$$u(A') = p_1 p_2, u(B') = (1 - p_1)(1 - p_2)$$

and so $\max\{p_1 p_2, (1 - p_1)(1 - p_2)\}$

By the pigeonhole principle, at least 2 of p_1, p_2, p_3 are greater than $1/2$ or less than $1/2$. Hence, if $p_i, p_j \geq 1/2$, then $p_i p_j \geq 1/4$. If $p_i, p_j \leq 1/2$ then $(1 - p_i)(1 - p_j) \geq 1/4$. So no matter what, at least one player can yield payoff greater than $1/4$ in one period.

Thus, for an average discounted payoff v , to sustain an SPE

$$v \geq (1 - \delta) \frac{1}{4} + \delta\alpha$$

The intuition for this is that we can have $1/4$ today and α for the continuation payoff using the recursive definition of the ADV. Then letting $v = \alpha$, we have

$$\alpha \geq (1 - \delta) \frac{1}{4} + \delta\alpha \implies \alpha \geq \frac{1}{4}$$

So no $v < \frac{1}{4}$ can be sustained.

- (c) Which assumption of the Perfect Folk Theorem is violated?

V^* is not n -dimensional i.e. fewer dimensions than the no. of players as $\dim(V) = 1 < n = 3$

Common Strategies

We present two common strategies in infinitely repeated games, and also show how to check for OSD at the same time.

Grim Trigger/Nash Reversion

The principle for a grim trigger/nash reversion strategy is to threaten a to play Nash equilibrium in every stage-game in the continuation after deviation.

Since this is covered in core micro we show a more complex Nash Reversion strategy.

Example 5.6. Consider the following normal-form game. P1 chooses row, P2 chooses column, and P1's payoff is listed first.

	<i>L</i>	<i>R</i>
<i>T</i>	5, 2	-5, 0
<i>M</i>	-2, 0	2, 2
<i>B</i>	-1, 6	-1, 6

- (a) Find each player's minmax payoff and represent graphically the set of feasible and individually rational payoffs.

Consider P1's minmax. Let P2 mix over (L, R) w.p. $(q, 1 - q)$. Then,

$$u_1(T) = 5q + (-5)(1 - q) = 10q - 5$$

$$u_1(M) = -2q + 2(1 - q) = 2 - 4q$$

$$u_1(B) = -1$$

P1 gets $\max\{10q - 5, 2 - 4q, -1\}$. Minmax is when

$$10q - 5 = 2 - 4q \implies q = \frac{1}{2}$$

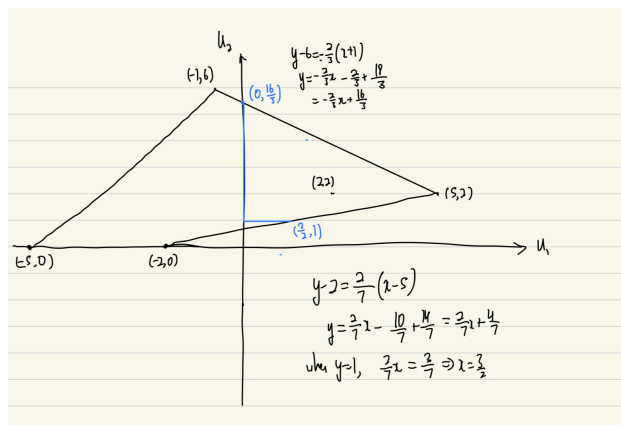
So, $\underline{v}_1 = \min\{0, 0, -1\} = 0$ when P2 mixes $(\frac{1}{2}L + \frac{1}{2}R)$.

Consider P2 minmax where P1 mixes w.p. (p_T, p_M, p_B) s.t. $p_T + p_M + p_B = 1$. Then

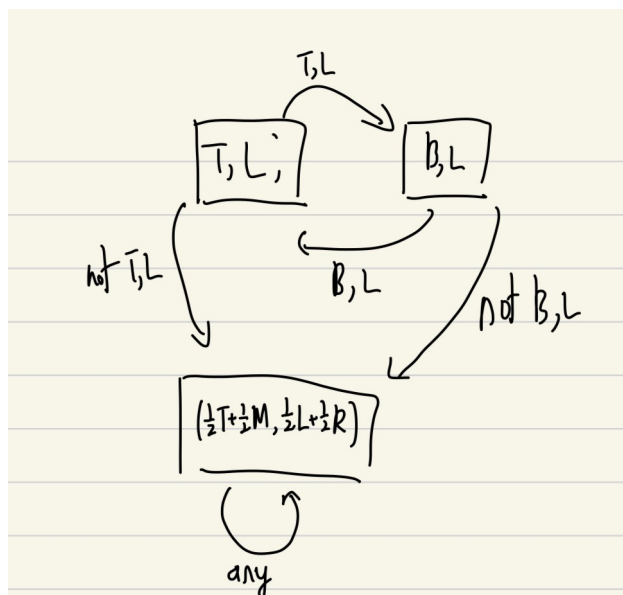
$$u_2(L) = 2p_T + 6p_B$$

$$u_2(R) = 2p_M + 6p_B$$

P2 gets $\max\{2p_T + 6p_B, 2p_M + 6p_B\}$. To minimise this is to allocate $(p_T, p_M, p_B) = (0.5, 0.5, 0)$. So, $\underline{v}_2 = \min\{1, 1\} = 1$. The set of strictly IR and feasible payoffs are



- (b) Find all Nash equilibria of the game and compute the players' payoffs.
 The pure NE is $\{(T, L), (M, R)\}$ with payoffs $(5, 2)$ and $(2, 2)$ respectively.
 Since B is a never best response, by Pearce's Lemma it is strictly dominated. Let P2 mix w.p. $(q, 1 - q)$. Then, $u_1(T) = 5q - 5(1 - q) = 10q - 5$, $u_1(M) = -2q + 2(1 - q) = 2 - 4q$. This gives $q = \frac{1}{2}$. Let P1 mix over (T, M) w.p. $(p, 1 - p)$. Then $u_2(L) = 2p$, $u_2(R) = 2(1 - p)$ which gives $p = \frac{1}{2}$. The mixed NE is thus $(\frac{1}{2}T + \frac{1}{2}M, \frac{1}{2}L + \frac{1}{2}R)$. The payoffs are $(0, 1)$.
- (c) Suppose that the game is repeated infinitely many times with a discount factor arbitrarily close to 1. Players maximize the discounted present value of their payoffs. Construct a subgame-perfect equilibrium in which P1's long-run average payoff is 2 and P2's is 4. [Hint: Alternate between two outcomes on the equilibrium path.]
 This is a slightly more complicated Nash reversion – basically, alternate between (T, L) and (B, L) and punish with the mixed NE if there is a deviation



On path long run average payoffs are $\left(\frac{5+(-1)}{2}, \frac{2+6}{2}\right) = (2, 4)$. Again, it suffices to check OSD. Since (T, L) is itself a Nash equilibrium, no need to check since no OSD for either player. Just need to check for state (B, L)

- P1

On-path:

$$-1 + 5\delta - \delta^2 + 5\delta^3 = 5 \sum_{i=0}^{\infty} \delta^{2i+1} - \sum_{i=0}^{\infty} \delta^{2i} = \frac{5\delta - 1}{1 - \delta^2}$$

The ADV is $\frac{5\delta - 1}{1 + \delta}$.

Off-path is

$$2 + 0 + \dots = 2$$

ADV is $(1 - \delta)2$

Check

$$\frac{5\delta - 1}{1 + \delta} \geq (1 - \delta)2 \implies 5\delta - 1 \geq (1 - \delta^2)2 \implies 2\delta^2 + 5\delta - 3 \geq 0$$

Solving the quadratic form gives us $\delta \geq \frac{1}{2}$. There is no need to check P2 since it has no profitable deviation.

Carrot and Stick

The carrot and stick strategy is to threaten hard punishment for a limited time before reverting back – we also punish failure to punish too.

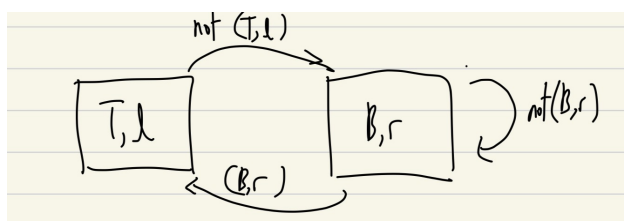
We show more for carrot-and-stick since it is harder to find.

Example 5.7. Consider the following normal form game. P1 chooses row, P2 chooses column.

	l	m	r
T	4, 1	2, 2	1, 0
M	5, 0	3, 3	1, 1
B	1, 0	2, 0	0, -1

We have already found the strict IR and feasible payoffs. We will now construct the SPE.

Notice that by IESDS $\{M, m\}$ is the only one-shot pure NE. Hence we can construct the following SPE



We check OSD

- At (T, l)
 - P1 On-path: $4 + 4\delta + 4\delta^2 + \dots$ P1 Off-path: $5 + 5 \cdot 0 + 4\delta^2 + \dots$
Checking $4 + 4\delta \geq 5 \implies \delta \geq \frac{1}{4}$
 - P2 On-path: $1 + \delta + \delta^2 + \dots$ P2 Off-path: $2 - \delta + \delta^2 + \dots$
Checking $1 + \delta \geq 2 - \delta \implies \delta \geq \frac{1}{2}$
- At (B, R)
 - P1 On-path: $0 + 4\delta + 4\delta^2 + \dots$ P1 Off-path: $2 + 0 + 4\delta^2 + \dots$
Checking $4\delta \geq 2 \implies \delta \geq \frac{1}{2}$
 - P2 On-path: $-1 + \delta + \delta^2$. P2 Off-path: $0 + (-1)\delta + \delta^2 + \dots$
Checking $-1 + \delta \geq -\delta \implies \delta \geq \frac{1}{2}$

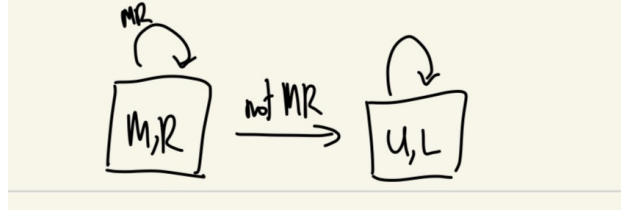
So, SPE is sustained for $\delta \geq \frac{1}{2}$

Example 5.8. Consider the following stage game

	L	R
U	1, 1	0, 0
M	0, 3	2, 2
D	0, 0	2, -1

We have already found the strict IR and feasible payoffs. We will now construct two SPE with different discounting factors.

1. SPE with $\delta \geq 1/2$. We can construct the following SPE

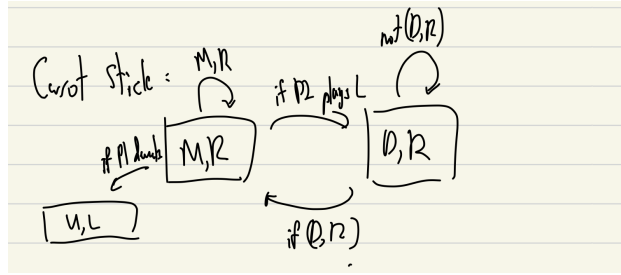


Payoff for both players on-path is $\sum_{i=0}^{\infty} 2\delta^i = \frac{2}{1-\delta}$. The ADV is 2.

Payoff off-path

- Do not need to consider P1 as there is no profitable deviation since M is its BR
- For P2: $3 + \sum_{i=1}^{\infty} \delta^i = 3 + \frac{\delta}{1-\delta}$. So the ADV is $3(1-\delta) + \delta = 3 - 2\delta$. Hence this gives us $2 \geq 3 - 2\delta \implies \delta \geq 1/2$.

2. SPE with $\delta = 1/3$. Here we construct a slightly more complicated SPE



Notice that at both states (M, R) and (D, R) P1 has no profitable deviation since it is best replies – so we do not need to check P1. Simply check P2

- At state (M, R) :
On-path is $(1-\delta)[2 + 2\delta + 2\delta^2 + \dots]$
Off-path is $(1-\delta)[3 + (-\delta) + 2\delta^2 + \dots]$
So, check $2 + 2\delta \geq 3 - \delta \implies \delta \geq 1/3$
- At state (D, R) :
On-path is $-1 + 2\delta + 2\delta^2$
Off-path is $0 - \delta + 2\delta^2$
Check $-1 + 2\delta \geq -\delta \implies \delta \leq \frac{1}{3}$

Hence only $\delta = \frac{1}{3}$ sustains this SPE.

5.2 Renegotiation-proofness, finite repetition, imperfect monitoring

Renegotiation-Proofness

Motivation: both players are worse off at the beginning of a punishment phase for grim-trigger or stick-carrot schemes, so they have strong incentive to renegotiate away from it, undermining the credibility of the equilibrium construction. We want to design collaborative repeated game equilibria that are both (i) subgame perfect and also (ii) renegotiation proof.

Definition 5.5 (Weak Renegotiation Proofness). A subgame perfect equilibrium consists of subgame-perfect continuation plays after every possible history; call this the ‘book of plays’ of the SPE.

Weak Renegotiation Proofness: no continuation play in the ‘book of plays’ is Pareto dominated by any other.

The intuition is that if a continuation play is pareto dominated, then after the history that triggers the dominated play, it can be renegotiated to a mutually preferred continuation. Note that weak renegotiation proofness involves a comparison **within** the playbook of given SPE, not across different SPEs.

Theorem 5.5 (Folk Theorem for renegotiation-proof SPE). Assume $n = 2$. Let $v = (v_1, v_2)$ be a feasible, strictly individually rational payoff-pair and assume it is generated by action profile $a = (a_1, a_2)$. If there exist stage game actions $a' = (a'_1, a'_2)$ and $a'' = (a''_1, a''_2)$ with

1. $v_1 > c_1 \equiv \max_{\tilde{a}_1} u_1(\tilde{a}_1, a'_2)$ and $v_2 < u_2(a'_1, a'_2)$
2. $v_1 < u_1(a''_1, a''_2)$ and $v_2 > c_2 \equiv \max_{\tilde{a}_2} u_2(a''_1, \tilde{a}_2)$

then for δ near 1 there is a weak negotiation proof subgame perfect equilibrium in the infinite repeated game with average discounted values (v_1, v_2) .

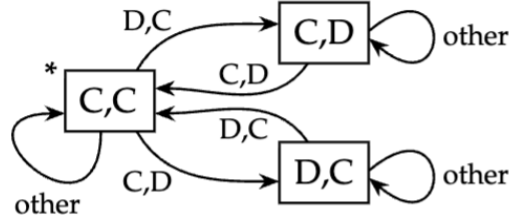
Furthermore, if an infinite repeated game has a weak renegotiation proof subgame-perfect equilibrium with ADV payoffs (v_1, v_2) , then there are action profiles a' and a'' satisfying (1) and (2) with weak inequalities.

There are two morals from the WRP Folk Theorem

- Possibility of renegotiation necessitates forgiveness so that players won't be stuck in a Pareto-suboptimal phase
- forgive but not forget: punishment must reward the punisher before cooperation can resume

We see that this is a good argument for using 'tit-for-tat' style strategies to support cooperation so that the punishment phase cannot be renegotiated

Example 5.9 (WRP in infinitely repeated prisoner's dilemma). In a prisoner's dilemma game, the WRP SPE can be characterised via the automaton



This is a tit-for-tat then forgive where

- Play (C, C) unless someone deviates
- If P1 defects then play (C, D) once (try again if failed) and return to (C, C) ; if P2 defects then (D, C) once and return

We can use the OSD to check SPE

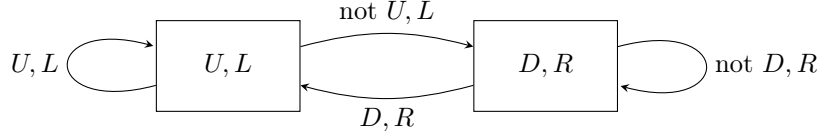
- At (C, C) : need $1 + \delta \geq 2 - \delta \iff \delta \geq 1/2$
- At (D, C) : need $-1\delta \geq 0 - \delta \iff \delta \geq 1/2$
- At (C, D) : analogous to at (D, C)

Notice that P1 prefers starting at (D, C) to (C, C) to (C, D) and P2 has opposite preferences so it satisfies the criterion intuitively.

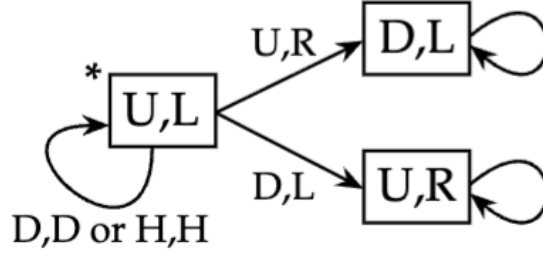
Example 5.10 (WRP in infinitely repeated Hawk-Dove). Consider the Hawk-Dove game

	L	R
U	6, 6	3, 7
D	7, 3	-1, -1

The stick-carrot scheme in the Hawk-dove game is an SPE for $\delta > 4/7$ but is not weakly renegotiation proof



It is not WRP because (D, R) is obviously pareto dominated by the continuation play (U, L) . Here is a WRP SPE for $\delta \geq 1/4$



To verify subgame perfection, it is enough to apply the one-deviation principle.

First consider the cooperative state in which the prescribed continuation play is (U, L) forever unless a deviation occurs. If player 1 complies, then her continuation payoff is

$$\frac{6}{1-\delta}.$$

If instead she deviates profitably in the current period, her highest possible current payoff is 7, and thereafter play moves to the punishment continuation (U, R) , which gives her payoff 3 in every future period. Hence her deviation payoff is

$$7 + \frac{3\delta}{1-\delta}.$$

Therefore player 1 does not want to deviate provided

$$\frac{6}{1-\delta} \geq 7 + \frac{3\delta}{1-\delta},$$

which is equivalent to

$$\delta \geq \frac{1}{4}.$$

By symmetry, the same condition ensures that player 2 does not want to deviate from (U, L) .

Next consider the punishment continuations (D, L) and (U, R) . Each of these is a stage-game Nash equilibrium, so once play reaches either of these states, no player can gain by a one-shot deviation. Hence there is no profitable deviation at those continuation histories.

It follows that the strategy profile is subgame perfect whenever $\delta \geq \frac{1}{4}$.

Finally, note the ranking of continuation payoffs. For player 1,

$$(D, L) \succ (U, L) \succ (U, R),$$

since these yield payoffs 7, 6, and 3 respectively. For player 2, the ranking is reversed:

$$(U, R) \succ (U, L) \succ (D, L),$$

since these yield payoffs 7, 6, and 3 respectively.

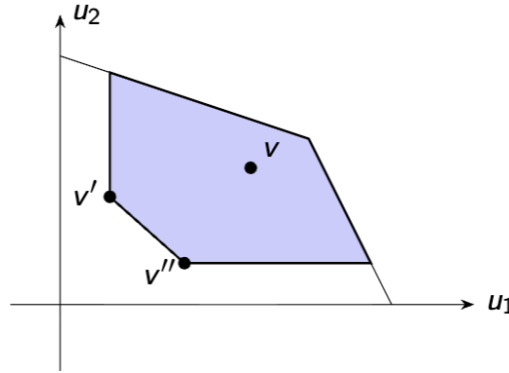
Finite Repetition

In many useful games e.g. coordination games but not in Prisoner's dilemma, finite repetition may expand the set of equilibria. The intuitive condition for this is that finite repetition expands the set of SPE if the stage game has multiple Nash equilibria ranked differently by the players

Theorem 5.6 (Finite Repetition Folk Theorem). Assume $n = 2$ and no discounting. Assume $v' = (v'_1, v'_2)$ and $v'' = (v''_1, v''_2)$ are Nash equilibrium payoffs in the stage game with $v'_1 < v''_1$ and $v'_2 > v''_2$.

If $v = (v_1, v_2)$ is feasible with $v_1 \geq v'_1$, $v_2 \geq v''_2$ and $v_2 - v'_2 \geq \frac{v''_2 - v'_2}{v'_1 - v''_1}(v_1 - v'_1)$, then for T sufficiently large, v can be approximated arbitrarily closely by the average payoffs of some SPE of the T -repeated game.

We can illustrate this as follows:



If v' and v'' are two Nash payoff vectors of the stage game such that P1 prefers v'' to v' and P2 prefers v' to v'' , then any feasible payoff in the shaded area (such as v) can be approximated as the average payoff of some SPE for T sufficiently large.

Example 5.11 (Finitely Repeated Hawk–Dove SPE). Consider the stage game

	L	R
U	6, 6	3, 7
D	7, 3	-1, -1

There are two pure Nash equilibria, (U, R) and (D, L) , and one mixed Nash equilibrium in which P1 plays U with probability $4/5$ and D with probability $1/5$, while P2 plays L with probability $4/5$ and R with probability $1/5$ i.e. $\{\frac{4}{5}U + \frac{1}{5}D, \frac{4}{5}L + \frac{1}{5}R\}$. The corresponding Nash payoffs are

$$(3, 7), \quad (7, 3), \quad (5.4, 5.4).$$

Suppose this stage game is repeated finitely many times, for $T < \infty$, with no discounting.

A simple subgame-perfect equilibrium that sustains (U, L) in all periods except the last one is the following:

- play (U, L) in every period up to period $T - 1$, unless someone deviates;
- if no one has deviated before period T , play the mixed Nash equilibrium in period T ;
- if P1 deviates at any earlier date, then from the next period onward revert forever to the pure Nash equilibrium (U, R) ;
- if P2 deviates at any earlier date, then from the next period onward revert forever to the pure Nash equilibrium (D, L) .

The logic is straightforward. Along the equilibrium path, both players enjoy the cooperative outcome $(U, L) = (6, 6)$. A deviation yields at most a one-period gain: P1 can raise her current payoff from 6 to 7 by switching from U to D , and P2 can similarly raise her current payoff from 6 to 7 by switching from L to R . But this short-run gain is offset by a long-run loss, because the continuation shifts to a Nash equilibrium that is particularly unfavorable to the deviator:

$$(U, R) \text{ gives P1 payoff 3,} \quad (D, L) \text{ gives P2 payoff 3.}$$

Thus each player's punishment is credible, since it is itself a Nash equilibrium of the stage game, and severe enough to deter deviation when there are enough periods left.

More generally, let (v_1, v_2) be a feasible payoff vector that we want to sustain for most of the repeated game, and let (v'_1, v'_2) and (v''_1, v''_2) be two Nash equilibrium payoffs such that

$$v'_1 > v''_1, \quad v''_2 > v'_2.$$

Let $\bar{v}_i$ denote player i 's maximal one-period payoff in the stage game. Choose T^* large enough that

$$T^* \frac{v''_1 - v'_1}{2} \geq \bar{v}_1 - v_1 \quad \text{and} \quad T^* \frac{v'_2 - v''_2}{2} \geq \bar{v}_2 - v_2.$$

Assume $T \gg T^*$.

Consider the following strategy profile:

- for $t \leq T - T^*$, play actions that generate (v_1, v_2) , unless someone deviates;
- if no one deviates, then in the final T^* periods alternate between the two Nash equilibrium payoffs (v'_1, v'_2) and (v''_1, v''_2) ;
- if P1 deviates at any $t \leq T - T^*$, revert for the rest of the game to the Nash equilibrium yielding (v'_1, v'_2) ;
- if P2 deviates at any $t \leq T - T^*$, revert for the rest of the game to the Nash equilibrium yielding (v''_1, v''_2) .

This is subgame perfect. In the final T^* periods, only Nash equilibria are played, so there is no profitable deviation there. At period $t = T - T^*$, consider P1. If she conforms, she gets

$$v_1 + T^* \frac{v'_1 + v''_1}{2}.$$

If she deviates, she gets at most

$$\bar{v}_1 + T^* v'_1.$$

Hence P1 prefers to conform if

$$v_1 + T^* \frac{v'_1 + v''_1}{2} \geq \bar{v}_1 + T^* v'_1,$$

which is equivalent to

$$T^* \frac{v''_1 - v'_1}{2} \geq \bar{v}_1 - v_1.$$

This holds by construction of T^* . The same argument applies to P2:

$$v_2 + T^* \frac{v'_2 + v''_2}{2} \geq \bar{v}_2 + T^* v''_2$$

whenever

$$T^* \frac{v'_2 - v''_2}{2} \geq \bar{v}_2 - v_2.$$

For earlier dates $t < T - T^*$, the one-shot gain from deviation is unchanged, but the punishment lasts even longer, so deviation is even less attractive.

The key intuition is that in a finitely repeated game one cannot use “grim trigger forever,” because the horizon is finite. However, one can still sustain desirable outcomes for a long initial segment of play by reserving the final block of periods for credible Nash equilibrium continuations. The last T^* periods serve as a punishment technology: they are chosen so that deviating today sacrifices enough future payoff to outweigh any current temptation. Thus, even with a finite horizon, cooperation can be sustained for most of the game.

Imperfect Monitoring

Relax the assumption that players observe each other's past moves.

Cooperative (collusive) equilibria may still be constructed under imperfect public monitoring using trigger strategies. For instance, Green and Porter showed that the Joint Executive Committee

- US railroad cartel controlling grain shipments from Chicago to the East Coast, before Sherman Act (1890) made it illegal.
- When the JEC (explicit cartel) thought that cheating had occurred, it cut prices for a limited time.
- Three price wars between 1880-1886; average length 10 weeks. Price was 66% higher and firms earned 11% more revenue in cooperative periods.

Imperfect public monitoring: At each t each player observes nothing but a private signal that agrees with an opponent's action with probability $1 - \varepsilon$ and differs from it with probability ε (players cannot observe their own payoff until the game is over).

Prisoner's dilemma with grim trigger cannot be sustained

- Suppose P1 observes that P2 has played D; but P1 knows that in equilibrium P2 never deviates, so she thinks it is a mistake (i.e. that she made an erroneous observation)
- P1 knows that if she carries out the punishment (plays D at least once), then poor innocent P2 likely observes D, which triggers punishment from P2
- P1 prefers cooperation, so she won't play D as required
- Hence P2 can deviate without risking punishment, and the SPE breaks down!

J. Ely and J. Välimäki (2002) and M. Piccione (2002): belief-free equilibrium that is robust to imperfect private monitoring.

Construct an SPE i.e. find continuation values after various histories so that

- P1 is indifferent between choosing C and D if P2 is playing C in the same round, and the same is true in case P2 is playing D in the same round
- so P1 is indifferent between C and D at all t , after any history
- if P1 sees that P2 has played D, then P1 decreases the probability of playing C
- the same also holds with P1's and P2's roles reserved

Neither player cares if they observed D by mistake or the other really deviated ('belief-free'), hence this SPE is robust to imperfect private monitoring.

Folk Theorems for all games building on this